12 kW Wind Turbine M8

Build book · Ø8.0 m rotor · 8 m tower · revision A

Item	M8 on the 8 m tower
Rated power	12 kW electric at 10.0 m/s (hub height)
Rotor	Ø8.0 m, 3 glass/epoxy blades with carbon caps, TSR 5.5, 131 rpm, collective pitch
Regulation	variable speed + one electric pitch cylinder (supercap backed); feather in storms
Generator	Rev D direct-drive PM: Rev C + 2 magnet rows, stack 104 mm, 28 turns
Brake	3 spring-applied calipers, 14.4 kNm, rotor lock pin
Hub height	8.0 m; lowest blade tip 4.0 m above ground
Tower	one-piece tilt-up monopole Ø400 → Ø400, t 10/8, 671 kg, f1 2.78 Hz
Foundation	spread footing 3.0–3.4 m square, 6.2–7.8 m³; rock plinth 2 m³
Head mass	1362 kg above the tower flange (CAD, every part)
Energy (vs 18 m tower)	sheltered 13.4 MWh (−42 %) · inland 24.4 MWh (−27 %) · coastal 35.9 MWh (−18 %) · highland 47.8 MWh (−16 %)



Contents: design summary (§1); general arrangement and BOM (§2); part catalogue with an axonometric view of every part (§3); the NM8 drawing set rev A (§4); reused drawings (§5); the assembly sequence in 13 steps (§6); erection, commissioning and inspection (§7); open items (§8).

Sources: docs 35 and 37, analysis/variant_m8 (dlc8_m8.py, tower8_m8.py, spindle_m8.py, hub_yaw_m8.py, generator_m8.py), cad/turbine_m8 (model8.py, drawings8.py, iso8.py, bom8.py), cad/hub_m8. Every number comes from those scripts; re-run them to update the book.

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How to use this book

- Every part has a card in §3 (view, part number, material, mass, drawing). The drawing follows in §4 (new and changed M8 parts) or §5 (reused unchanged).
- Build in the order of §6. Each step lists the parts, the torques (NM8-900) and the checks that must be signed before the next step.
- Site work (footing, erection, commissioning) is §7. Keep the signed checklists with the turbine file.
- All drawings are NOT CHECKED. Release for the prototype only after the review and the checks of §8.

1 Design summary: 12 kW M8 on an 8 m tower

What the 8 m tower changes

The M8 turbine of doc 35 is unchanged above the yaw bearing except for what the detail design of doc 37 found: the third brake caliper moved to 135° (the rotor lock holds 90°), the hub rear gussets were shortened to clear the drum web, the spinner grew to Ø1 040 to enclose the pitch bearings, and the nacelle cover was lengthened 15 mm at the front.

Tower. A soft-stiff tower is impossible at 8 m, so the tower is stiff: f1 above 1.15 × 1P at 131 rpm and below 3P at rated. The 3P excitation crosses f1 at low speed, so the controller passes that speed band under torque.

Tower check	Value
Shell	Ø400 → Ø400, t 10/8, 671 kg
f1 rigid / soft soil + ice	2.78 / 2.65 Hz (window 2.51–5.95)
3P exclusion zone	48–61 rpm
Governing case	earthquake 0.5 g (ground D) + operation, utilisation 0.768
Fatigue damage (base)	0.53
Base bolts	16 × M24 10.9 on Ø500
Tilt-up	150.0 kNm, line pull 66.2 kN (gin pole 0.33 H)

Loads. The whole aeroelastic load set was rerun at 8 m hub height with stronger shear and turbulence and the stiff tower. Simulated at 8 m (103 cases, shear 0.25, turbulence +20 %, stiff tower): Mtilt 16.0, thrust 6.8 (design values, doc 37 §2). The 18 m controller tripped often in the stronger turbulence and once reached 220 rpm, so the 8 m controller has a faster speed loop (6 rad/s), a faster normal pitch rate (20°/s) and the trip at 165 rpm; peak 181 rpm (design 184). Blade, spindle, bearings, brake and tower were re-checked with the larger of the 18 m and 8 m envelopes; the brake grows to 14.4 kNm.

Energy at 8 m

Slower, more turbulent wind near the ground. Log wind profile per site roughness:

Site (roughness z0)	Mean wind 18 m → 8 m	AEP 18 m	AEP 8 m	Loss	Turbulence
Sheltered valley / farm (0.25 m)	5.0 → 4.05 m/s	23.0 MWh	13.4 MWh	−42 %	+23 %
Open inland lowland (0.05 m)	6.0 → 5.17 m/s	33.4 MWh	24.4 MWh	−27 %	+16 %
Coast / exposed lowland (0.01 m)	7.0 → 6.24 m/s	43.8 MWh	35.9 MWh	−18 %	+12 %
Highland / headland (0.03 m)	8.5 → 7.42 m/s	57.0 MWh	47.8 MWh	−16 %	+14 %

- Coast, highland and open land lose 16–27 %; a sheltered farm loses 42 % and should use the 18 m tower.
- Safety: lowest blade tip 4.0 m above ground. Fenced keep-out radius 5 m; ice-fall zone 24 m when icing is possible.

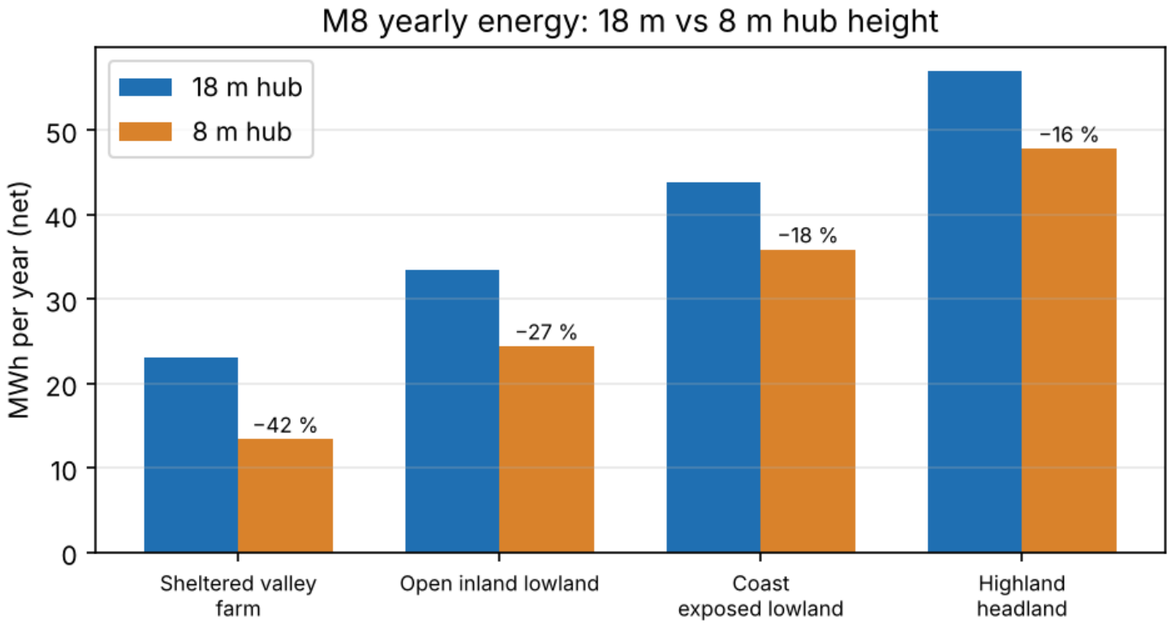
Decisions used in this book (doc 35 defaults, open for Martin)

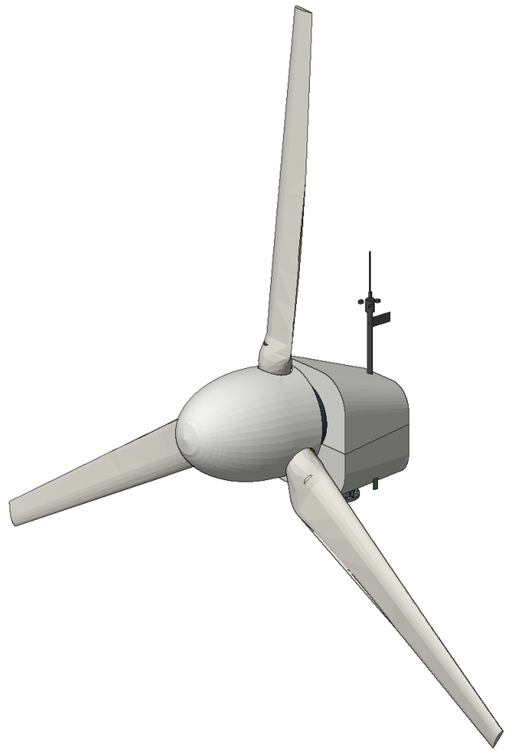
- D1 bedplate NP8-300 (136 kg) instead of NP1-300 (338 kg).
- D2 three calipers 4.8 kNm (third at 135°) instead of two stronger ones.
- D3 plain carbon caps; D4 keep 28 turns; D5 crank r140 (pin plane moved to 150, clash-free).

Main parts

- Blades 3 × 34.1 kg; hub and pitch 337 kg budget; generator Rev D; NP8-300 bedplate; yaw ring Dpw 410.
- Spindle reserve 2.5, bearing s0 6.7/6.2, life factor 5.6.

1 Design summary: figures and masses





Mass by group (assembled turbine)

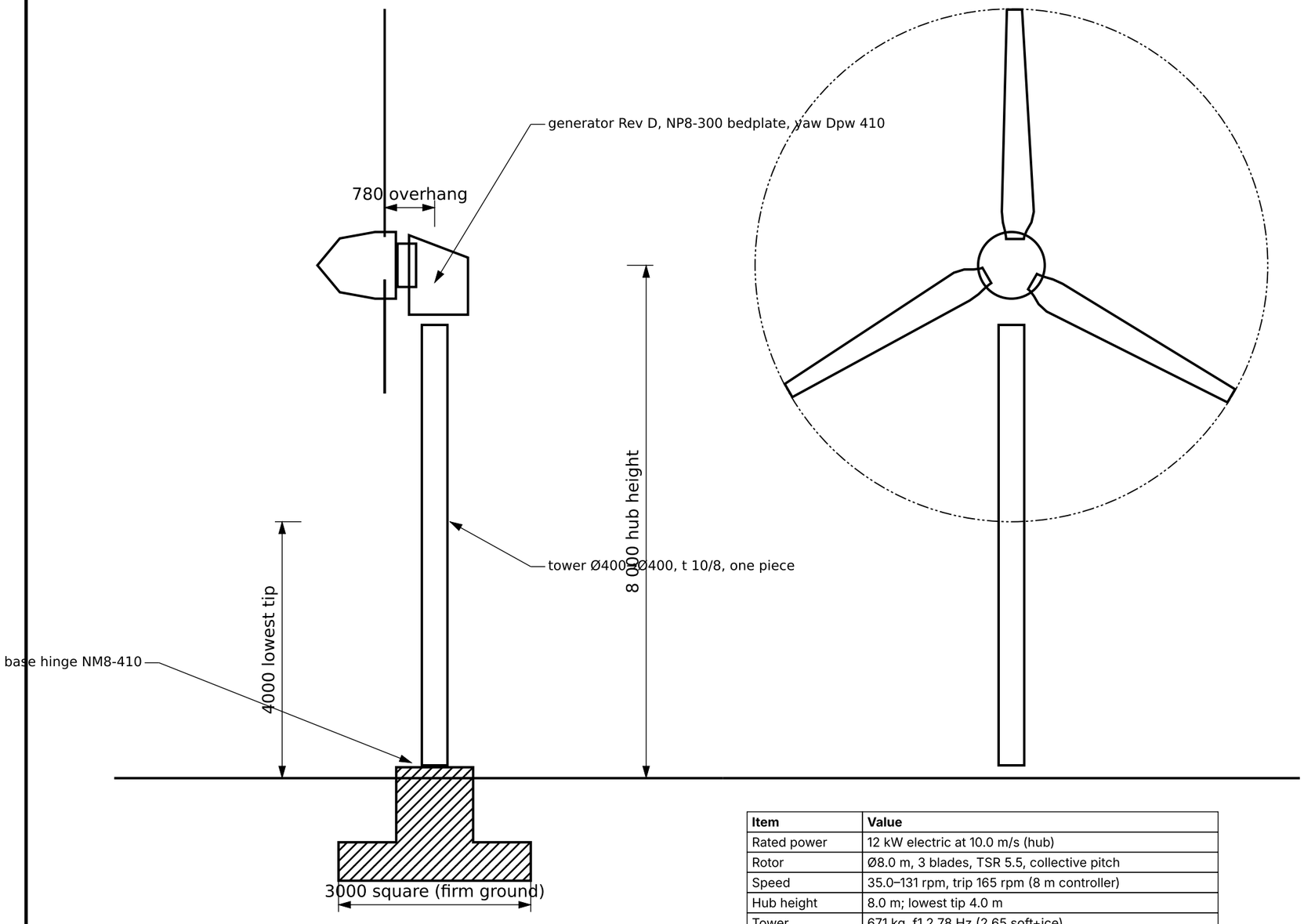
Group	kg (CAD)
nacelle	170
electrical	134
yaw	288
drivetrain	440
rotor	455
tower	741
foundation	15042
Head above the tower flange	1362

CAD head mass includes every modelled part (cover, frame, adapter, lock holder, junction box, spinner); the doc 35 budget was 1,280 kg. The tower was checked with the larger value.

2 General arrangement and BOM

NM8-000 … NM8-020

SIDE VIEW (wind from the left)



VIEW FROM UPWIND

AXONOMETRIC VIEW (not to scale)



Item	Value
Rated power	12 kW electric at 10.0 m/s (hub)
Rotor	Ø8.0 m, 3 blades, TSR 5.5, collective pitch
Speed	35.0–131 rpm, trip 165 rpm (8 m controller)
Hub height	8.0 m; lowest tip 4.0 m
Tower	671 kg, f1 2.78 Hz (2.65 soft+ice)
Exclusion zone	48–61 rpm (3P × f1)
Head mass	1362 kg above the tower flange (CAD)
Energy vs 18 m	−16 % highland … −27 % inland
Stopping	pitch to feather (supercap) + 3 spring calipers 14.4 kNm
Storm	feathered, idling or braked, yaw active

NOTES

1. Docs 35 and 37. Shared 12 kW parts keep their drawings (index NM8-001). Generator Rev D change sheet NM8-215.
2. Keep people and vehicles out of the rotor disc: fenced zone of radius 5 m round the tower base; lowest tip 4.0 m. Ice-fall zone 1.5 × (H + D) = 24 m when icing is possible.
3. Siting at 8 m: rotor bottom must be clear of the wake of buildings and shelter belts (≥ 20 × obstacle height upwind in the main wind directions); snow drift must stay below 2.5 m.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

GENERAL ARRANGEMENT · 12 kW M8 ON 8 m TOWER

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-000	A · 2026-11-06	1:75
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM NM8-002/003	≈ 1362 kg head + 671 kg tower	Claude / not checked

Drawing	Title	Source
NM8-000	GENERAL ARRANGEMENT · 12 kW M8 ON 8 m TOWER	M8 rev A
NM8-001	DRAWING INDEX	M8 rev A
NM8-002	BILL OF MATERIALS (1/2)	M8 rev A
NM8-003	BILL OF MATERIALS (2/2)	M8 rev A
NM8-010	GENERAL ARRANGEMENT · NACELLE AND DRIVETRAIN	M8 rev A
NM8-020	ROTOR ASSEMBLY · HUB, PITCH SYSTEM, BLADES, SPINNER	M8 rev A
NM8-100	BLADE Ø8.0 m · PLANFORM, TWIST AND ROOT	M8 rev A
NM8-105	BLADE ROOT INSERT M16 (BONDED)	M8 rev A
NM8-110	BLADE MOULD SET (PLUG, TWO HALVES, FRAME)	M8 rev A
NM8-200	HUB WELDMENT (COLLECTIVE PITCH)	M8 rev A
NM8-210	SPINNER (GRP) WITH BLADE BOOTS	M8 rev A
NM8-215	GENERATOR REV D · CHANGE SHEET (STACK 74 → 104 mm)	M8 rev A
NM8-250	COLLECTIVE PITCH · BEARINGS, CRANKS, LINKS, YOKE, CYLINDER	M8 rev A
NP8-300	BEDPLATE WELDMENT 12 kW NP8 (LIGHT)	M8 rev A
NM8-311	ROTOR BRAKE · THREE CALIPERS AND THIRD CALIPER BRACKET	M8 rev A
NM8-911	YAW SYSTEM LAYOUT · SLEWING RING Dpw 410	M8 rev A
NM8-400	TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE	M8 rev A
NM8-410	BASE HINGE, GIN POLE INTERFACE AND ERECTION	M8 rev A
NM8-500	FOUNDATION · SPREAD FOOTING (8 m TOWER, M8)	M8 rev A
NM8-900	FASTENER AND TORQUE SCHEDULE	M8 rev A
NM8-910	BOUGHT PARTS SPECIFICATION (ENVELOPES AND INTERFACES)	M8 rev A
NP1-101/102	Front spacer, rear seal sleeve (Rev B)	12 kW P1 rev B
NP1-202	Stator front clamp ring	12 kW P1 rev B
NP1-211	Bearing seal covers front / rear	12 kW P1 rev B
NP1-212C	Drum rear cover with 24 fan blades	Rev C (doc 23)
NP1-230	Brake disc 16 mm Ø880 / Ø652	12 kW P1 rev B
NP1-310	Brake caliper brackets 3 and 9 o'clock	12 kW P1 rev B
NP1-320/321	Rotor lock pin and holder	12 kW P1 rev B
NP1-420	Yaw drive brackets (radius +30)	12 kW doc 25
NY-400…460	Adapter, pad units, lock, encoder, cable hanger (radii +25, see NM8-410)	12 kW doc 25
NM8-700/710/720/730	Nacelle cover, frame, met mast, junction box (S6 parts, cover front 15 mm longer	S6 rev B
NM8-420	Gin pole and tilt-up rigging (S6 tool, line pull see NM8-400)	S6 rev B
NM8-600/620/800	Resistor box, base cabinet, cabling routes (S6)	S6 rev B

NOTES

1. Rev A (2026-11-06): first issue for the 12 kW M8 on the 8 m tower (doc 37). Every new or changed part has an axonometric view from the 3D model (cad/turbine_m8/model8.py).
2. Reused drawings are used unchanged except where this index says otherwise. Their PDFs are in the earlier sets (cad/nacelle_p1, cad/yaw, cad/turbine_6kw).
3. All sheets NOT CHECKED: prototype release only after review, the open checks of doc 37 §9 and the supplier drawings of the bought parts.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

DRAWING INDEX (NM8 SET REV A + REUSED DRAWINGS)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-001	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
–	–	Claude / not checked

Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
		A ROTOR				
1	NM8-100	Blade Ø8.0 m, glass/epoxy, carbon UD caps (B1–B3)	3	make	E-glass/epoxy, carbon UD caps, PVC core	34.1
2	NM8-105	Root insert M16 (12 per blade, bonded)	36	make	C45E+QT zinc-flake coated	0.8
3	NM8-200	Hub weldment: shell Ø530 × 10, 3 necks Ø360, drum flange	1	make	S355NL, welded, stress relieved, machined	179.8
4	NM8-210	Spinner (GRP 4 mm) with blade boots	1	make	glass/polyester, gelcoat RAL 9010	27.6
5	NM8-250	Pitch bearings Dpw 300 four-point (3, bought)	1	buy	bought: 42CrMo4 rings	76.2
6	NM8-260	Pitch cranks r140 with pins (3)	1	make	42CrMo4+QT / S355 tube	30.0
7	NM8-261	Pitch links 220 mm with spherical eyes (3)	1	make	S355 / bought rod ends	4.5
8	NM8-262	Yoke + 3 guide rods	1	make	S355 plate / C45 chromed rods	8.0
9	NM8-270	Electric pitch cylinder ~28.5 kN rated, 204 mm stroke + motor	1	buy	bought	14.0
10	NM8-280	Supercap module + pitch drive/controller (bought)	1	buy	bought	6.5
11	NM8-290	Nose frame (actuator + electronics carrier)	1	make	S355 rings + tube struts	6.0
		B DRIVETRAIN AND GENERATOR REV D				
12	NM8-311	Third caliper bracket (135°)	1	make	S355J2+N 25 plate, welded	7.0
13	NM8-913	Tapered roller bearing 32028 X (front + rear)	2	buy	bought 32028 X	4.0
14	NM8-914	Overspeed sensors M18 inductive (2, bought)	1	buy	bought M18 PNP	0.4
15	NM8-915	Spring-applied brake caliper 4.8 kNm (3 on the disc)	3	buy	bought, spring-applied	9.0
16	NP1-100D	Spindle Rev D (30 mm longer between the rear seat and the flan	1	make	42CrMo4+QT forged	59.2
17	NP1-101	Front spacer / seal counterface Ø180 (Rev B)	1	make	42CrMo4+QT, induction hardened	1.4
18	NP1-102	Rear seal sleeve Ø195 on the Ø155 shoulder (Rev B)	1	make	42CrMo4+QT, induction hardened	1.3
19	NP1-200D	Stator carrier Rev D, 45 dovetail grooves, rim 104 mm	1	make	S355J2+N	45.4
20	NP1-201D	Stator Rev D: 45 tooth-segment stacks, 104 mm (7 magnet rows)	1	make	M270-35A 0.35 mm, self-bonding	71.0
21	NP1-202	Stator front clamp ring	1	make	1.4404 (non-magnetic)	2.2
22	NP1-203D	Coils on segments, 45 × 28 turns, 104 mm stack	1	make	Cu enamelled Ø1.55 class 200, NKN 0.30	20.0
23	NP1-210D	Rotor drum weldment Rev D (shell +30; Rev C fins and ribs, see	1	make	S355J2+N	95.5
24	NP1-211-F	Front bearing seal cover	1	make	S355J2	1.5
25	NP1-211-R	Rear bearing seal cover, bore Ø198 (Rev B)	1	make	S355J2	1.4
26	NP1-212C	Drum rear cover with 24 fan blades (Rev C; blades not modelled	1	make	EN AW-5083-H111	7.2
27	NP1-220F	Magnets 280 × N45SH 30×15×12 in 7 rows + comb + sleeve	1	make	NdFeB N45SH	15.0
28	NP1-230	Brake disc Ø880/Ø652 with 12 lock holes (Rev B)	1	make	C45E+N 16 mm	33.6
29	NP1-310	Rotor brake caliper bracket (left, Rev B)	1	make	S355J2+N 25 plate	18.1
30	NP1-320	Rotor lock pin Ø20 (engaged, top)	1	make	42CrMo4+QT	0.5
31	NP1-321	Lock pin holder with 2 overspeed sensor seats	1	make	S355J2	24.5
		C NACELLE				
32	NM8-700-1	Nacelle cover, lower tray (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	15.7
33	NM8-700-2	Nacelle cover, upper lid (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	17.7
34	NM8-710	Cover support arms + rear rail	1	make	S235 galv.	8.3
35	NM8-720	Met mast: anemometer, vane, air terminal	1	make	1.4404 tube 42.4 × 2 + bought sensors	4.5
36	NP8-300	Bedplate weldment 12 kW NP8 (plates 20 / 30 / 10)	1	make	S355J2+N plates, stress relieved, machined	124.3
		D YAW SYSTEM				
37	NM8-911	Yaw slewing ring Dpw 410, z165 m4	1	buy	bought: 42CrMo4, induction hardened teeth	66.0
38	NM8-912	Yaw gearmotor 0.25 kW, output ≥ 2.3 kNm static, motor brake	2	buy	bought	19.0
39	NP1-420	Yaw drive bracket	2	make	S355J2+N	7.7
40	NY-400	Tower-top adapter with friction track	1	make	S355J2+N weldment	118.3
41	NY-410	Friction pad unit	2	make	S355 / DIN 2093 springs / organic pads	14.0
42	NY-430	Yaw lock (screw block)	1	make	42CrMo4 / S355	6.5
43	NY-440	Yaw encoder + cam limit switch unit	1	make	S355 + absolute encoder + cam switch	6.0
44	NY-460	Cable hanger tube + loop cleat	1	make	S355 tube 168.3 × 4.5	10.0
		E ELECTRICAL				
45	NM8-600	Braking / dump resistor box (vented)	1	make	1.4301 box + steel-grid resistors	18.0
46	NM8-620	Tower base cabinet IP55 800 × 600 × 300 on stand	1	make	stand S235 galv.; bought enclosure	95.0
47	NM8-730	Nacelle junction box IP66 (400 × 340 × 150)	1	make	bought: 1.4301 enclosure	9.0

NOTES

1. Masses from the CAD model (cad/turbine_m8/model8.py) for made parts; supplier data or the doc 35 budget for bought and pitch parts. Full BOM: bom8.csv.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BILL OF MATERIALS (1/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-002	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
48	NM8-800	Drop cable + base duct (cabling route)	1	make	H07RN-F 4G10 + 2 × LiYCY signal; PE duct Ø11	12.0
		F TOWER				
49	NM8-400	Tower 8 m, Ø400 → Ø400, t 10/8, one piece	1	make	S355J2+N, hot-dip galvanised	699.8
50	NM8-401	Hand-hole cover 200 × 350 × 6 with gasket	1	make	S355 galv.	3.3
51	NM8-410	Base hinge brackets + hinge pin Ø50	1	make	S355J2+N galv.; pin 42CrMo4+QT	37.4
		G FOUNDATION				
52	NM8-500	Spread footing (firm ground) with pedestal	1	make	C30/37 XF4 air-entrained, B500C	14880.0
53	NM8-510	Anchor cage 16 × M24 + 4 hinge anchors	1	make	M24 8.8 HDG threaded rod, S355 template ring	161.7
		H TOOLS AND MOULDS				
54	NM8-110-1	Blade mould, lower (pressure side) half on frame	1	make	GRP tooling laminate 12 mm, vinylester tooli	118.1
55	NM8-110-2	Blade mould, upper (suction side) half	1	make	GRP tooling laminate 12 mm	118.1
56	NM8-110-3	Mould frame (RHS 80 × 80 × 4)	1	make	S235 RHS 80×80×4	178.6
57	NM8-420	Gin pole 3.0 m (erection tool)	1	make	S355 tube 168.3 × 6.3	94.0
		J FASTENERS, CABLES, SITE MATERIALS				
58	NM8-900-01	Bolt M16 × 90 10.9 zinc-flake + washer (blade roots into the p	36	buy	10.9 zinc-flake	0.20
59	NM8-900-02	Bolt M16 × 70 10.9 (hub to drum)	12	buy	10.9 zinc-flake	0.17
60	NM8-900-03	Bolt M16 × 80 10.9 (pitch bearing outer rings to hub necks, 20	60	buy	10.9 zinc-flake	0.18
61	NM8-900-04	Bolt M20 × 90 10.9 (spindle to bedplate)	12	buy	10.9 zinc-flake	0.32
62	NM8-900-05	HV set M16 × 110 10.9 (yaw inner ring, PCD 350)	16	buy	EN 14399-4	0.26
63	NM8-900-06	Bolt M16 × 70 10.9 (yaw outer ring PCD 490, pads, calipers, br	40	buy	10.9 zinc-flake	0.17
64	NM8-900-07	HV set M20 × 90 10.9 (adapter to tower)	16	buy	EN 14399-4	0.40
65	NM8-900-08	Nut M24 8.8 HDG + plate washer (base 16 × 3, hinge 8)	56	buy	8.8 HDG	0.15
66	NM8-900-09	Small fasteners kit M6–M12 A4 (cover, spinner, mast, nose fram	1	buy	A4-70	3.00
67	NM8-800-W1	Cable H07RN-F 4G10 (generator → box)	3	buy	m	0.60
68	NM8-800-W2	Cable H07RN-F 4G10 (box → loop → hand hole)	12	buy	m	0.60
69	NM8-800-W3	Cable NYY-J 4 × 10 (hand hole → cabinet)	4	buy	m	0.50
70	NM8-800-W4	Cable ÖLFLEX 4G1.5 shielded (3 brake calipers)	20	buy	m	0.10
71	NM8-800-W5	Cable LiYCY 12 × 0.5 (sensors)	16	buy	m	0.12
72	NM8-800-W6	Cable ÖLFLEX 7G1.5 (yaw motors)	16	buy	m	0.15
73	NM8-800-W7	Slip ring 6 power + 6 signal (pitch supply and bus, in the nos	1	buy		1.50
74	NM8-800-W8	Earthing conductor Cu 50 mm² + ring earth + 2 rods 3 m	1	buy	Cu	18.00
75	NM8-500-M1	Concrete C30/37 XF4 air-entrained (firm ground)	6.2 m³	buy	ready-mix	
76	NM8-500-M2	Rebar B500C (mesh top + bottom, pedestal cage)	1	buy	≈ 460 kg	460.00
77	NM8-500-M3	Non-shrink grout ≥ 40 MPa, 30 mm under the base flange	70 l	buy		
78	NM8-600-K1	Contactor K1 3-pole NC (mirror contacts) + safety relay PL d	1	buy		2.00
79	NM8-620-PE	Converter 12 kW, dump chopper, AC module, 48 V module, control	1	make	in-house electronics	60.00

NOTES

1. Masses from the CAD model (cad/turbine_m8/model8.py) for made parts; supplier data or the doc 35 budget for bought and pitch parts. Full BOM: bom8.csv.

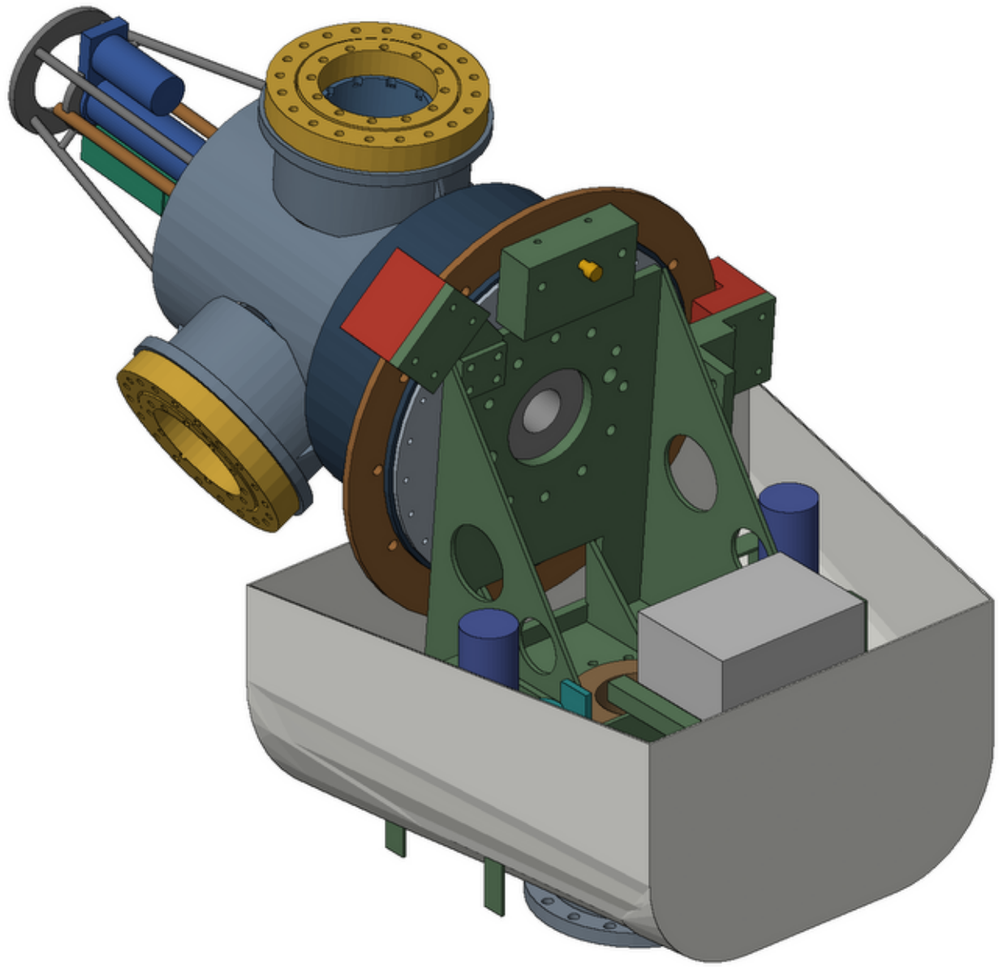
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

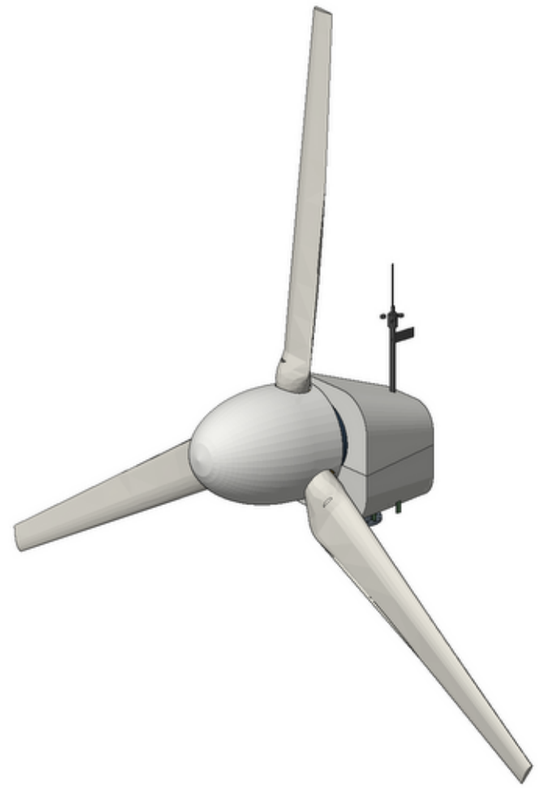
BILL OF MATERIALS (2/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-003	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Interface	Value
Rotor plane to yaw axis	780 mm (doc 35 overhang; 750 + 30 Rev D)
Rotor centre above yaw bearing top	570 mm; shaft tilt 5°
Generator Rev D	stack 104 mm, 7 rows, 280 blocks, 28 turns
Stretch plane	spindle x 410: parts ahead move 30 mm forward
Bearings	2 × 32028 X, span 240 (unchanged)
Brake	3 spring calipers at 0°, 135°, 180° (lock pin at 90°)
Bedplate	NP8-300, 136 kg (plate model), util 0.68
Yaw	ring Dpw 410 z165 m4, 2 gearmotors, 2 pad units
Hub to drum	12 × M16 10.9 PCD 400, Ø360 spigot
Spindle to bedplate	12 × M20 10.9 PCD 350, Ø200 spigot

NOTES

1. A03: nacelle with the upper cover lid, met mast, spinner and blades removed, seen from the rear left. A02: complete head.
2. Whole-turbine interference check (cad/turbine_m8/clash8.py): no clashes left except intended fits (spigots, pins in eyes, bolts in tapped holes, embedded anchors, gear mesh contact) - list in doc 37 §6.
3. The hub, drum, stator and spindle front move 30 mm forward together, so all front interfaces (hub spigot, bearing seats, seal covers) are as drawn on the Rev B/C sheets; only the lengths in NM8-215 change.

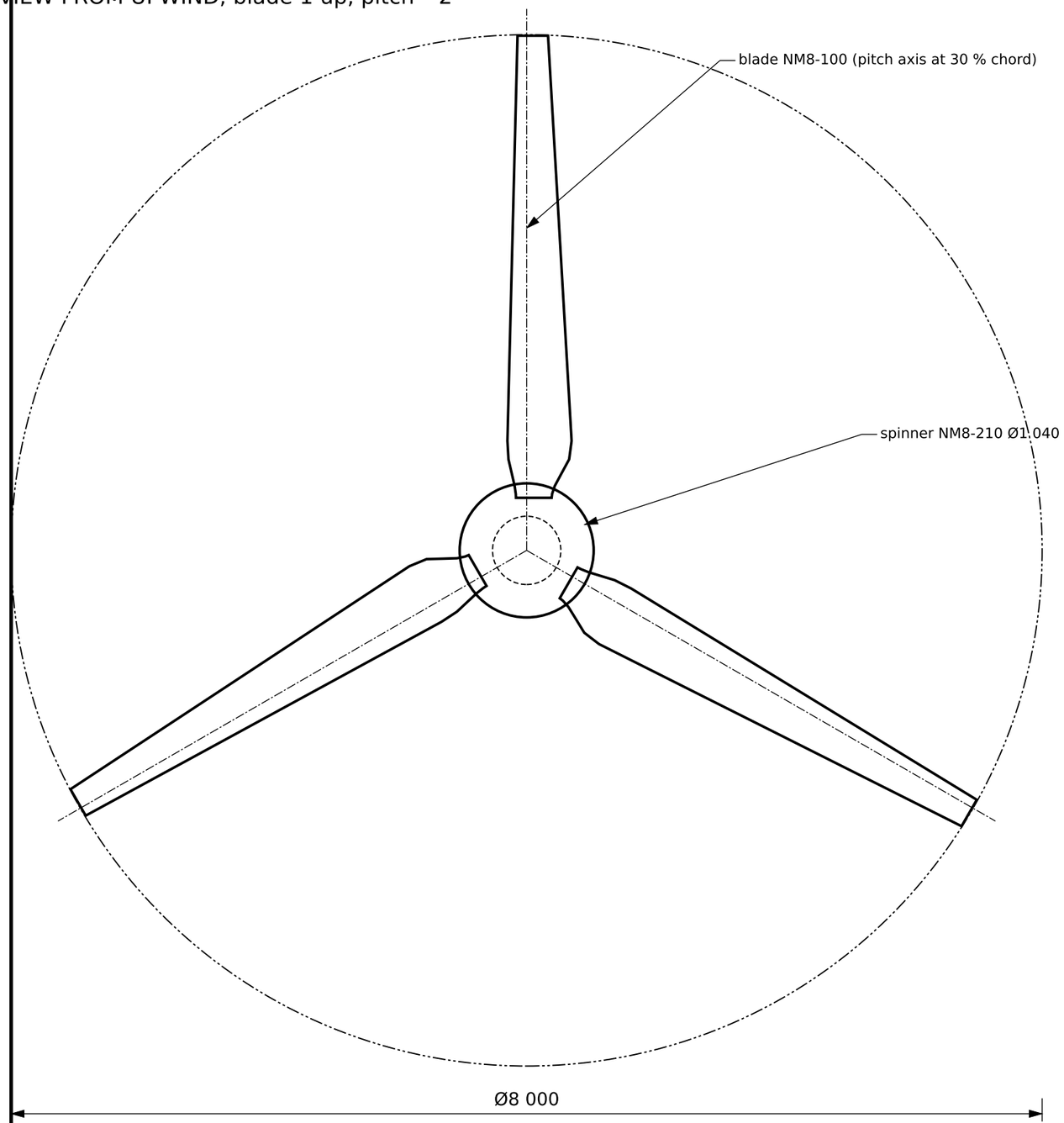
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

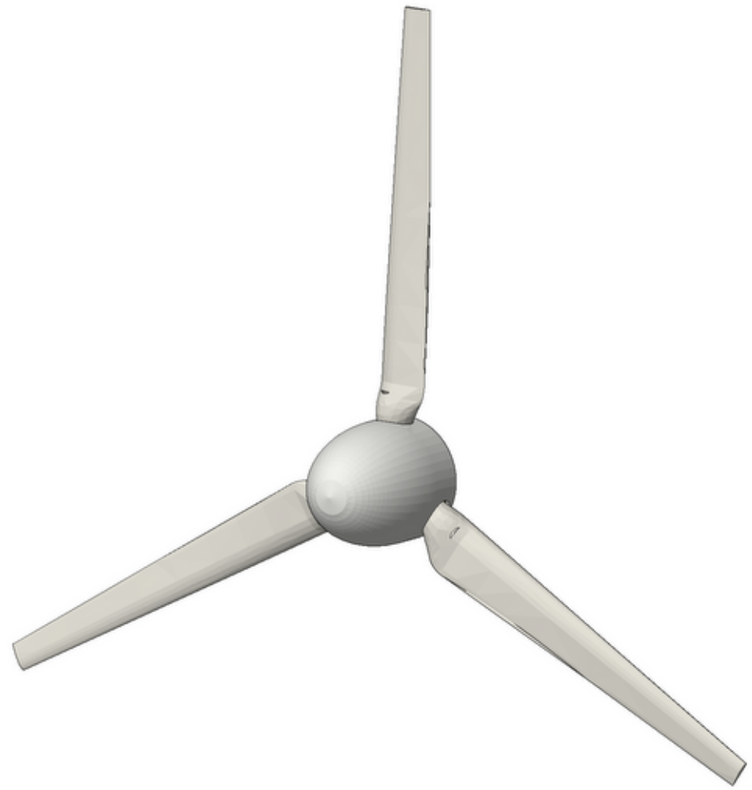
GENERAL ARRANGEMENT · NACELLE AND DRIVETRAIN

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-010	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 1362 kg above the tower flange	Claude / not checked

VIEW FROM UPWIND, blade 1 up, pitch −2°



AXONOMETRIC VIEW (not to scale)



Item	Value
Blades	3 × 34.1 kg, carbon UD caps (D3)
Blade root	Ø280, 12 × M16 inserts PCD 245
Pitch bearings	4-point Dpw 300, balls 30, Z 28
Pitch drive	one electric cylinder, crank r140, collective
Hub	S355NL weldment 180 kg (budget)
Rotor inertia	≈ 385 kg m²
Pitch range	−2° (fine) … 92° (feather)

NOTES

1. Assembly order (NM8-020 = build book steps 8–10): hub pre-assembled with bearings, cranks, links, yoke, cylinder and nose frame on the bench; mounted on the drum; blades fitted with the cylinder at the −2° end stop; spinner last.
2. Static-moment matched blade set: moments within 0.5 % (balance by root ballast pocket); mark each blade with its hub position.
3. Bolt torques NM8-900. Blade root bolts M16 10.9: 280 Nm, re-torque after 50 h.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

ROTOR ASSEMBLY · HUB, PITCH SYSTEM, BLADES, SPINNER

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-020	A · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 455 kg	Claude / not checked

3 Part catalogue

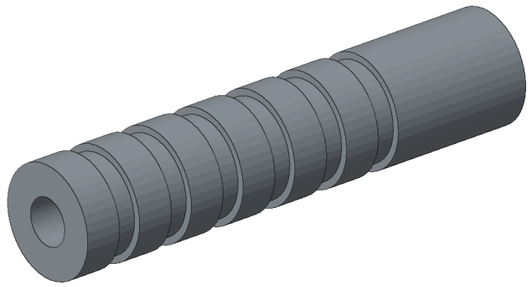
Axonometric view, part number, material and mass of every part

Axonometric view, part number, material and mass of every part

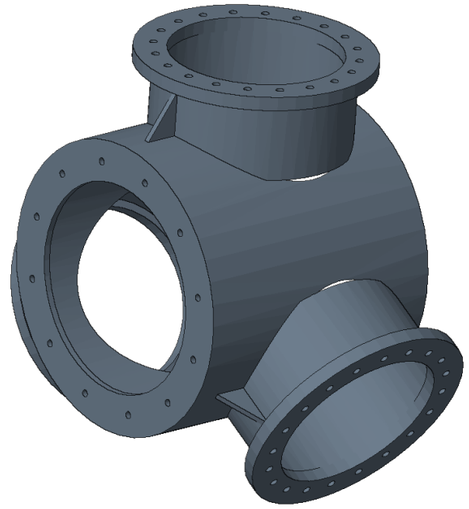
3 Part catalogue (axonometric views)



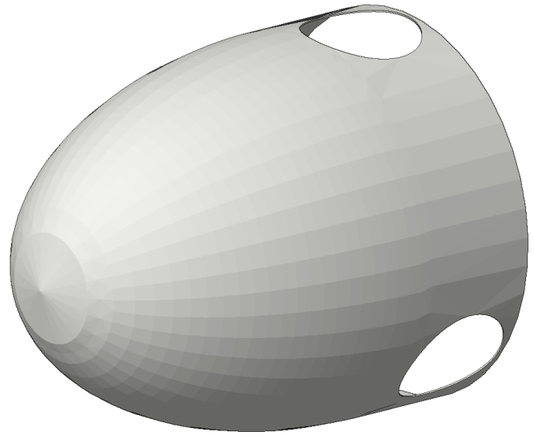
NM8-100 ×3
Blade Ø8.0 m, glass/epoxy, carbon UD caps (B1–B3)
E-glass/epoxy, carbon UD caps, PVC core
make · 34.1 kg each · drawing NM8-100



NM8-105 ×36
Root insert M16 (12 per blade, bonded)
C45E+QT zinc-flake coated
make · 0.8 kg each · drawing NM8-105



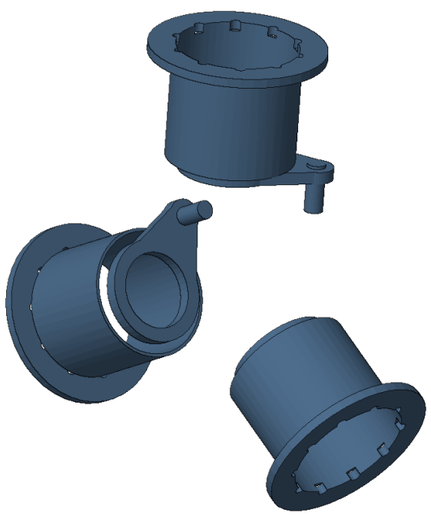
NM8-200 ×1
Hub weldment: shell Ø530 × 10, 3 necks Ø360, drum flange
S355NL, welded, stress relieved, machined
make · 179.8 kg each · drawing NM8-200



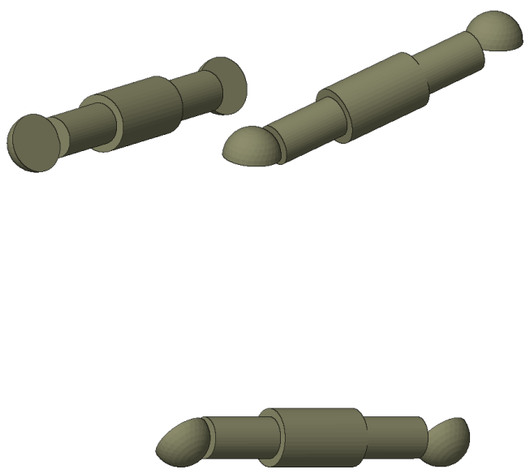
NM8-210 ×1
Spinner (GRP 4 mm) with blade boots
glass/polyester, gelcoat RAL 9010
make · 27.6 kg each · drawing NM8-210



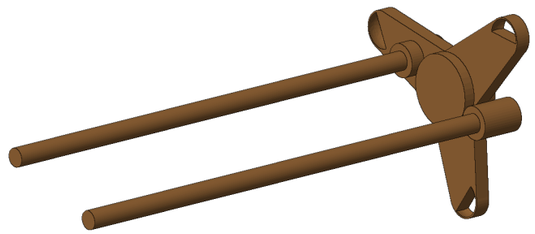
NM8-250 ×1
Pitch bearings Dpw 300 four-point (3, bought)
bought: 42CrMo4 rings
buy · 76.2 kg each · drawing NM8-250



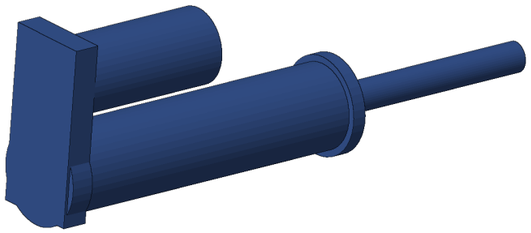
NM8-260 ×1
Pitch cranks r140 with pins (3)
42CrMo4+QT / S355 tube
make · 30.0 kg each · drawing NM8-260



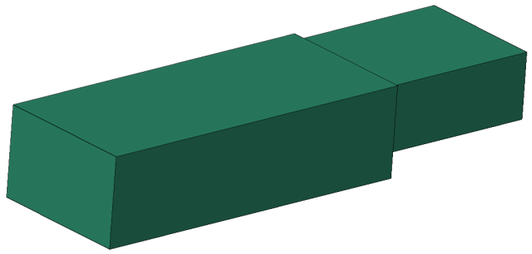
NM8-261 ×1
Pitch links 220 mm with spherical eyes (3)
S355 / bought rod ends
make · 4.5 kg each · drawing NM8-261



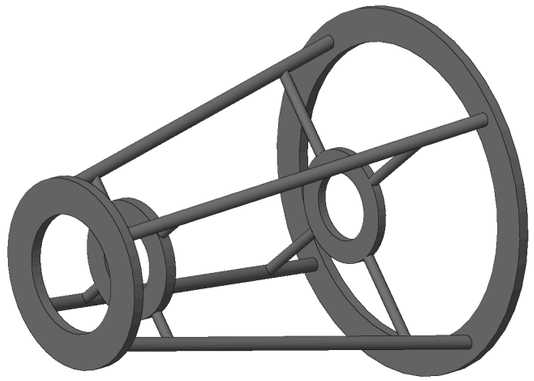
NM8-262 ×1
Yoke + 3 guide rods
S355 plate / C45 chromed rods
make · 8.0 kg each · drawing NM8-262



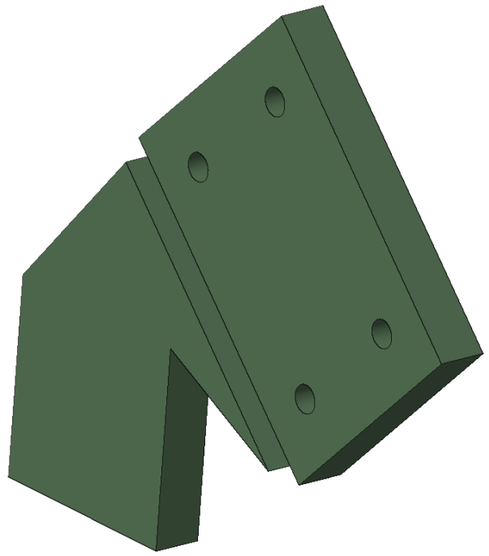
NM8-270 ×1
Electric pitch cylinder ~28.5 kN rated, 204 mm stroke + motor (bought)
bought
buy · 14.0 kg each · drawing NM8-250



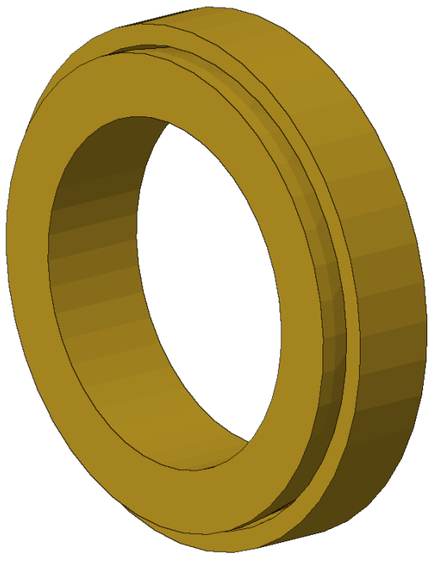
NM8-280 ×1
Supercap module + pitch drive/controller (bought)
bought
buy · 6.5 kg each · drawing NM8-250



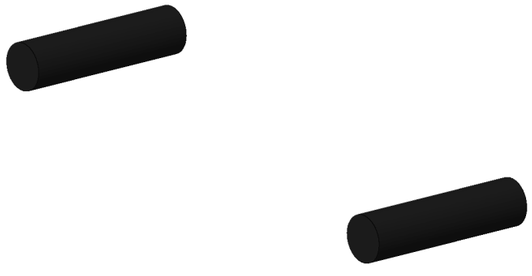
NM8-290 ×1
Nose frame (actuator + electronics carrier)
S355 rings + tube struts
make · 6.0 kg each · drawing NM8-290



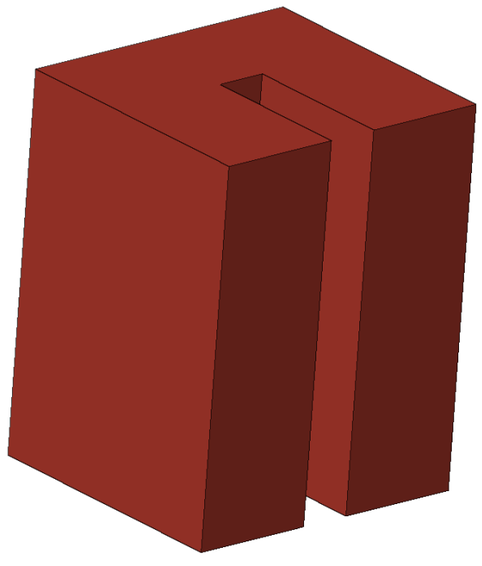
NM8-311 ×1
Third caliper bracket (135°)
S355J2+N 25 plate, welded
make · 7.0 kg each · drawing NM8-311



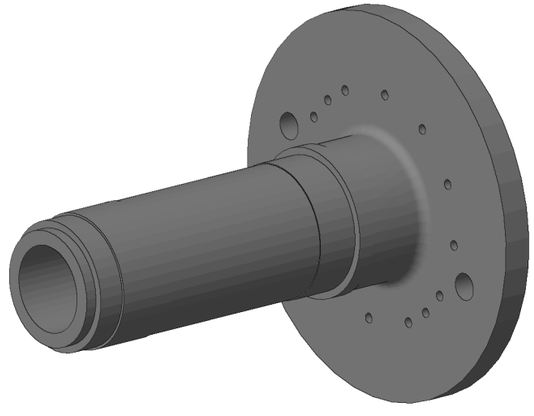
NM8-913 ×2
Tapered roller bearing 32028 X (front + rear)
bought 32028 X
buy · 4.0 kg each · drawing NM8-910



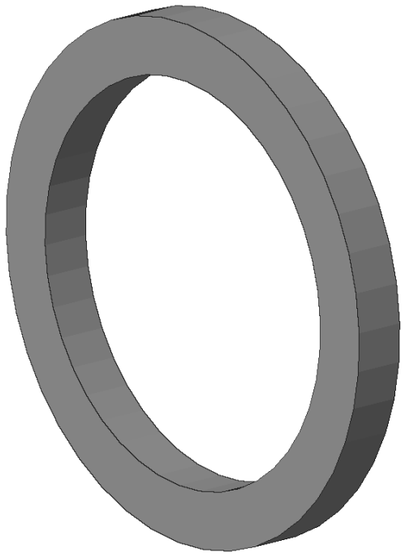
NM8-914 ×1
Overspeed sensors M18 inductive (2, bought)
bought M18 PNP
buy · 0.4 kg each · drawing NM8-910



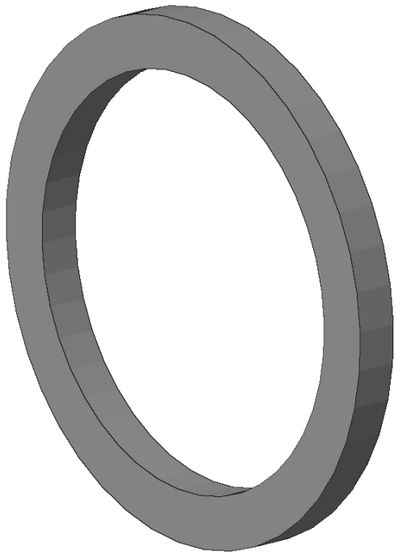
NM8-915 ×3
Spring-applied brake caliper 4.8 kNm (3 on the disc)
bought, spring-applied
buy · 9.0 kg each · drawing NM8-910



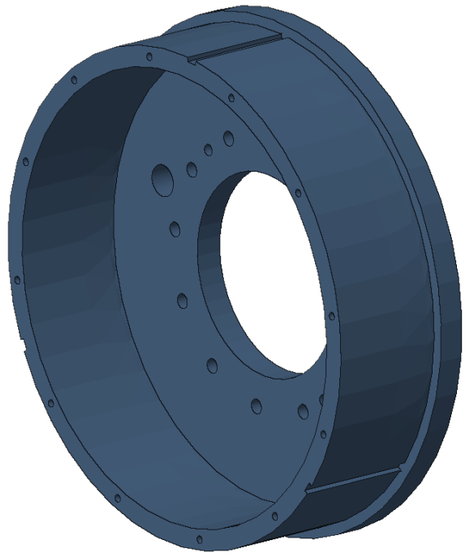
NP1-100D ×1
Spindle Rev D (30 mm longer between the rear seat and the flange)
42CrMo4+QT forged
make · 59.2 kg each · drawing NP1-100D



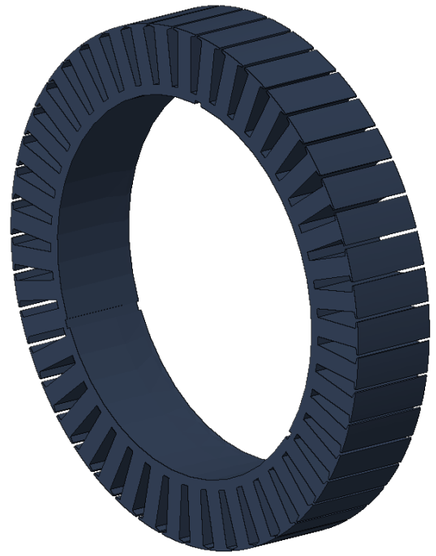
NP1-101 ×1
Front spacer / seal counterface Ø180 (Rev B)
42CrMo4+QT, induction hardened
make · 1.4 kg each · drawing NP1-101



NP1-102 ×1
Rear seal sleeve Ø195 on the Ø155 shoulder (Rev B)
42CrMo4+QT, induction hardened
make · 1.3 kg each · drawing NP1-102



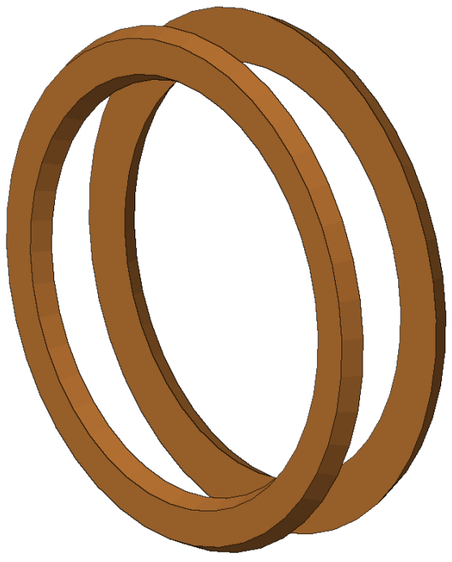
NP1-200D ×1
Stator carrier Rev D, 45 dovetail grooves, rim 104 mm
S355J2+N
make · 45.4 kg each · drawing NP1-200D



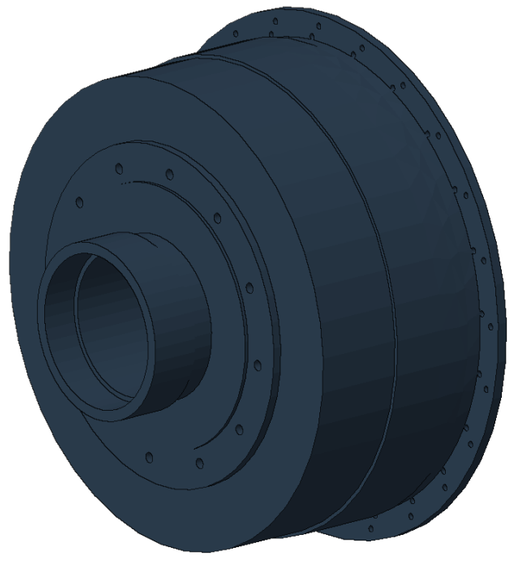
NP1-201D ×1
Stator Rev D: 45 tooth-segment stacks, 104 mm (7 magnet rows)
M270-35A 0.35 mm, self-bonding
make · 71.0 kg each · drawing NP1-201D



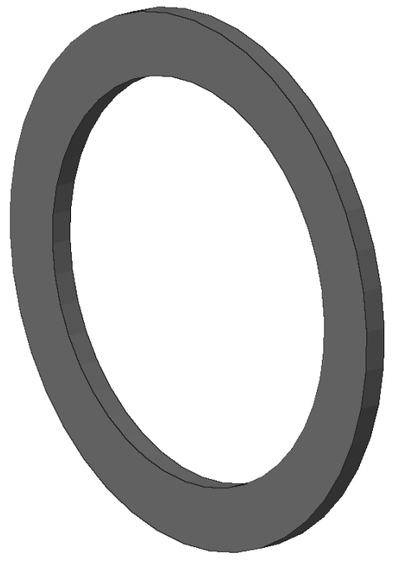
NP1-202 ×1
Stator front clamp ring
1.4404 (non-magnetic)
make · 2.2 kg each · drawing NP1-202



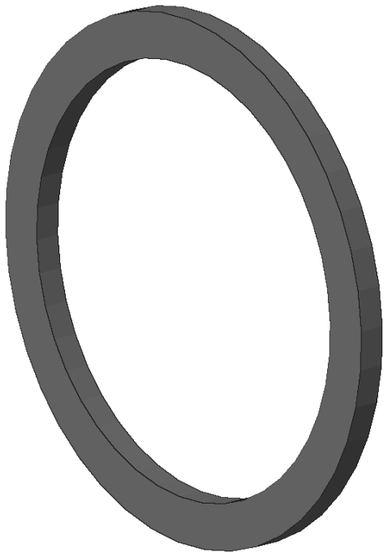
NP1-203D ×1
Coils on segments, 45 × 28 turns, 104 mm stack
Cu enamelled Ø1.55 class 200, NKN 0.30
make · 20.0 kg each · drawing NP1-203D



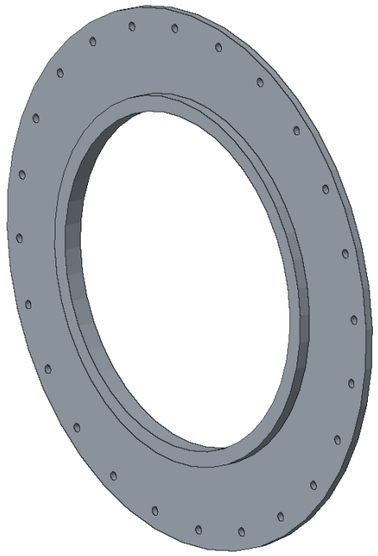
NP1-210D ×1
Rotor drum weldment Rev D (shell +30; Rev C fins and ribs, see NP1-210C)
S355J2+N
make · 95.5 kg each · drawing NP1-210D



NP1-211-F ×1
Front bearing seal cover
S355J2
make · 1.5 kg each · drawing NP1-211



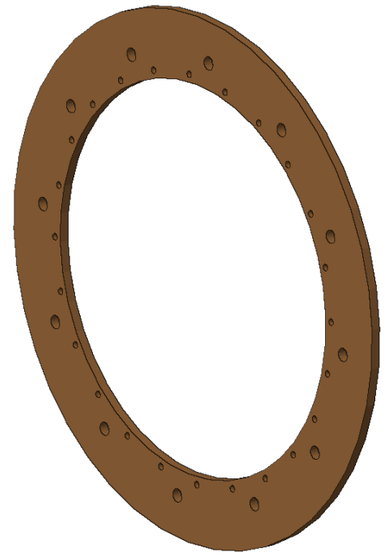
NP1-211-R ×1
Rear bearing seal cover, bore Ø198 (Rev B)
S355J2
make · 1.4 kg each · drawing NP1-211



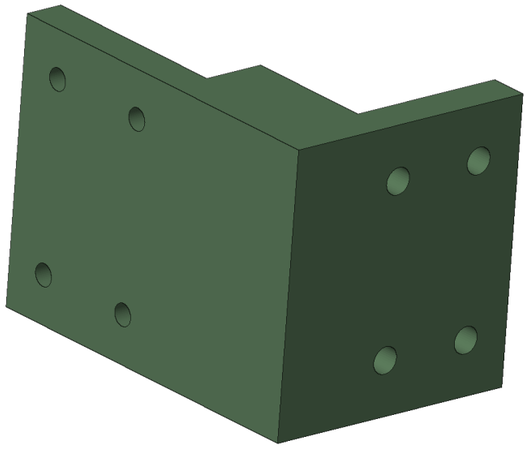
NP1-212C ×1
Drum rear cover with 24 fan blades (Rev C; blades not modelled)
EN AW-5083-H111
make · 7.2 kg each · drawing NP1-212C



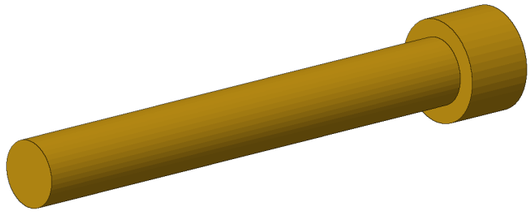
NP1-220F ×1
Magnets 280 × N45SH 30×15×12 in 7 rows + comb + sleeve
NdFeB N45SH
make · 15.0 kg each · drawing NP1-220F



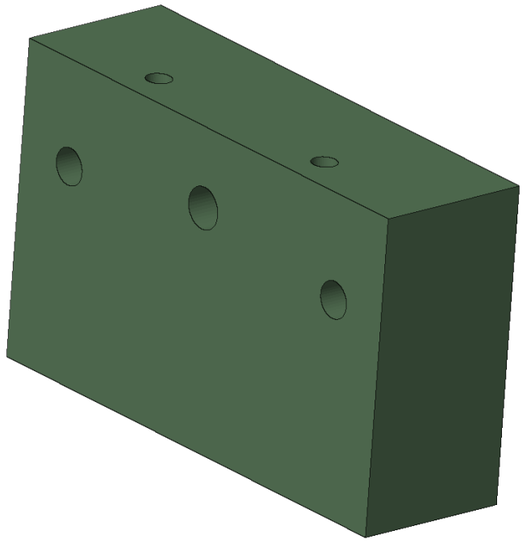
NP1-230 ×1
Brake disc Ø880/Ø652 with 12 lock holes (Rev B)
C45E+N 16 mm
make · 33.6 kg each · drawing NP1-230



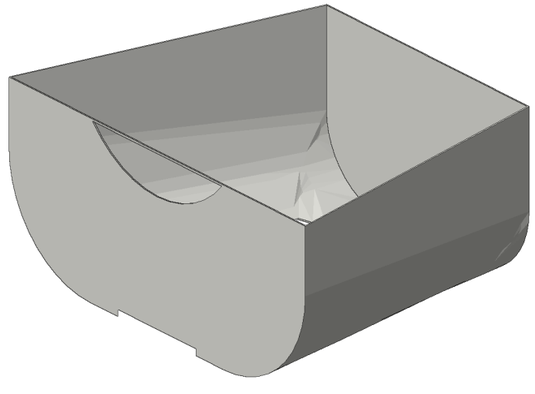
NP1-310 ×1
Rotor brake caliper bracket (left, Rev B)
S355J2+N 25 plate
make · 18.1 kg each · drawing NP1-310



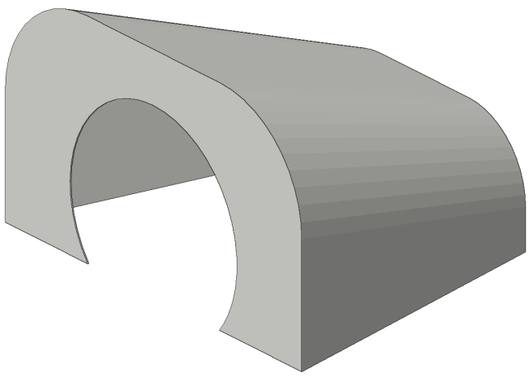
NP1-320 ×1
Rotor lock pin Ø20 (engaged, top)
42CrMo4+QT
make · 0.5 kg each · drawing NP1-320



NP1-321 ×1
Lock pin holder with 2 overspeed sensor seats
S355J2
make · 24.5 kg each · drawing NP1-321

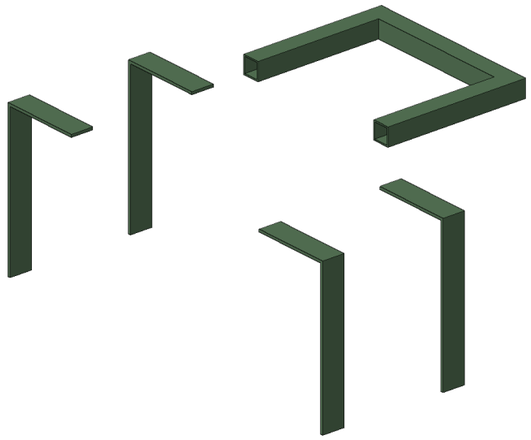


NM8-700-1 ×1
Nacelle cover, lower tray (GRP 5 mm)
glass/polyester, gelcoat RAL 9010
make · 15.7 kg each · drawing NM8-700



NM8-700-2 ×1

Nacelle cover, upper lid (GRP 5 mm)
glass/polyester, gelcoat RAL 9010
make · 17.7 kg each · drawing NM8-700



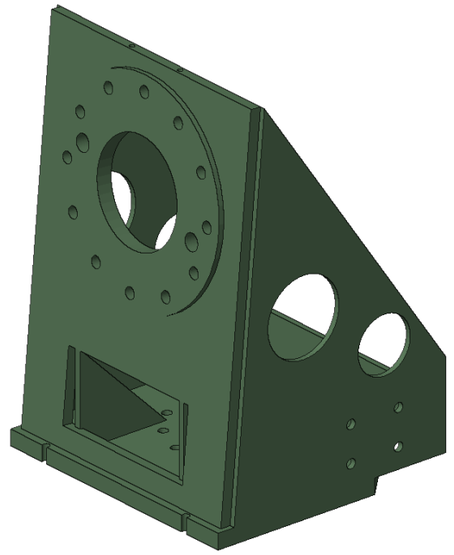
NM8-710 ×1

Cover support arms + rear rail
S235 galv.
make · 8.3 kg each · drawing NM8-710



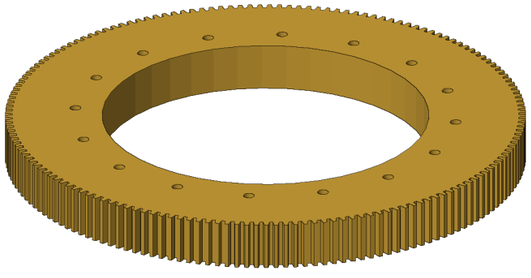
NM8-720 ×1

Met mast: anemometer, vane, air terminal
1.4404 tube 42.4 × 2 + bought sensors
make · 4.5 kg each · drawing NM8-720



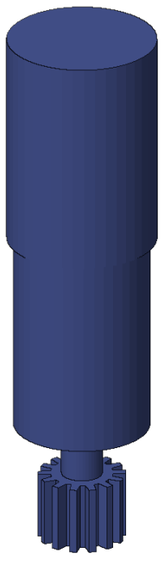
NP8-300 ×1

Bedplate weldment 12 kW NP8 (plates 20 / 30 / 10)
S355J2+N plates, stress relieved, machined
make · 124.3 kg each · drawing NP8-300



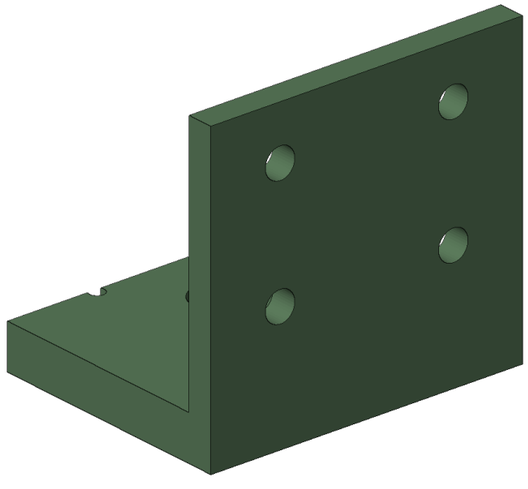
NM8-911 ×1

Yaw slewing ring Dpw 410, z165 m4
bought: 42CrMo4, induction hardened teeth
buy · 66.0 kg each · drawing NM8-911



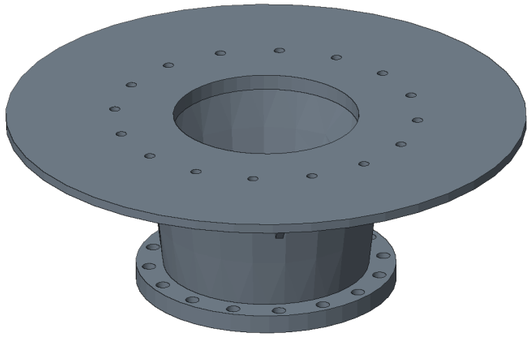
NM8-912 ×2

Yaw gearmotor 0.25 kW, output ≥ 2.3 kNm static, motor brake
bought
buy · 19.0 kg each · drawing NM8-912



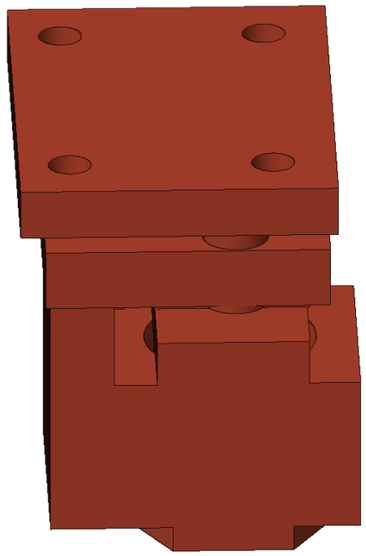
NP1-420 ×2

Yaw drive bracket
S355J2+N
make · 7.7 kg each · drawing NP1-420

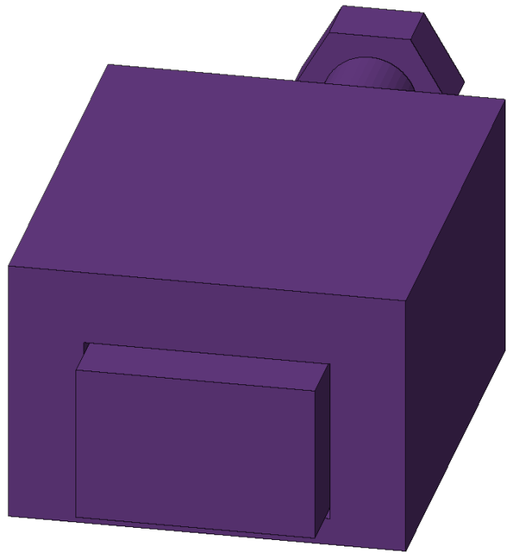


NY-400 ×1

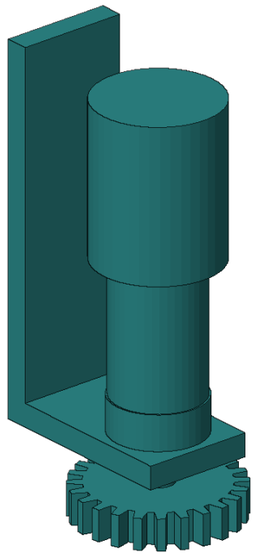
Tower-top adapter with friction track
S355J2+N weldment
make · 118.3 kg each · drawing NY-400



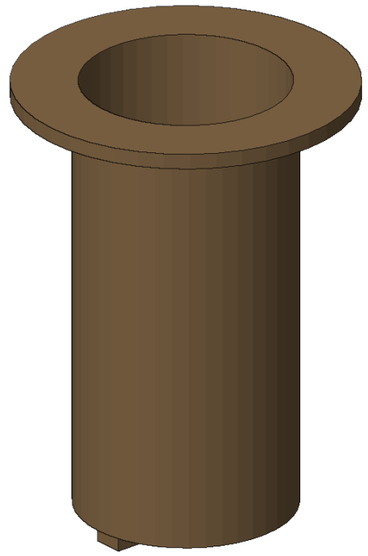
NY-410 ×2
Friction pad unit
S355 / DIN 2093 springs / organic pads
make · 14.0 kg each · drawing NY-410



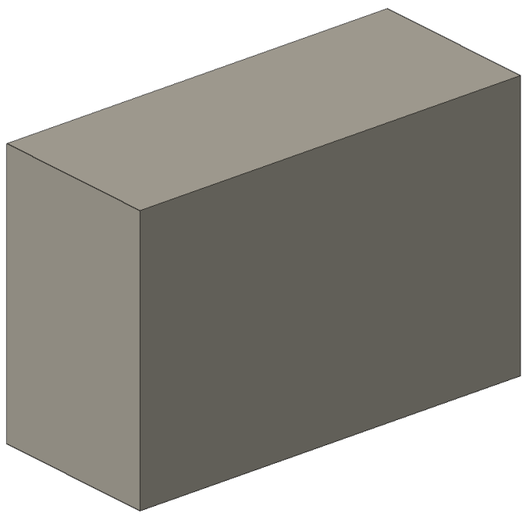
NY-430 ×1
Yaw lock (screw block)
42CrMo4 / S355
make · 6.5 kg each · drawing NY-430



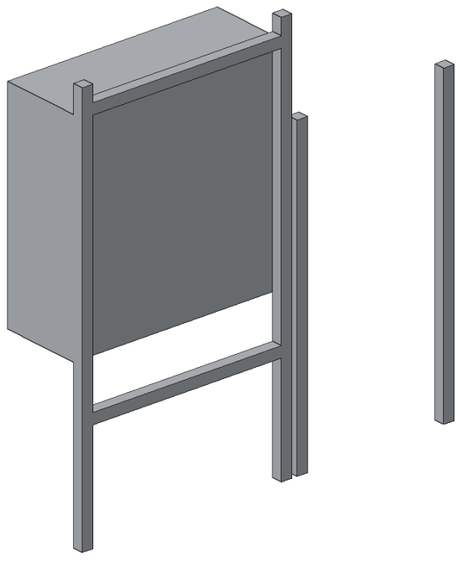
NY-440 ×1
Yaw encoder + cam limit switch unit
S355 + absolute encoder + cam switch
make · 6.0 kg each · drawing NY-440



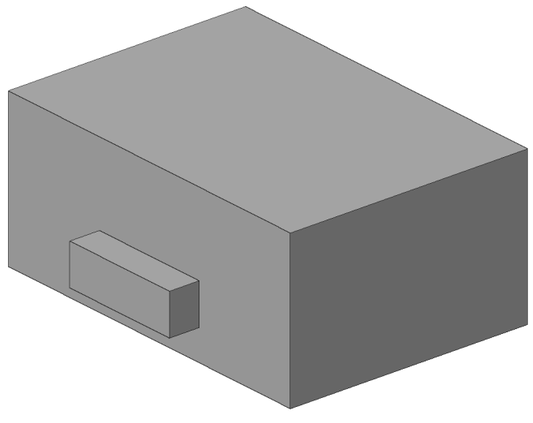
NY-460 ×1
Cable hanger tube + loop cleat
S355 tube 168.3 × 4.5
make · 10.0 kg each · drawing NY-460



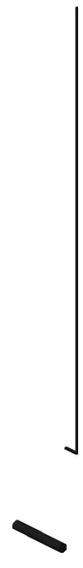
NM8-600 ×1
Braking / dump resistor box (vented)
1.4301 box + steel-grid resistors
make · 18.0 kg each · drawing NM8-600



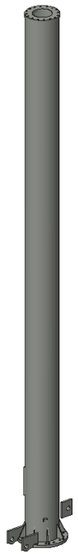
NM8-620 ×1
Tower base cabinet IP55 800 × 600 × 300 on stand
stand S235 galv.; bought enclosure
make · 95.0 kg each · drawing NM8-620



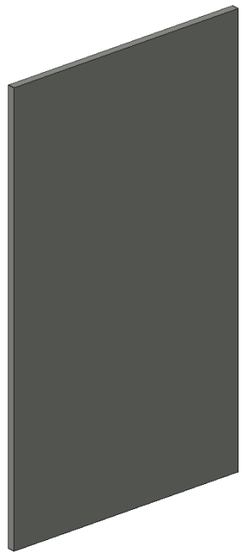
NM8-730 ×1
Nacelle junction box IP66 (400 × 340 × 150)
bought: 1.4301 enclosure
make · 9.0 kg each · drawing NM8-800



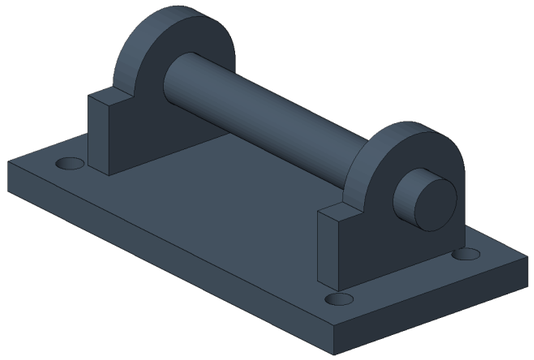
NM8-800 ×1
Drop cable + base duct (cabling route)
H07RN-F 4G10 + 2 × LiYCY signal; PE duct Ø110
make · 12.0 kg each · drawing NM8-800



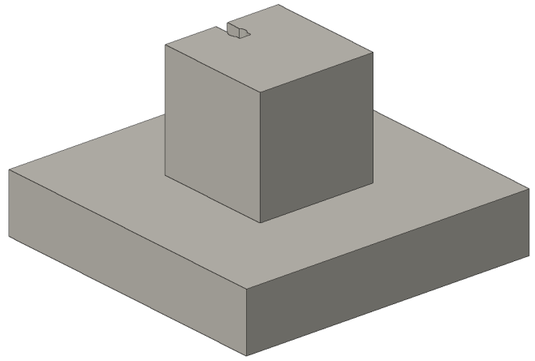
NM8-400 ×1
Tower 8 m, Ø400 → Ø400, t 10/8, one piece
S355J2+N, hot-dip galvanised
make · 699.8 kg each · drawing NM8-400



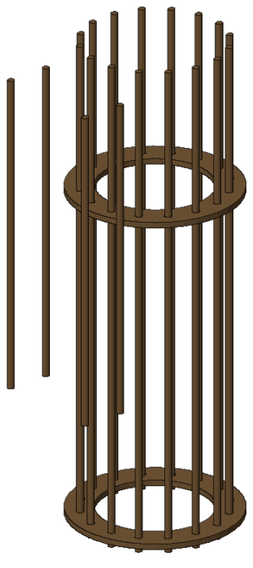
NM8-401 ×1
Hand-hole cover 200 × 350 × 6 with gasket
S355 galv.
make · 3.3 kg each · drawing NM8-400



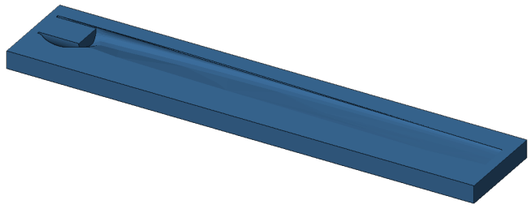
NM8-410 ×1
Base hinge brackets + hinge pin Ø50
S355J2+N galv.; pin 42CrMo4+QT
make · 37.4 kg each · drawing NM8-410



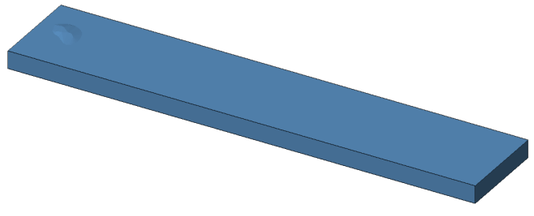
NM8-500 ×1
Spread footing (firm ground) with pedestal
C30/37 XF4 air-entrained, B500C
make · 14880.0 kg each · drawing NM8-500



NM8-510 ×1
Anchor cage 16 × M24 + 4 hinge anchors
M24 8.8 HDG threaded rod, S355 template rings
make · 161.7 kg each · drawing NM8-510



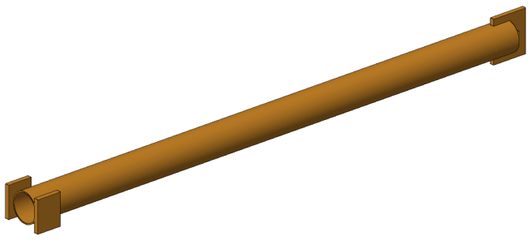
NM8-110-1 ×1
Blade mould, lower (pressure side) half on frame
GRP tooling laminate 12 mm, vinylester tooling gelcoat
make · 118.1 kg each · drawing NM8-110



NM8-110-2 ×1
Blade mould, upper (suction side) half
GRP tooling laminate 12 mm
make · 118.1 kg each · drawing NM8-110



NM8-110-3 ×1
Mould frame (RHS 80 × 80 × 4)
S235 RHS 80×80×4
make · 178.6 kg each · drawing NM8-110

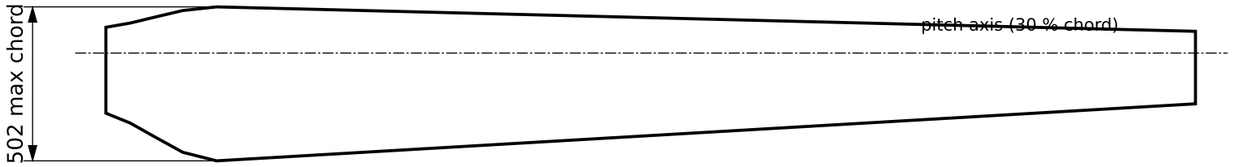


NM8-420 ×1
Gin pole 3.0 m (erection tool)
S355 tube 168.3 × 6.3
make · 94.0 kg each · drawing NM8-420

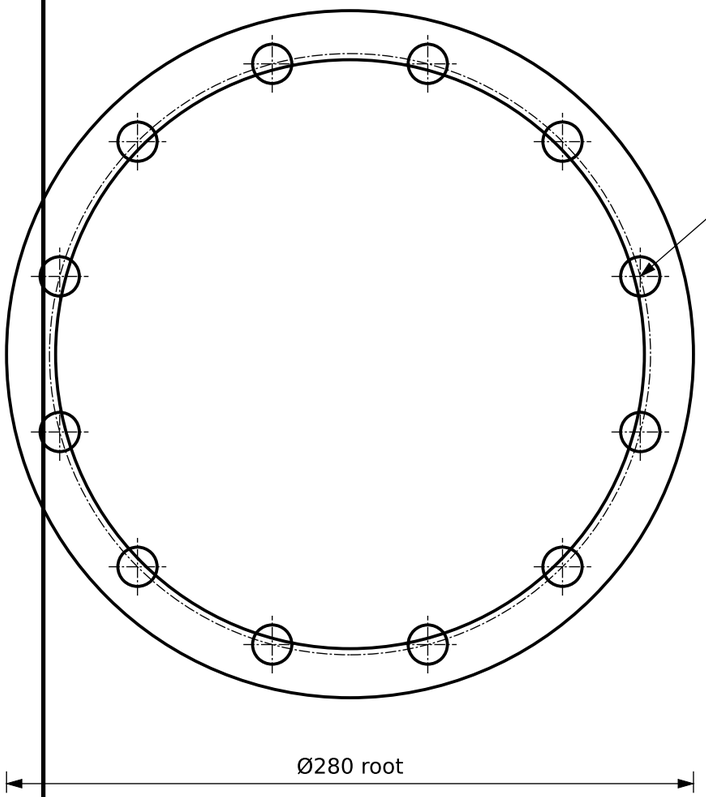
4 NM8 drawing set, revision A

New and changed M8 parts, each with its axonometric view

PLANFORM (suction side up, leading edge up)



FACE



AXONOMETRIC VIEW (not to scale)



r mm	chord mm	twist °	t/c
450	280	20.0	0.91
600	389	20.0	0.64
800	501	20.0	0.30
1000	486	18.4	0.29
1400	452	12.6	0.26
1800	419	9.0	0.24
2200	386	6.5	0.22
2600	352	4.7	0.20
3000	319	3.4	0.19
3400	286	2.3	0.18
3800	253	1.5	0.18
4000	236	1.2	0.18

NOTES

1. Twist relative to the pitch reference at −2° setting (blade at its fine end stop). Aerofoil family as doc 10, thickness scaled to the 4.0 m span (structure_m8.py).
2. Laminate: doc 10 construction scaled; carbon UD caps sized by buckling. Mass 34.1 kg; flap frequency ≈ 18.8 Hz at rated (3P 6.6 Hz).
3. Root: 12 bonded inserts M16 on PCD 245 (bolt force 64 kN design, F_t,Rd 113 kN). Root face machined flat 0.1 after cure, square to the pitch axis 0.1°.
4. First-article static test SLM H × 1.35 flap, edge test, insert pull-out (3 per batch). Balance set to 0.5 % static moment.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

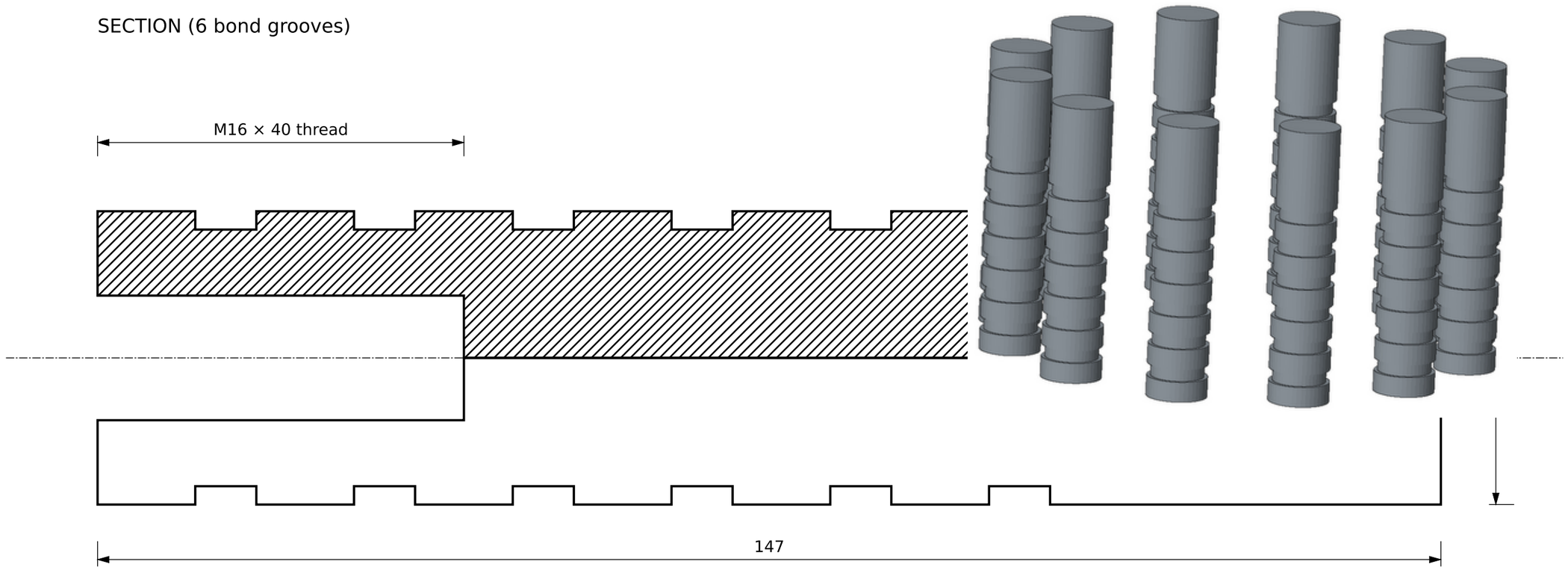
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BLADE Ø8.0 m · PLANFORM, TWIST AND ROOT

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-100	A · 2026-11-06	1:20
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
E-glass/epoxy infused; carbon UD caps; PVC core 34.1 kg		Claude / not checked

AXONOMETRIC VIEW (not to scale)

SECTION (6 bond grooves)



NOTES

1. The doc 34 M12 insert scaled 4/3: Ø32 × 147, internal M16-6H 40 deep, 6 grooves 2 deep for the bond.
2. Bonded into the root laminate with the NM8-115 fixture (12 spigots on PCD 245, as NS6-115 with the M8 pattern).
3. Pull-out test 3 per batch to 1.5 × 64 kN; thread gauge every insert.

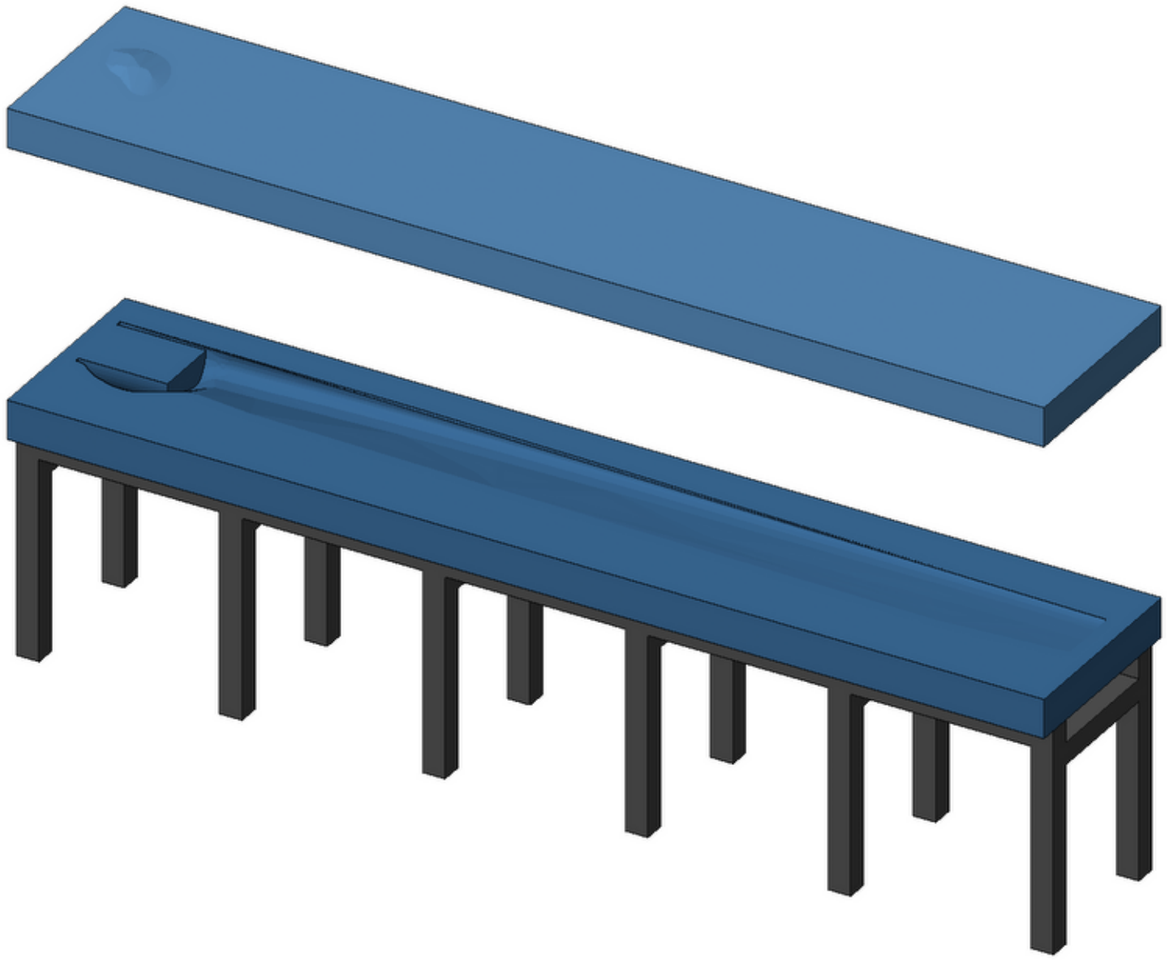
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BLADE ROOT INSERT M16 (BONDED)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-105	A · 2026-11-06	2:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
C45E+QT, zinc-flake coated (Geomet 500)	0.78 kg	Claude / not checked

AXONOMETRIC VIEW (not to scale)



NOTES

1. Same construction as NS6-110 (doc 34), lengthened for the 3.56 m M8 blade: CNC-machined plug from the NM8-100 surfaces, GRP halves 12 mm with vinylester tooling gelcoat, steel frame RHS 80 × 80 × 4.
2. Parting line on the leading/trailing edge; root end with the NM8-115 insert fixture flange; 6 locating cones.
3. Mould acceptance: blade profile within 0.5 mm of the plug at 10 stations; twist within 0.2°.

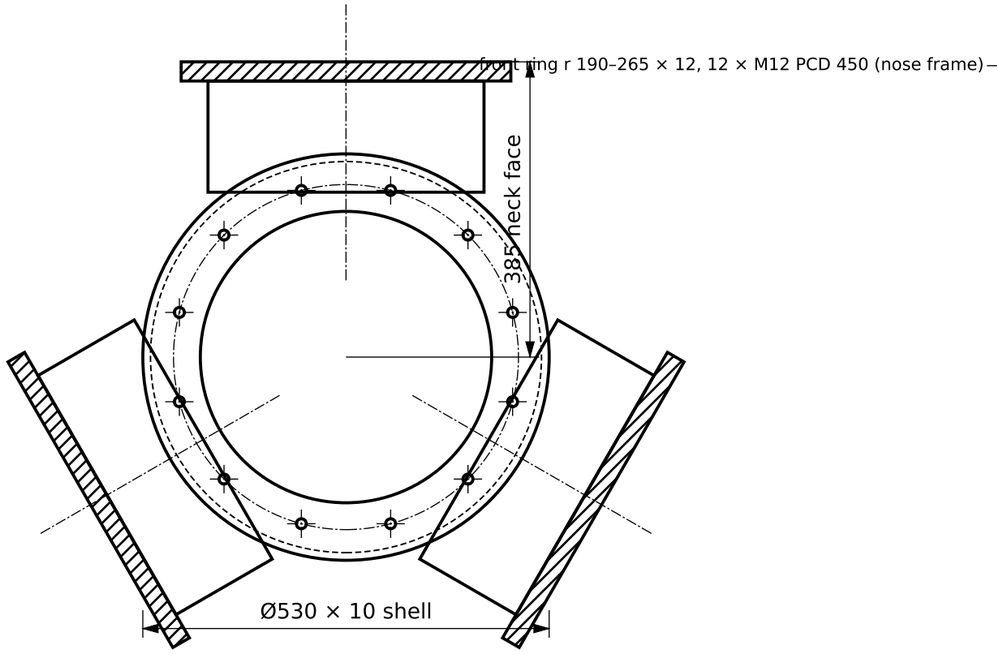
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

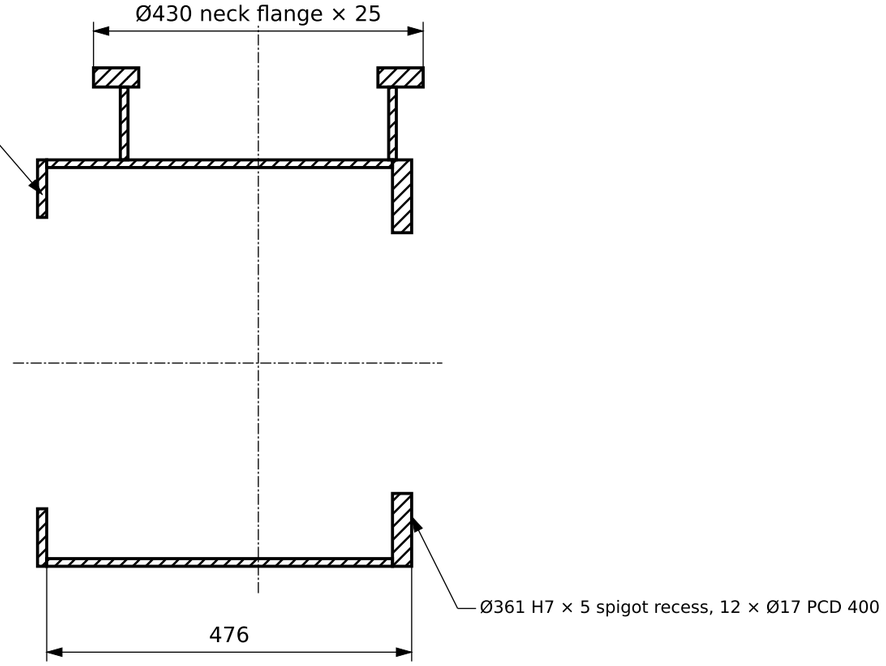
BLADE MOULD SET (PLUG, TWO HALVES, FRAME)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-110	A · 2026-11-06	–
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
GRP tooling laminate on S235 RHS frame	≈ 2 × 118 kg + frame 179 kg	Claude / not checked

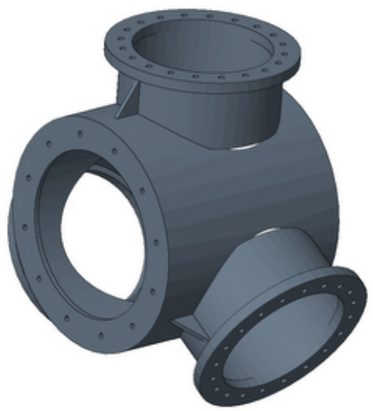
VIEW FROM UPWIND (spinner and nose frame removed)



SECTION THROUGH BLADE 1 AXIS



AXONOMETRIC VIEW (not to scale)



NOTES

1. Necks Ø360 × 10 on the Ø530 × 10 shell: parked SLM stress 117 MPa (allow 323); fatigue 31.5 MPa (allow 42.1).
2. Neck flanges Ø430 × 25 carry the pitch bearing outer rings: 20 × M16 tapped on PCD 360; faces machined after stress relief, flat 0.05, square to the neck axis 0.05.
3. Rear gussets stop on the drum flange ring (x 200) so they clear the drum web; front gussets 68 long. Full penetration welds ISO 5817 B, 100 % MT, UT on neck welds.
4. Drum joint as Rev C: Ø360 spigot, 12 × M16 10.9 PCD 400, 280 Nm. FE check of the neck-to-shell weld before release (doc 37 §9).

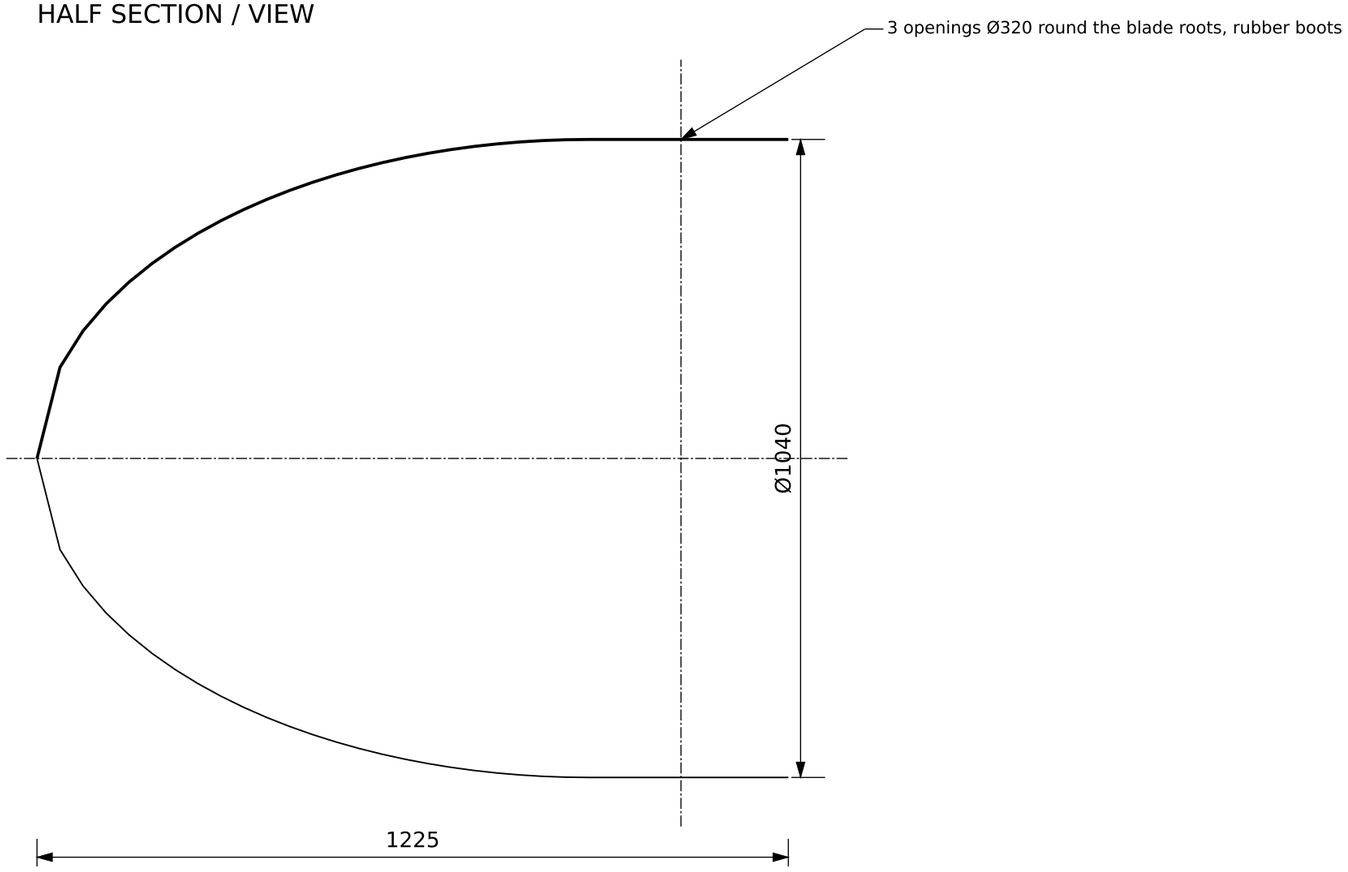
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unles eview

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

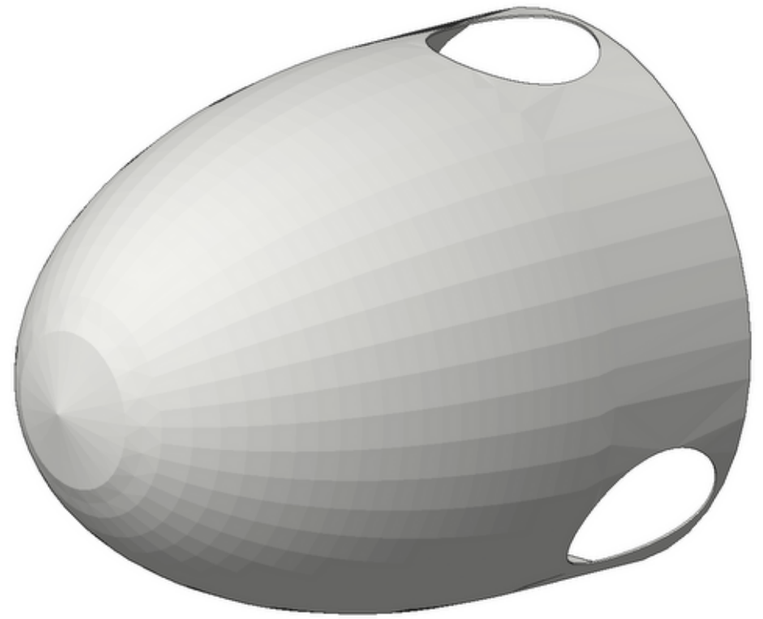
HUB WELDMENT (COLLECTIVE PITCH)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-200	A · 2026-11-06	1:8
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355NL, welded, stress relieved, machined	180 kg (budget)	Claude / not checked

HALF SECTION / VIEW



AXONOMETRIC VIEW (not to scale)



NOTES

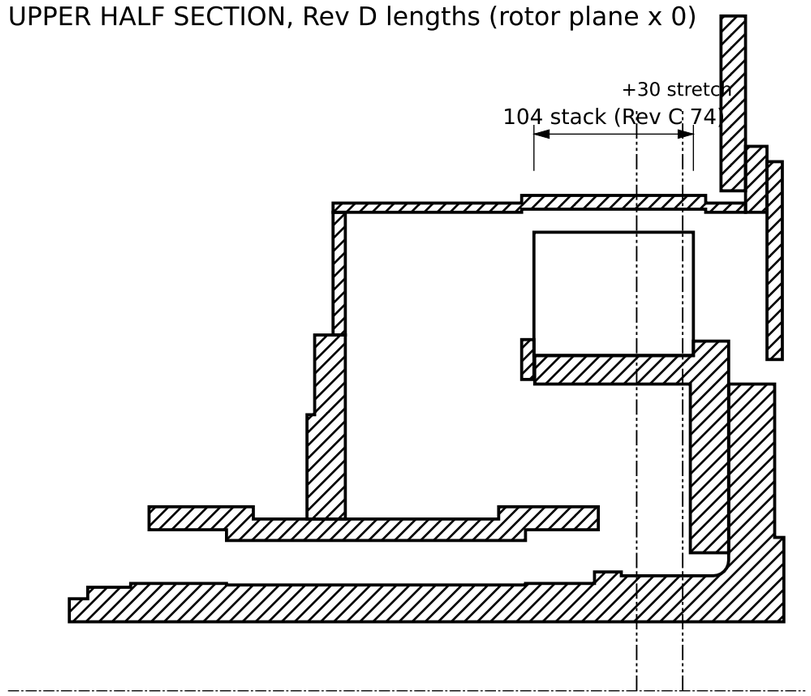
1. Ogive nose to r 520 at x −150, cylindrical to the base ring at x 175 (spindle frame, rotor plane x 0). 4 mm glass/polyester, 3 split segments with flanged joints.
2. Encloses the pitch bearings (outer edge r 485), the nose frame and the cylinder; inspection hatch Ø200 at the nose for the slip ring and the pitch electronics.
3. Fixed to the hub front ring and neck flanges with 9 brackets (A4 M8), 10 mm clearance to all moving pitch parts.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

SPINNER (GRP) WITH BLADE BOOTS

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-210	A · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
glass/polyester 4 mm, gelcoat RAL 9010	27.6 kg	Claude / not checked

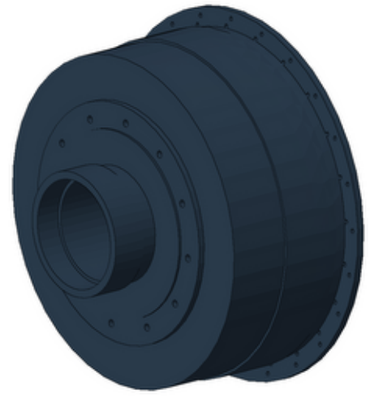


Part	Rev C	Rev D
NP1-100D spindle	x 40–476	x 40–506 (Ø150 section 400–460)
NP1-200D stator carrier	rim 74	rim 104, web unchanged
NP1-201D stator	5 rows, 74 mm	7 rows, 104 mm
NP1-203D coils	28 turns	28 turns, longer side (turn +60 mm)
NP1-210D drum	shell 239	shell 269; fins/ribs as Rev C
NP1-220F magnets	200 blocks	280 blocks (+14 spare)
Torque rated	730 Nm	991 Nm
Line EMF 100 °C	372.0 V	384.8 V
Efficiency rated	0.9334	0.9296
Drum stability margin	302	237

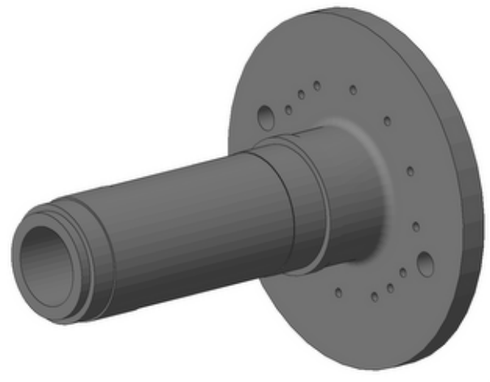
NOTES

1. All Rev C drawings stay valid with the lengths of this sheet: every feature ahead of spindle x 410 keeps its Rev C distance to the rotor plane, every feature behind x 410 moves 30 mm aft (bedplate, disc, calipers, rear cover).
2. Spindle check with Rev D lengths and M8 loads: static reserve 2.5, bearings s0 6.7 / 6.2, life factor 5.6 (spindle_m8.py).
3. Cooling as Rev C (36 inner fins, 48 outer ribs, 24 cover blades); hot spot 117 °C at 110 % (generator_m8_cooling.json). Fins and ribs are not in the 3D model.
4. Magnet comb and sleeve lengthened by 30; skew step 0.143° between the 7 rows (±0.43°).

AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



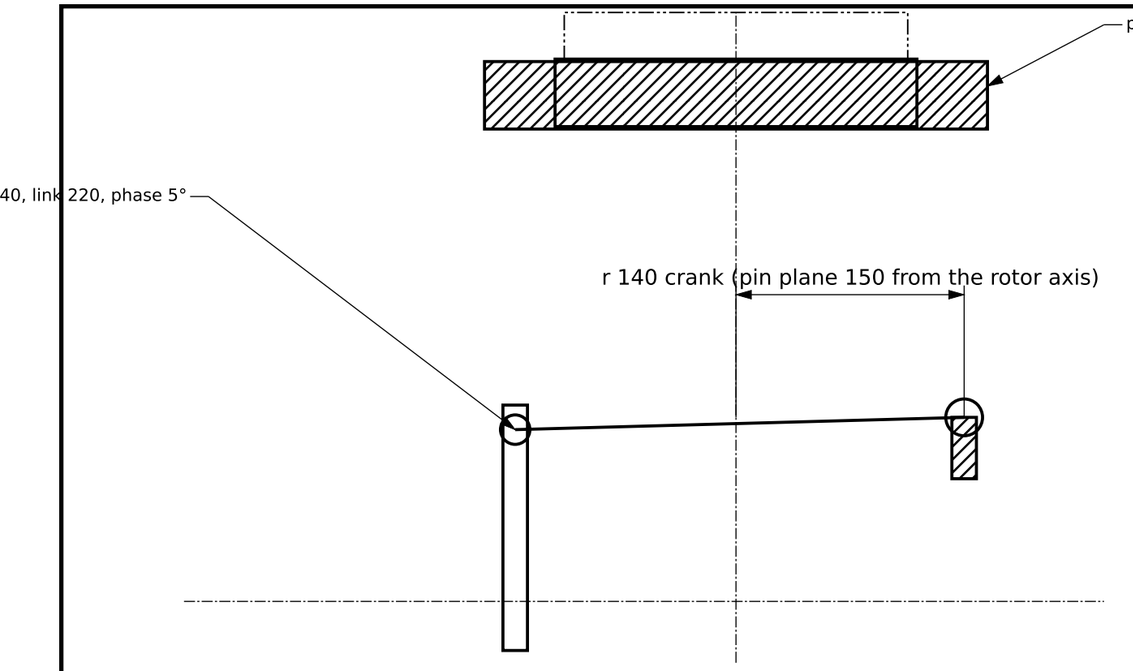
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

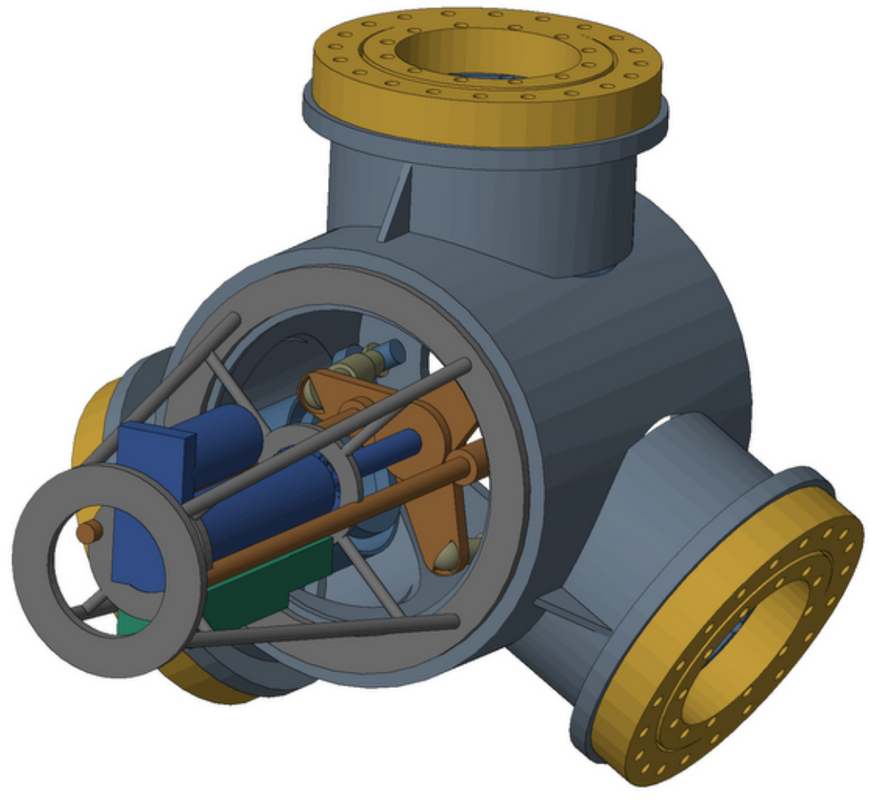
GENERATOR REV D · CHANGE SHEET (STACK 74 → 104 mm)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-215	A · 2026-11-06	1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
as Rev C sheets	322 kg generator	Claude / not checked

BLADE 1 PLANE, PITCH −2° (schematic)



AXONOMETRIC VIEW (not to scale)



Pitch system (hub_yaw_m8, crank r140)	Value
Cylinder stroke	204 mm
Force at trip (design)	23.5 kN
Rated force	28.5 kN
Motor peak torque	12.6 Nm
Energy per feather stroke	4.14 kJ
Strokes from one module (EOL)	3.6
Storm hold force on screw	37.1 kN
Motor brake needed	14.2 Nm
Pitch bearing	Dpw 300, balls 30, Z 28
Bearing s0 (SLM H)	1.37
Clash check	clean at −2, 10, 30, 45, 60, 80, 92°

NOTES

1. Kinematics from analysis/hub/collective.py with R_C 140, S_P 150, R_Y 140, L 220, phase 5° (cad/hub_m8/model_m8.py). The pin plane was moved out from 118 to 150 so the crank pin clears the drum bearing housing (r ≤ 120).
2. Bearings: 4-point ungeared, 20 × M16 outer to the neck flange (PCD 360), 12 × M16 inner to the blade root (PCD 245). Grease, sealed, −40 °C grease.
3. Cylinder ≈ 28.5 kN rated, 204 mm stroke, motor brake ≥ 14.2 Nm; supercap module sized for 5.0 feather strokes at end of life; nose frame carries both.
4. Mechanical end stops at −2.5° and 93° on crank 1; limit switches at −2° and 92°.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

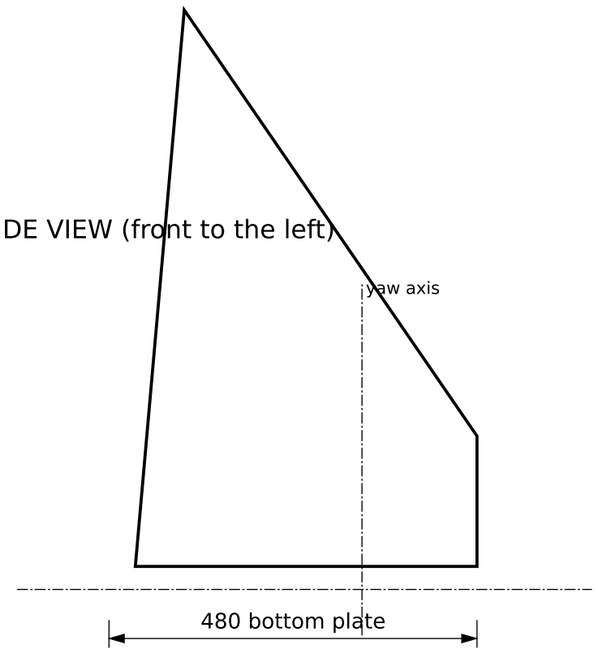
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

COLLECTIVE PITCH · BEARINGS, CRANKS, LINKS, YOKE, CYLINDER

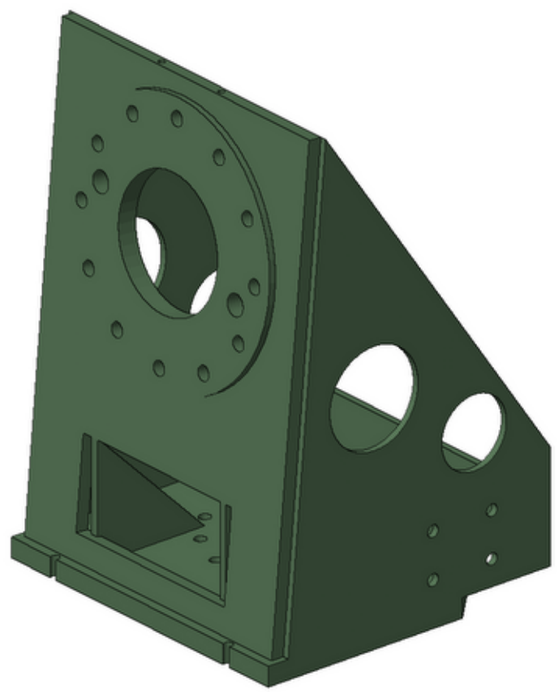
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-250	A · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 140 kg pitch parts (budget)	Claude / not checked

Plate	t mm	kg
Front plate	20	44.8
Bottom plate (yaw flange)	30	43.6
Side plates (2)	10	28.3
Check	util	
Root section	0.12	
Front plate	0.31	
Bottom plate	0.68	
Yaw bolts	0.42	
Fatigue root	0.14	

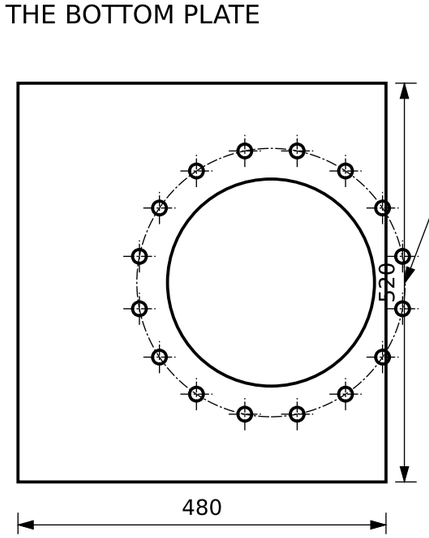
SIDE VIEW (front to the left)



AXONOMETRIC VIEW (not to scale)



PLAN OF THE BOTTOM PLATE



NOTES

1. NS6-300 geometry (doc 32) with the M8 plate thicknesses from bedplate_m8.py (decision D1): front 20, bottom 30, sides 10. Interfaces as NP1-300: spindle flange 12 × M20 PCD 350 + Ø200 spigot, pad unit and gearmotor brackets.
2. Changed for the Dpw 410 yaw ring: 16 × Ø18 on PCD 350 (was 300), central bore Ø270, pad unit bracket holes 25 mm further out. Third caliper bracket NM8-311 bolts through the front plate (4 × Ø17).
3. Stress relieve before machining; spigot bore and the yaw face machined in one set-up (parallel 0.05, square 0.05).

Dimensions mm · ISO 2768-mK · ISO 8015 · first ar

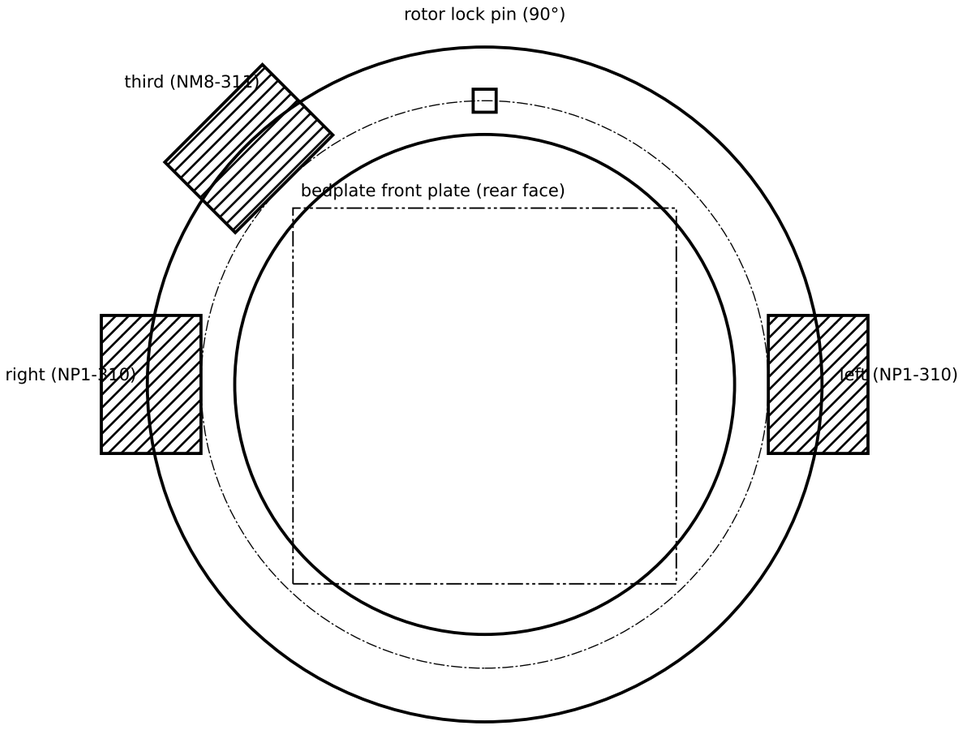
only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

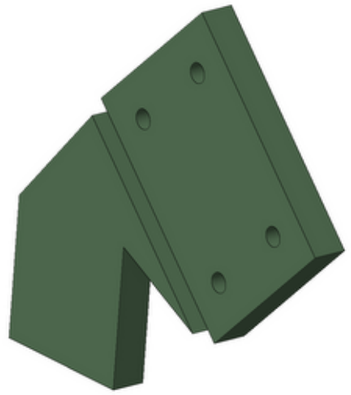
BEDPLATE WELDMENT 12 kW NP8 (LIGHT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP8-300	A · 2026-11-06	1:8
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plates, stress relieved, machined	136 kg (plate model), 124 kg CAD	Claude / not checked

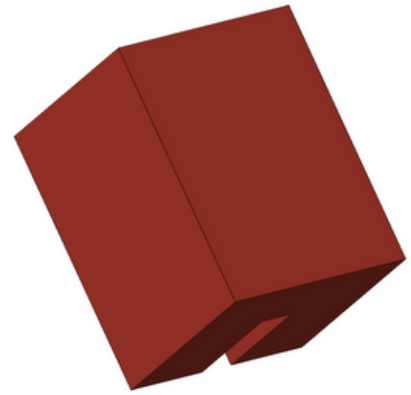
VIEW FROM THE REAR (downwind), angles from +y towards +z



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Brake (spindle_m8 / hub_yaw_m8)	Value
Required holding torque	14.4 kNm (parked aero torque at 8 m, design; 18 m: 13.1)
Calipers	3 × spring-applied, 4.8 kNm each
Clamp force per caliper	16.2 kN at μ 0.40, r_eff 0.37 m
Parked aero torque (sim, char.)	9.32 kNm at 18 m, 10.59 kNm at 8 m
Disc	NP1-230 Rev B, 16 mm, unchanged

NOTES

1. The third caliper sits at 135° (upper right seen from the rear) because the rotor lock pin holder occupies 90°. It uses the same caliper and mounting plate as NP1-310 (4 × M16 axial).
2. Bracket: 25 mm mounting plate + 25 mm web + 25 mm foot, bolted 4 × M16 10.9 through the bedplate front plate outside the spindle flange bolts (r ≥ 205). Weld a5, MT 100 %.
3. Alternative (decision D2 of doc 35): two calipers with a stronger spring pack (7.2 kNm each) in the existing positions; the third bracket then goes.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

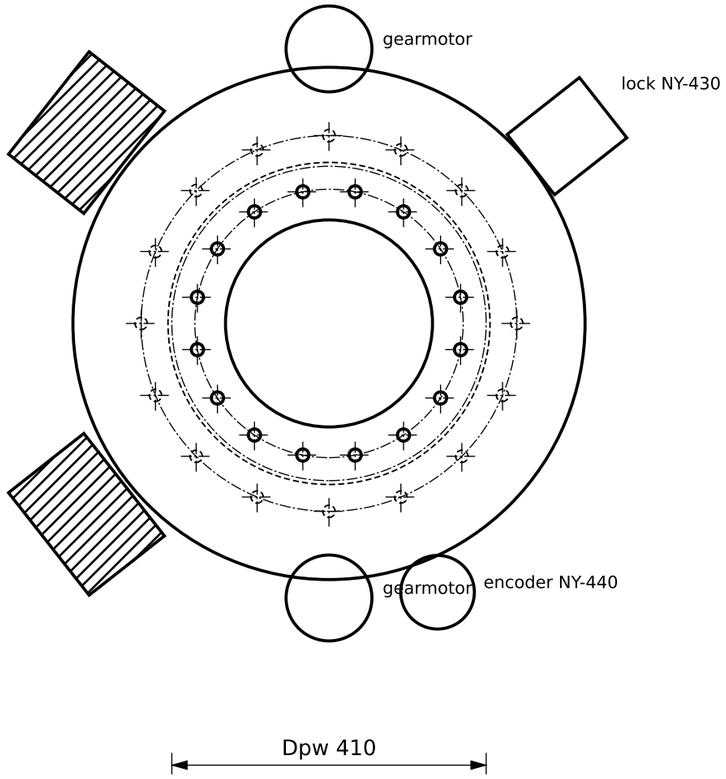
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

ROTOR BRAKE · THREE CALIPERS AND THIRD CALIPER BRACKET

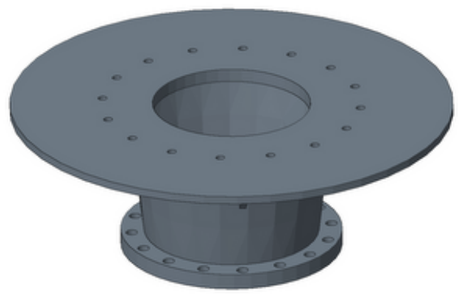
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-311	A · 2026-11-06	1:8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
bracket S355J2+N 25 plate, welded	bracket 7.0 kg	Claude / not checked

Yaw (hub_yaw_m8)	Value
Ring	Dpw 410, gear z165 m4, face 60
s0 (SLM H + shaft moment)	1.41 (Dpw 360: 0.95)
Tilt moment design	52.4 kNm
Gearbox static need	2308 Nm
Gear static S_F pinion / ring	3.39 / 2.69
Gear static S_H pinion / ring	1.54 / 1.18
Inner ring bolts	16 × M16 10.9 PCD 350 (bedplate)
Outer ring bolts	16 × M16 10.9 PCD 490 (adapter NY-400)

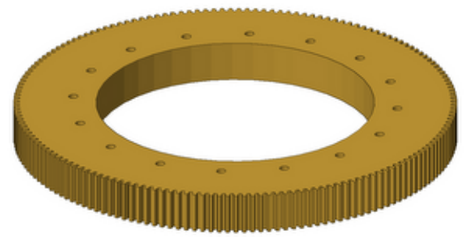
PLAN (X downwind to the right), bedplate removed



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

1. Doc 25 yaw system with the ring one size up (Dpw 360 fails s0 0.95 under the M8 SLM). Every radial NY dimension moves out 25 mm: adapter seat r 150–275, friction track r 275–420, pad units, lock and encoder.
2. Gearmotor centre r 358 (pinion z14 m4 on z165); NP1-420 brackets moved out 30 mm. Gearbox output ≥ 2.3 kNm static, motor brake on the input.
3. Ring bolt circles are preliminary: the supplier drawing of the chosen ring replaces this sheet's PCDs and hole sizes; bolt checks in hub_yaw_m8 used the Dpw 360 PCDs and are conservative.

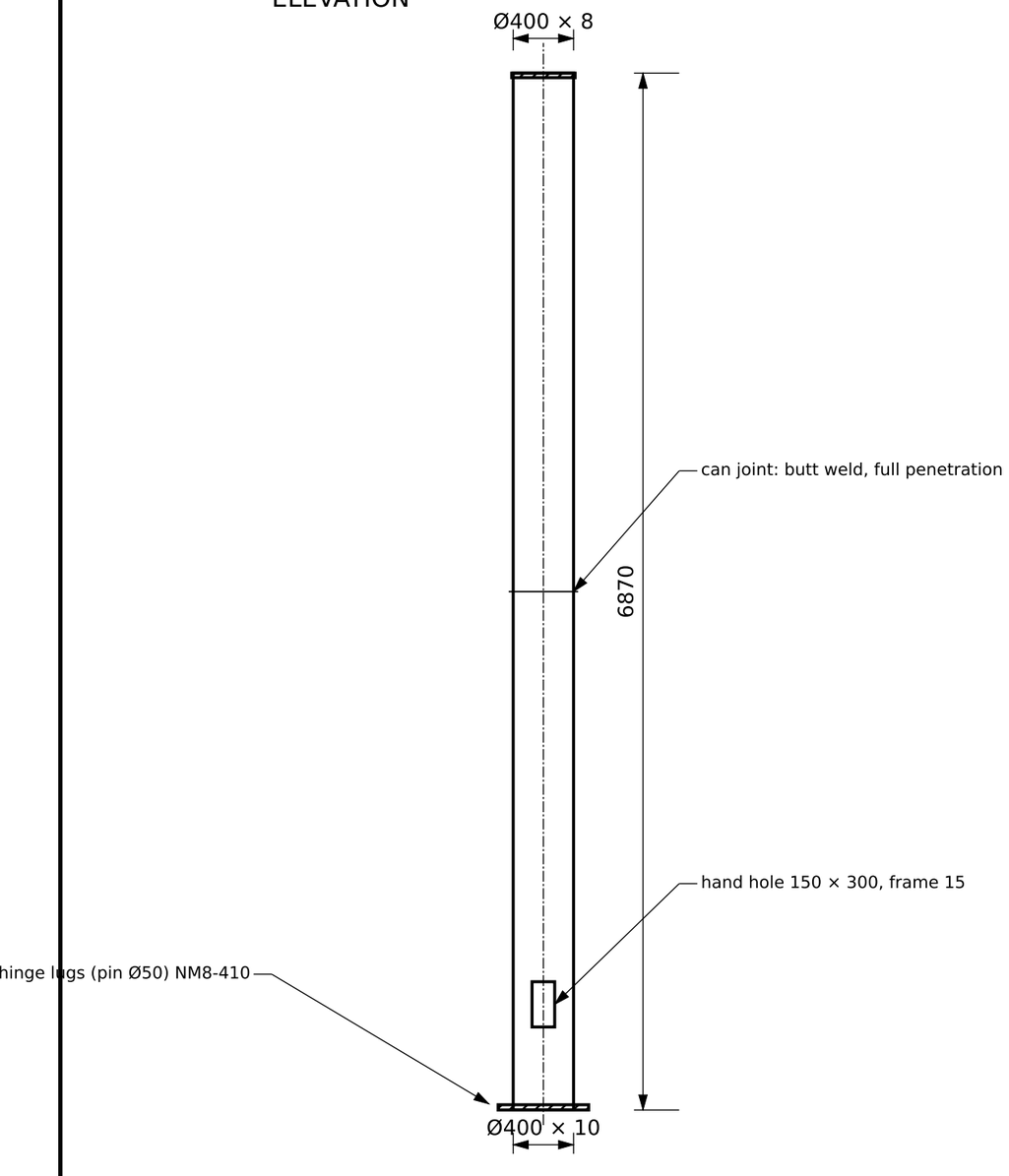
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

YAW SYSTEM LAYOUT · SLEWING RING Dpw 410

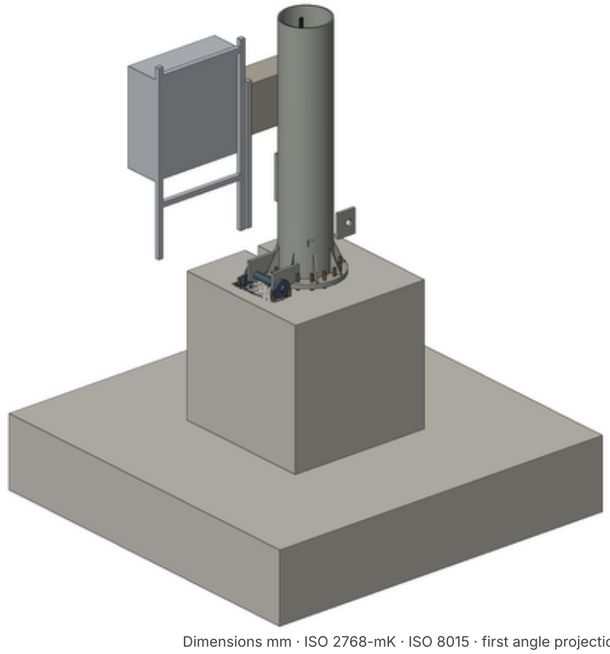
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-911	A · 2026-11-06	1:8
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
ring bought; NY parts per doc 25	ring ≈ 66 kg	Claude / not checked

Tower (tower8_m8, stiff_final_sim8)	Value
Mass	671 kg
f1 rigid / soft soil + ice	2.78 / 2.65 Hz
f2	22.28 Hz
3P exclusion zone	48–61 rpm
Utilisation (governing)	0.768 (earthquake 0.5 g (ground D) + operation)
Fatigue damage base / mid	0.53 / 0.07
Base flange	16 × M24 10.9 on Ø500, F_bolt 202.5 kN (F_t,Rd 254.2)
Top flange	Ø420 × 30, 16 × M20 PCD 380 to NY-400
Tilt-up base moment	150.0 kNm, line pull 66.2 kN
D/t max	50.0

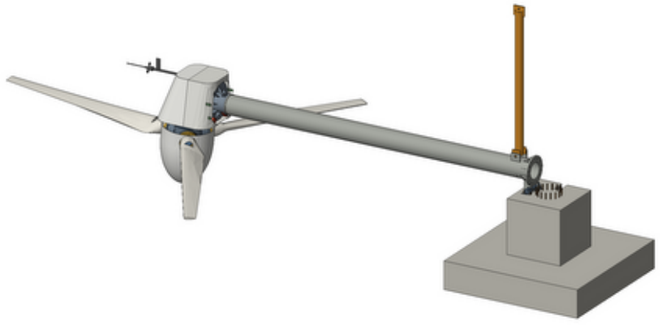
ELEVATION



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Stiff tower: f1 above 1.15 × 1P at 131 rpm (2.51 Hz) and below 3P/1.1 at rated (5.95 Hz). The

nferential welds full

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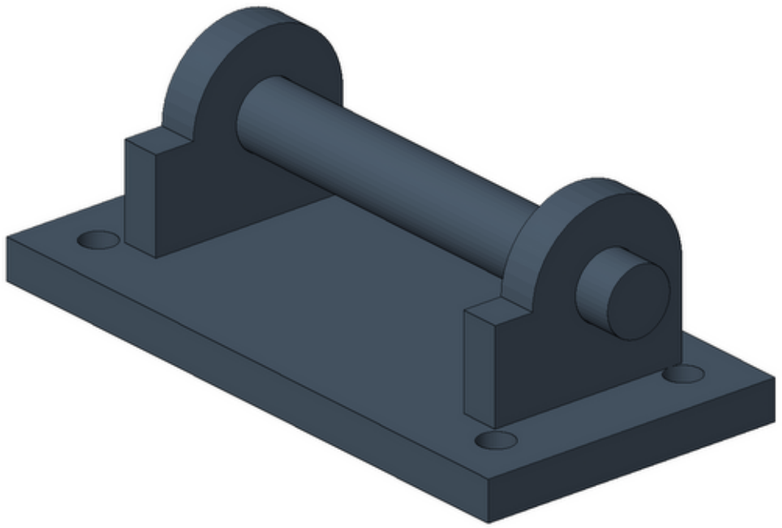
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

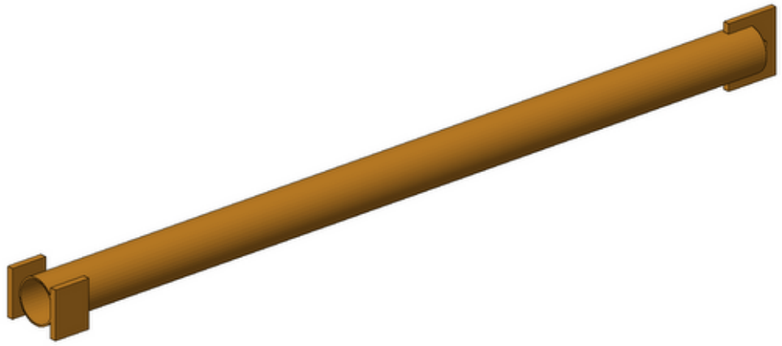
TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-400	A · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, rolled and seam welded; hot-dip galvanized	671 kg (sizing), 700 kg CAD	Claude / not checked

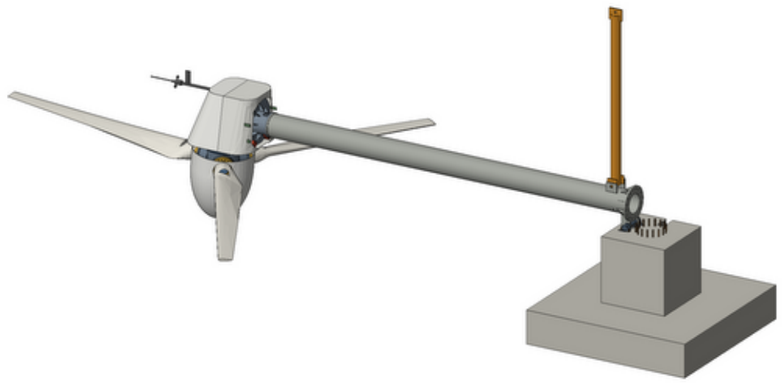
AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Erection (tower8_m8)	Value
Tilt-up base moment	150.0 kNm
Winch line pull (gin pole 0.33 H)	66.2 kN
Tower utilisation in tilt-up	0.37
Hinge pin	Ø50 h9, 42CrMo4+QT, double shear
Gin pole	Ø168.3 × 6.3, 3.0 m (S6 tool NS6-420)

NOTES

1. The S6 hinge and gin pole (doc 34) carry the M8 head: line pull rises with the 1.39 t head; use a 10 t winch with a 2-part line (≈ 32 kN on the drum) or a 5 t winch with a 4-part line.
2. Hinge brackets on 4 × M24 anchors in the pedestal; pin greased, secured by R-clips; tower lies on a trestle at 2.5 m during head mounting.
3. No one inside the fall zone (1.5 × tower height) while lifting; wind below 8 m/s; tag lines on two blade tips.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

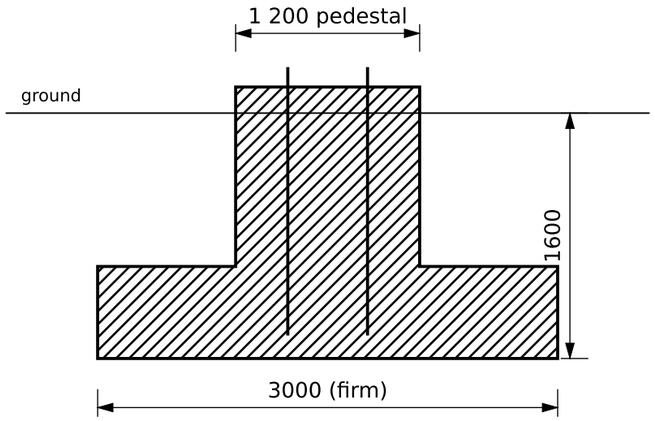
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BASE HINGE, GIN POLE INTERFACE AND ERECTION

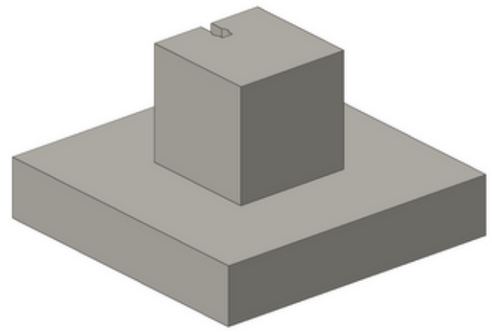
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-410	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N 30 mm plate, galv.; pin 42CrMo4+QT Ø50 h9	37 kg	Claude / not checked

Ground class	B m	depth m	concrete m³	bearing kPa (q_Rd)
firm soil (dense gravel, moraine)	3.0	1.6	6.2	234.0 (300.0)
soft volcanic soil (loose ash, pumice sand)	3.4	1.6	7.8	97.3 (100.0)
frost-susceptible silt (thawed)	3.2	1.6	7.0	126.3 (150.0)
rock (anchored plinth)	1.6 × 1.6 × 0.8	–	2.0	8 anchors Ø20

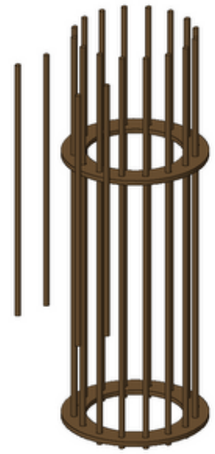
SECTION (firm ground)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Anchor cage NM8-510: 16 × M24 8.8 HDG on PCD 500, two template rings, embedded 1.45 m; 4 hinge anchors M24.
2. Pedestal top 170 above ground; 30 mm grout under the base flange. Concrete C30/37 XF4 air-entrained, cover 50, rebar ≈ 75 kg/m³.
3. Ground class from a trial pit at the site (depth 2 m); frost-susceptible silt: replace 0.5 m under the slab with compacted gravel. Rock: drilled plinth with 8 anchors.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

FOUNDATION · SPREAD FOOTING (8 m TOWER, M8)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-500	A · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C30/37 XF4 air-entrained; B500C rebar	6.2–7.8 m³	Claude / not checked

Joint	Fasteners	Torque / preload	Check
Blade root → pitch bearing inner ring	12 × M16 10.9 per blade	280 Nm (μ 0.12), re-torque at 50 h	F_bolt 64 kN design vs 113
Pitch bearing outer ring → hub neck	20 × M16 10.9 per blade	280 Nm	doc 35 §5
Hub → drum web	12 × M16 10.9 PCD 400	280 Nm	slip 1.7, separation 5.3
Spindle → bedplate	12 × M20 10.9 PCD 350	540 Nm	slip 6.4, separation 2.3
Stator carrier → spindle flange	12 × M12 10.9 PCD 240	110 Nm	slip 1.4
Disc → drum ring	24 × M12 10.9 PCD 662	110 Nm	slip 1.7
Caliper → bracket	4 × M16 10.9 each (3 calipers)	280 Nm	–
Third caliper bracket → bedplate	4 × M16 10.9 + nuts	280 Nm	–
Yaw inner ring → bedplate	16 × M16 10.9 HV PCD 350	280 Nm	util 0.42
Yaw outer ring → adapter	16 × M16 10.9 PCD 490	280 Nm	–
Adapter → tower top flange	16 × M20 10.9 HV PCD 380	540 Nm	ratio 0.45
Tower base flange → anchors	16 × M24 8.8 HDG, 3 nuts each	tension to 150 kN (tool), re-check at 3 months	F_bolt 202.5 kN
Hinge brackets → pedestal	4 × M24 8.8 HDG	350 Nm	–
Cover, spinner, mast, nose frame	M6–M12 A4-70	A4 table	–

NOTES

1. Torques for 10.9 zinc-flake with μ 0.12 (lubricated); mark every bolt after torquing; check 10 % of each joint at the first service (3 months) and yearly.
2. HV sets (EN 14399-4) where the joint carries the yaw moment; no washers under the nut side of HV sets other than those in the set.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

FASTENER AND TORQUE SCHEDULE

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-900	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
ISO 4014/4017, EN 14399-4 (HV), 10.9 zinc-flake unless stated	–	Claude / not checked

Part	Specification	Interface	Qty
Pitch bearing (NM8-250)	4-point, Dpw 300, balls 30, Z 28, s0 ≥ 1.25 under SLM H	outer 20 holes PCD 360, inner 12 × M16 PCD 245	3
Pitch cylinder + motor (NM8-270)	≥ 28.5 kN rated, 37 kN static, 204 mm stroke, motor brake ≥ 14.2 Nm	trunnion in the nose frame	1
Supercap module (NM8-280)	≥ 4.1 kJ per stroke × 3.6 at EOL, −30 °C heater	nose frame tray	1
Main bearings (NM8-913)	32028 X tapered roller, C 280 kN, C0 430 kN	Ø140 j6 / f6, Ø210 P7	2
Brake calipers (NM8-915)	spring-applied, 4.8 kNm each on the Ø880 × 16 disc, ≥ 16.2 kN clamp	4 × M16 axial	3
Yaw slewing ring (NM8-911)	Dpw 410, external gear z165 m4, face 60, s0 ≥ 1.25	PCD 350 / 490, 16 × M16 each	1
Yaw gearmotor (NM8-912)	≥ 2.3 kNm static output, pinion z14 m4, motor brake	NP1-420 bracket	2
Magnets (NP1-220F)	N45SH 30 × 15 × 12, NiCuNi, 280 + spares	comb + sleeve	280
Slip ring	6 power + 6 signal, IP65, −30 °C	spindle nose Ø120	1
Overspeed sensors	M18 inductive PNP, 2 channels	NP1-321 seats	2
Converter + controller	12 kW, doc 12 / doc 35 §8	base cabinet	1

NOTES

1. Replace the 3D envelopes with the supplier models before release and rerun cad/turbine_m8/clash8.py.
2. Low-temperature grades (−30 °C operation, −40 °C storage) for all lubricants, seals and electronics.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BOUGHT PARTS SPECIFICATION (ENVELOPES AND INTERFACES)

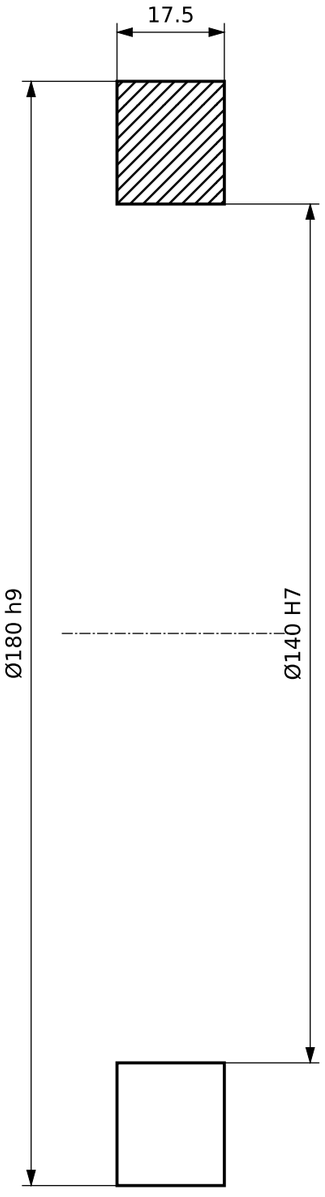
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-910	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

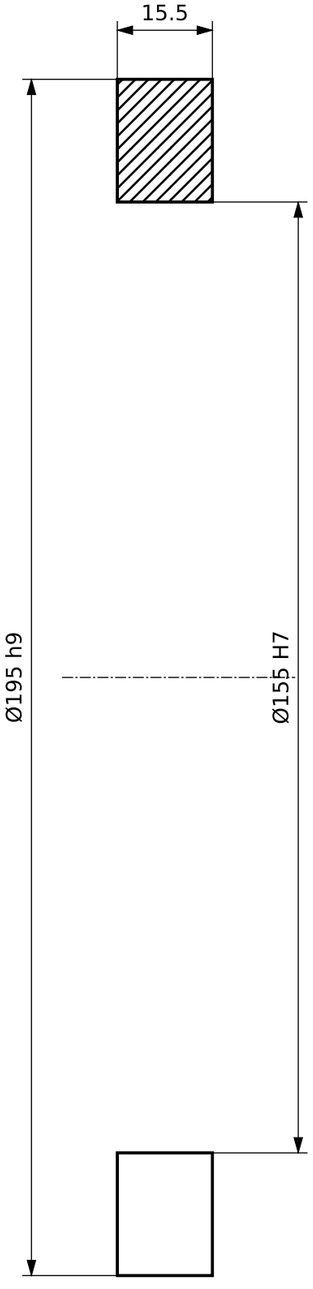
5 Reused drawings

NP1 rev B · generator Rev C (lengths per NM8-215) · yaw NY (radii per NM8-911) · S6 cover and electrical

NP1-102 REAR SEAL SLEEVE (Rev B)

NP1-101 SECTION





NOTES

1. Faces parallel within 0.005 (it sets the bearing preload path).
2. NP1-102: same material and hardening as NP1-101; O-ring 155 × 3 in a bore groove (static seal); slides onto the Ø155 shoulder before the rear cone, front face against the cone back face.
3. Rev B: OD 160 → 180 so the front seal cover clears the KMT 27 nut (doc 17).
4. Ø180 surface induction hardened 55 to 60 HRC, depth 1 mm, ground Ra 0.2 to 0.4, no lead.
5. Bore chamfer 1.5×30° on the cone side.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

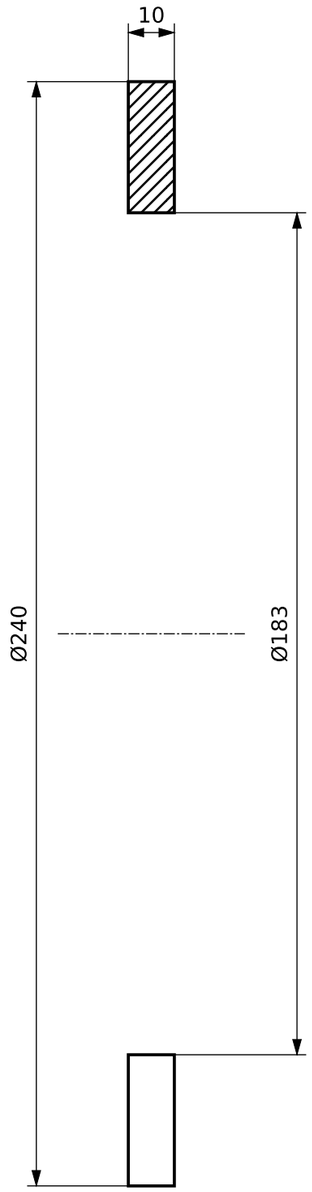
12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

FRONT SPACER + REAR SEAL SLEEVE NP1-102

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-101	B · 2026-11-03	1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
42CrMo4+QT, surface induction hardened	1.4 kg	Claude / not checked

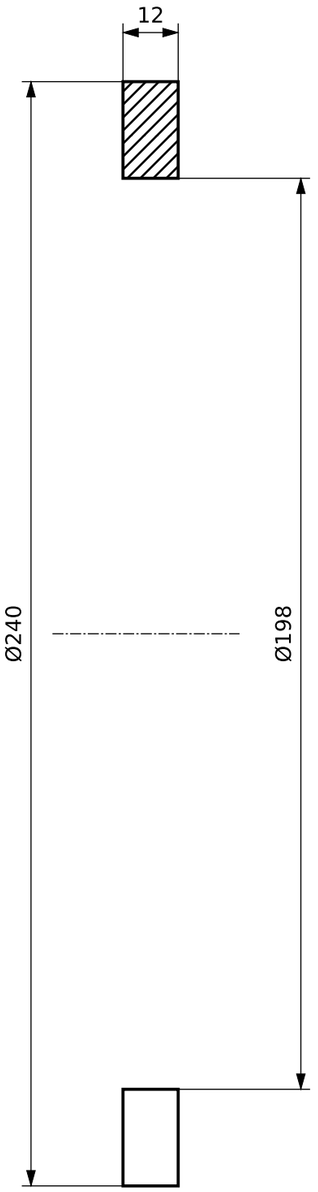
FRONT COVER: lip seal 180×200×10 HNBR + V-ring

6× Ø9 PCD 225; seal bore H8, Ra 1.6



REAR COVER: lip seal 195×220×12 HNBR on NP1-102

6× Ø9 PCD 225; seal bore H8, Ra 1.6



NOTES

1. Rev B (doc 17): both bores are larger than every part they pass. Front cover goes on after the KMT 27 nut and the rolling-torque check; rear cover is bolted to the drum before the drum goes over the fitted rear cone (Ø191 envelope) onto the sleeve NP1-102.
2. Seal lips face the bearing (grease retention); V-ring outside on the front cover runs against the spacer face.
3. Purge: grease exits under the lip; collect with absorbent ring at service.
4. Pilot on the housing bore Ø210, 2 mm deep (not shown), for concentric seals.

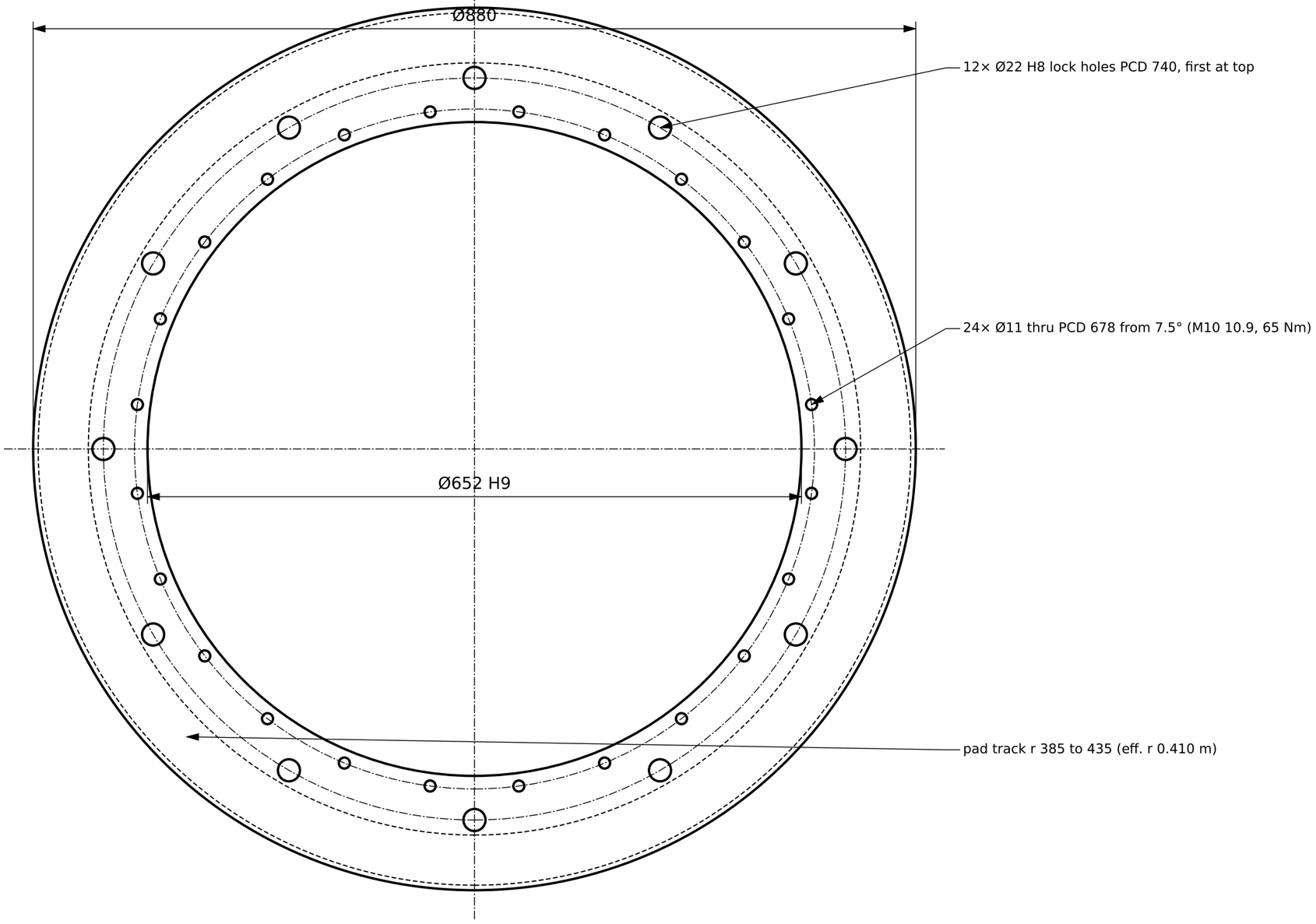
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

MAIN BEARING SEAL COVERS (FRONT / REAR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-211	B · 2026-11-03	1:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2	2.8 kg	Claude / not checked

VIEW · Rev B · thickness 16 · slides on from the front over the Ø646 drum shell



NOTES

1. Faces ground parallel 0.03, flatness 0.05, Ra 1.6. Axial run-out mounted on the drum ≤ 0.10.
2. Two spring-applied hydraulically released calipers (bought), at 90° and 270° (horizontal), each ≥ 13.3 kN clamp force on 16 mm disc; brake torque 8.7 kNm at μ 0.4.
3. Rev B (doc 17): bore Ø640 → Ø652 (fits over the drum shell from the front), OD Ø800 → Ø880 (pad track outside the Ø710 flange ring, room for the caliper behind the disc), lock holes moved here from the ring.
4. Corrosion: zinc-flake coating outside the pad track; pad track left bare (it is cleaned by the weekly brake test, doc 11).

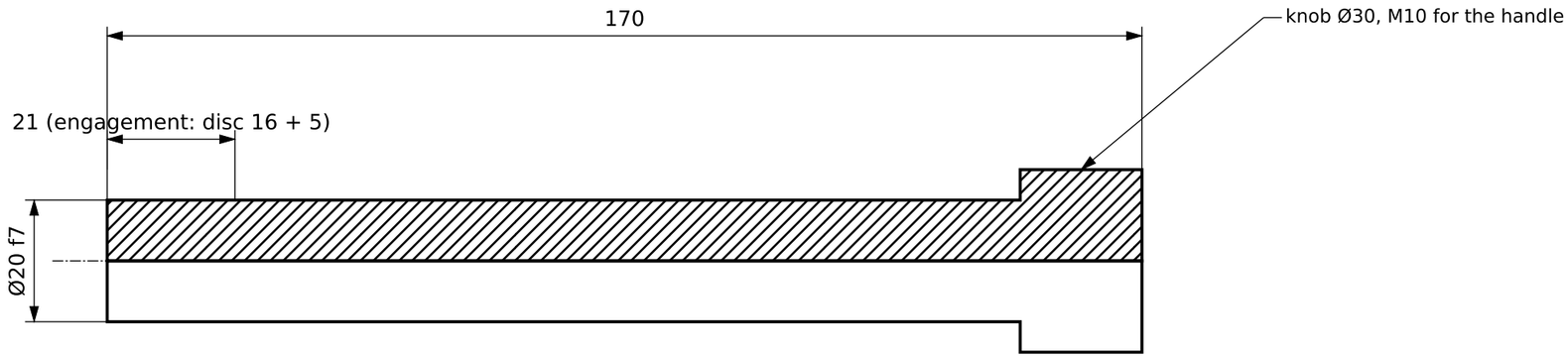
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

BRAKE DISC AND ROTOR LOCK HOLES

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-230	B · 2026-11-03	1:4.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C45E+N (or S355J2+N), 16 mm	34.5 kg	Claude / not checked

LOCK PIN



NOTES

1. Rev B (doc 17): pin Ø20 at the top (12 o'clock) on PCD 740, entering the disc from behind; holder NP1-321 sits on the top edge of the bedplate front plate and also carries the two overspeed sensors. Shear at 8.7 kNm: 23.5 kN, τ = 75 MPa (allowable 300); bearing on the 16 mm disc 73 MPa.
2. Stroke 40: retracted, the nose is 19 mm behind the disc and clear of the Ø710 ring and the rear cover (r ≥ 360).
3. Holder has a proximity switch that reports 'pin engaged' to the safety chain (no rotor release while engaged, doc 11).
4. Insert only at standstill with the brake applied.

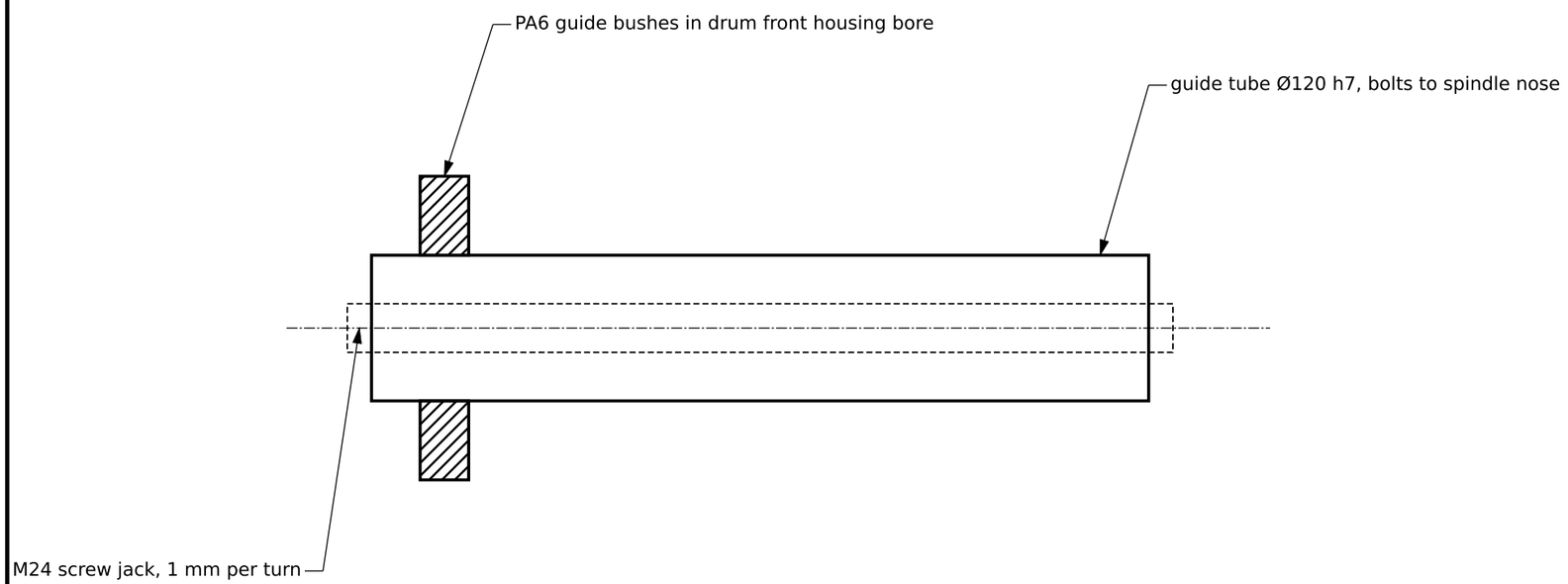
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

ROTOR LOCK PIN AND HOLDER

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-320	B · 2026-11-03	1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
pin 42CrMo4+QT; holder S355J2	≈ 4 kg	Claude / not checked

GUIDE TUBE ON SPINDLE NOSE



NOTES

1. Radial magnetic pull with the drum off-centre is ≈ 1.9 kN per mm of eccentricity (doc 07) and grows quickly; never let the magnet ring approach the stator by hand.
2. Procedure: spindle vertical, nose up; guide tube bolted to the nose; drum hung from the crane, centred by two PA6 bushes on the tube before the magnets reach the stator; lowered by the screw jack until the rear cup reaches the rear cone.
3. Gap check during lowering: 4 non-magnetic feelers (G10, 1.5 mm) in the gap at 90°.

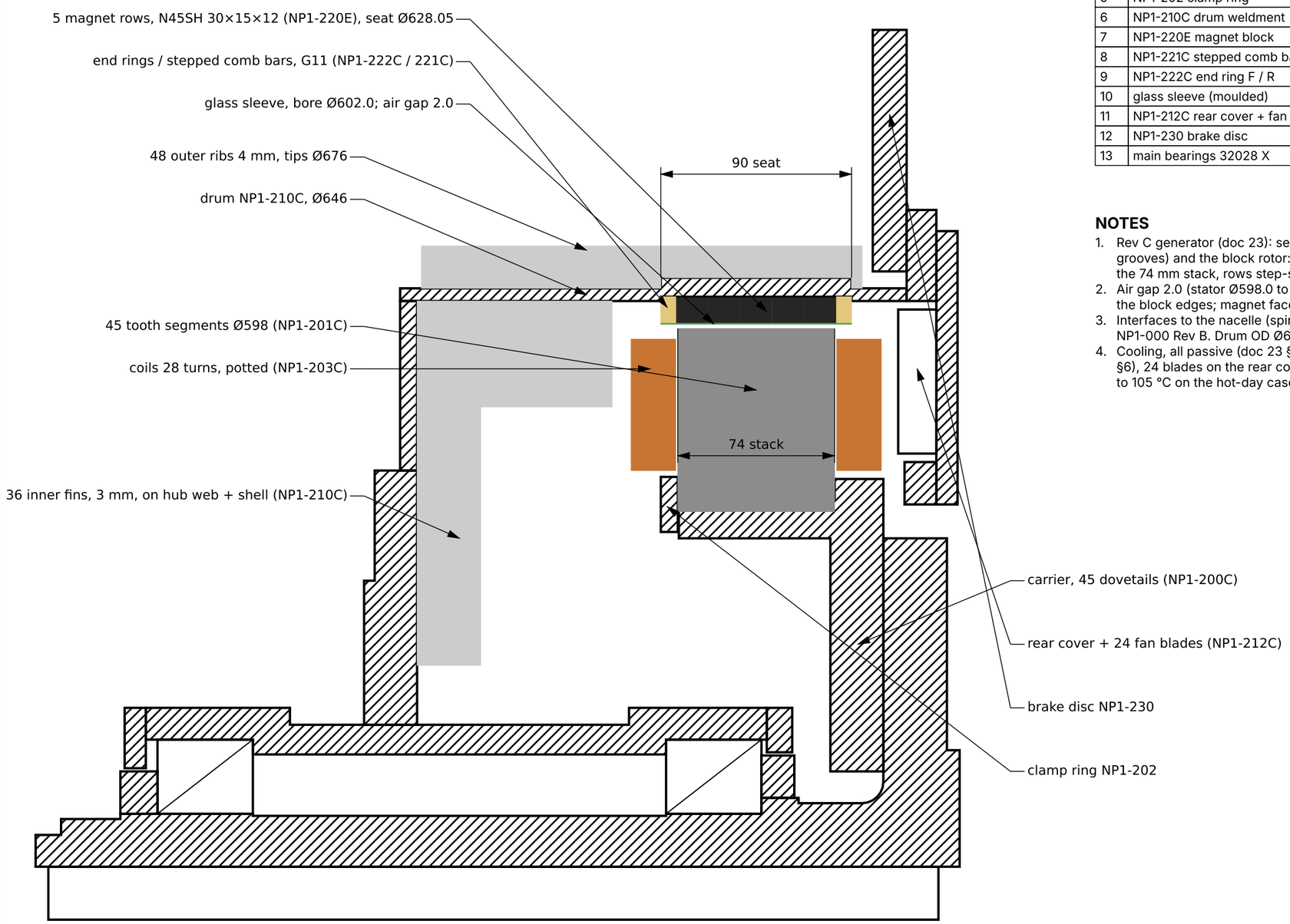
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

DRUM ASSEMBLY GUIDE (MAGNET SAFETY TOOL)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-T01	B · 2026-11-03	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355 / PA6 bushes	≈ 30 kg	Claude / not checked

GENERATOR SECTION (upper half, 1:2.2)



#	Part	Qty	Material / note
1	NP1-100 spindle (stationary)	1	42CrMo4+QT
2	NP1-200C stator carrier	1	S355J2+N, 45 dovetails
3	NP1-201C tooth segment stack	45+3	M270-35A bonded, 205 sh.
4	NP1-203C coil on segment	45+3	28 t × 4 × Ø1.55, cl. 200
5	NP1-202 clamp ring	1	1.4404
6	NP1-210C drum weldment	1	S355J2+N, seat Ø628.05
7	NP1-220E magnet block	200+10	N45SH 30×15×12
8	NP1-221C stepped comb bar	40+4	G11
9	NP1-222C end ring F / R	1+1	G11
10	glass sleeve (moulded)	1	E-glass/epoxy 0.5–0.9
11	NP1-212C rear cover + fan	1	5083, 24 blades
12	NP1-230 brake disc	1	C45E+N
13	main bearings 32028 X	2	bought

NOTES

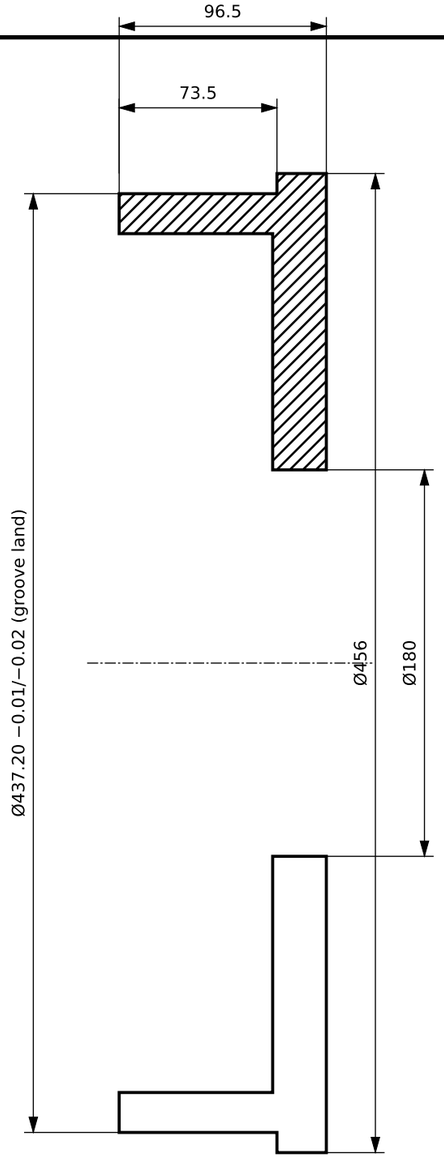
1. Rev C generator (doc 23): segmented stator of doc 18 (45 tooth segments in carrier dovetail grooves) and the block rotor: one N45SH block 30 × 15 × 12 per pole across, five rows along the 74 mm stack, rows step-skewed 0.20° by the stepped comb bars.
2. Air gap 2.0 (stator Ø598.0 to sleeve bore Ø602.0); sleeve 0.52 at the block centre, 0.89 at the block edges; magnet face r 301.52.
3. Interfaces to the nacelle (spindle, bearings, hub flange, disc, bedplate) unchanged from NP1-000 Rev B. Drum OD Ø646 unchanged.
4. Cooling, all passive (doc 23 §4 and §4a): carrier → spindle → finned insert path (doc 14 §6), 24 blades on the rear cover, 36 inner fins and 48 outer ribs on the drum; hot spot 94 to 105 °C on the hot-day cases (limit 130).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

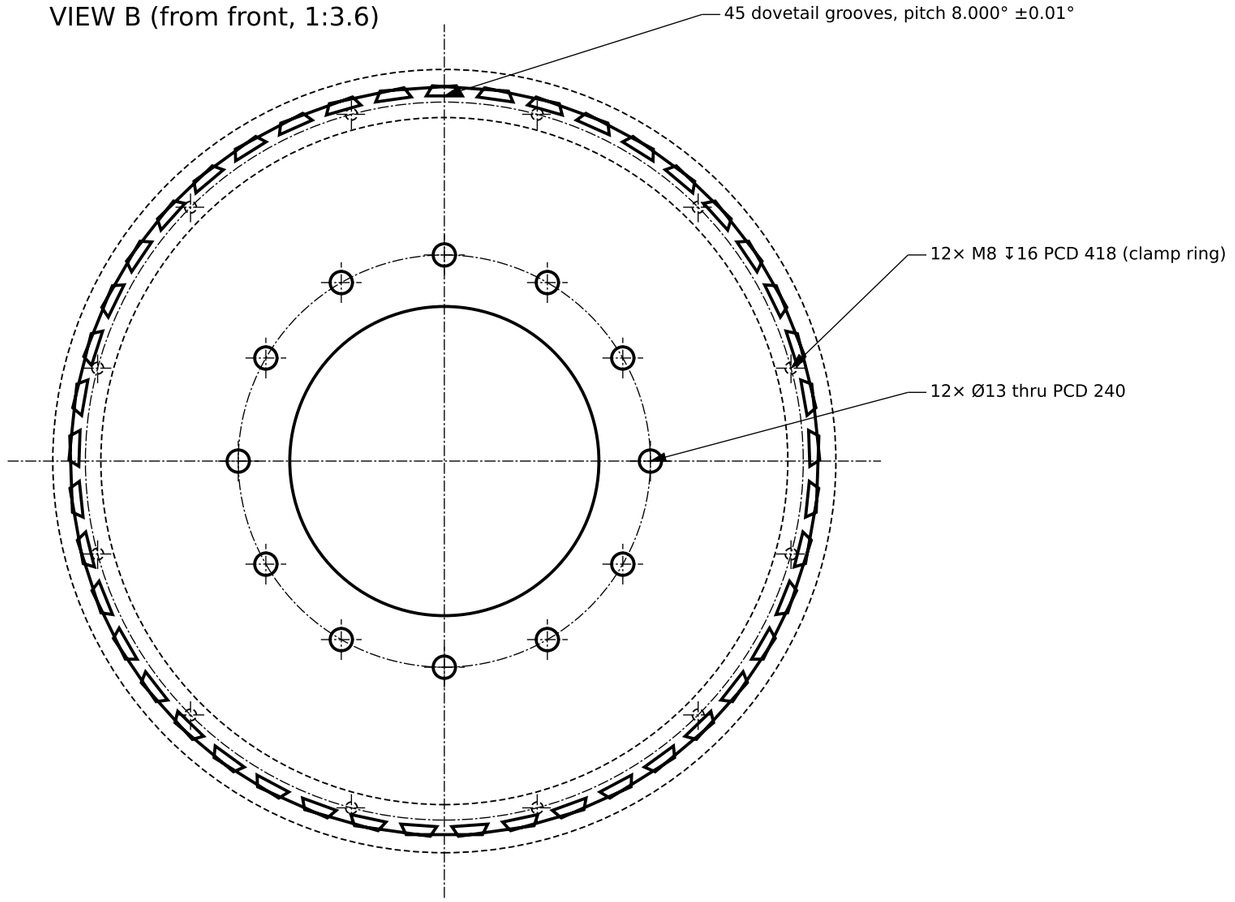
GENERATOR REV C · ASSEMBLY SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-G00	C · 2026-11-05	1:2.2
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
see parts list	≈ 330 kg drivetrain (no hub)	Claude / not checked

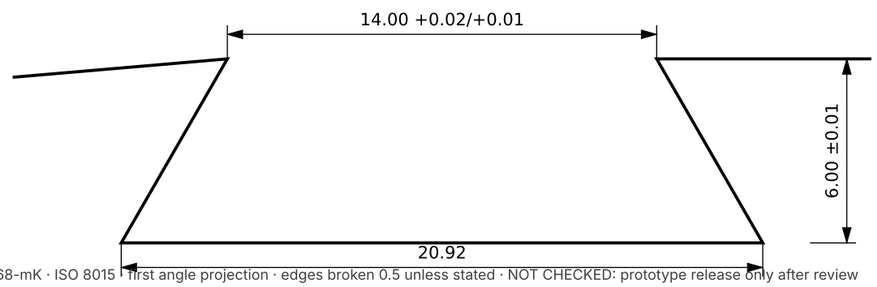


SECTION A-A (1:2.86)

VIEW B (from front, 1:3.6)



DETAIL C: GROOVE (5:1), flanks 60°



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

1. Rev C (doc 18 §5.1): grooves replace the shrink fit and the 3 keys. Grooves open to the front (x 343.5), closed by the rear stack stop (x 417); broached or wire-EDM.
2. Groove angular position ±0.01°, groove seat radius R212.60 ±0.01 to the bolting-face axis; foot clearance 0.01 to 0.03 (slides by hand when clean), fixed by VPI resin, clamped by NP1-202 (12× M8 A4-80, 20 Nm).
3. Run-out of groove lands and rear stop face to the bolting face and Ø180 spigot ≤ 0.02; bolting face flatness 0.02.
4. Bolt to the spindle flange: 12× M12 10.9 + hardened washers, 110 Nm, 2 fitted dowels Ø10 H7 PCD 240; Al-filled paste 0.05 on the full face (heat path, doc 14 §6).
5. Stress relieve 580 to 620 °C, 2 h, before final machining. No paint on grooves and bolting face; elsewhere epoxy primer 60 µm.

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

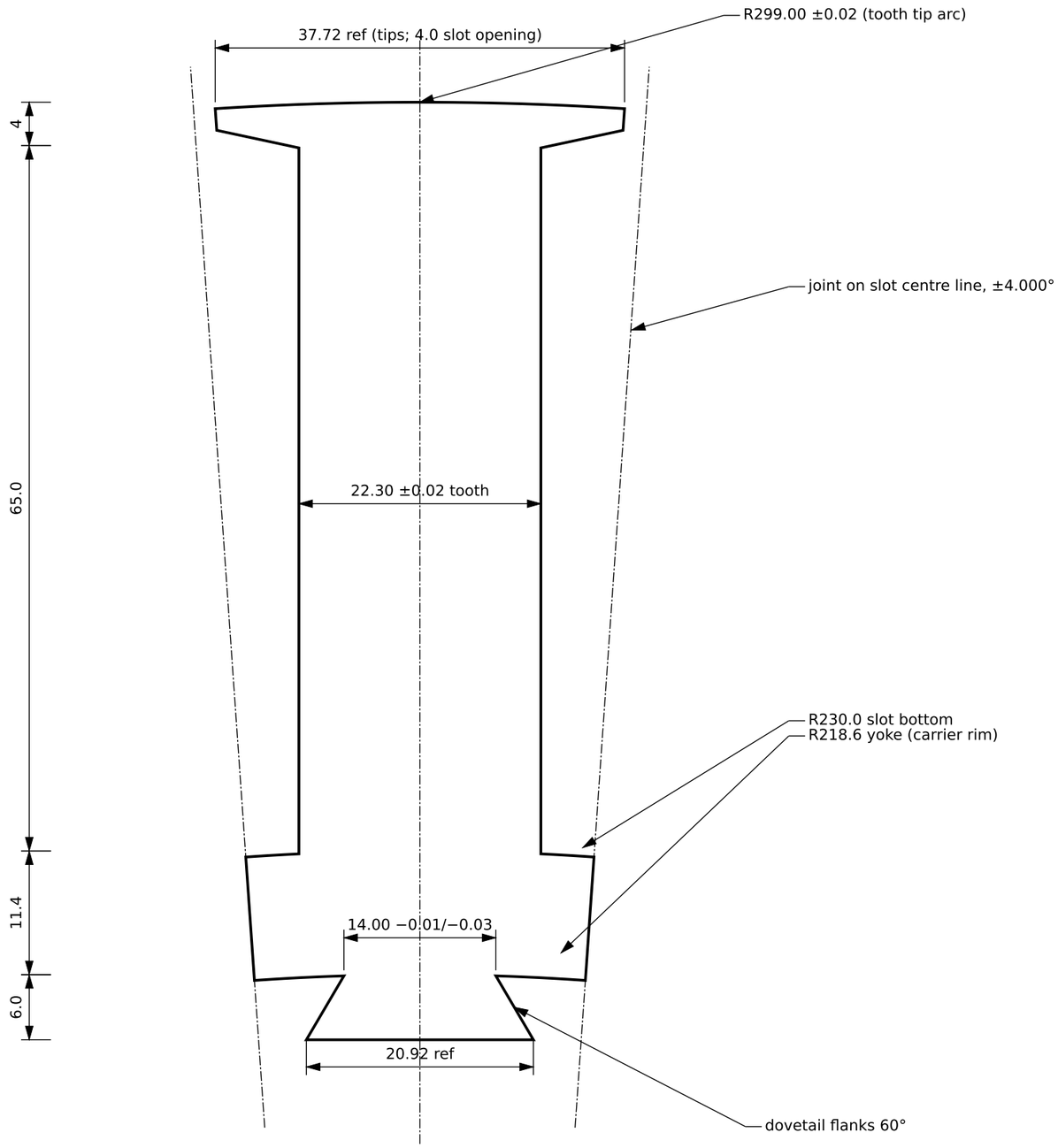
STATOR CARRIER WITH 45 DOVETAIL GROOVES

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-200C	C · 2026-11-05	1:2.86 / 1:3.6 / 5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N ring forging or plate, stress relieved	40.6 kg	Claude / not checked

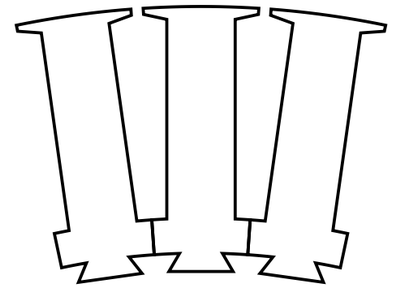
NOTES

1. DXF layer OUTLINE = cutting contour. Laser cut (P1), burr ≤ 0.02, heat zone ≤ 0.1; or progressive die (series).
2. Magnetic contour identical to the doc 14 one-piece lamination; joints on the slot centre lines in the yoke (8.000° pitch).
3. Stack: 205 sheets, 74.0 ±0.1 (sheets counted, height measured per segment), bonded in fixture NP1-T03 at 1.0 to 1.5 MPa, varnish cure per supplier.
4. Checks per stack: tip radius R299.00 ±0.02 on gauge, foot width 14.00 −0.01/−0.03, slot opening edges straight 0.05 over 74; Franklin test ≥ 10 Ω·cm².
5. Do not machine tooth tips after stacking. 45 + 3 spare segments per stator; all identical.

SEGMENT (2:1), radial direction up



3 OF 45 SEGMENTS (1:2)



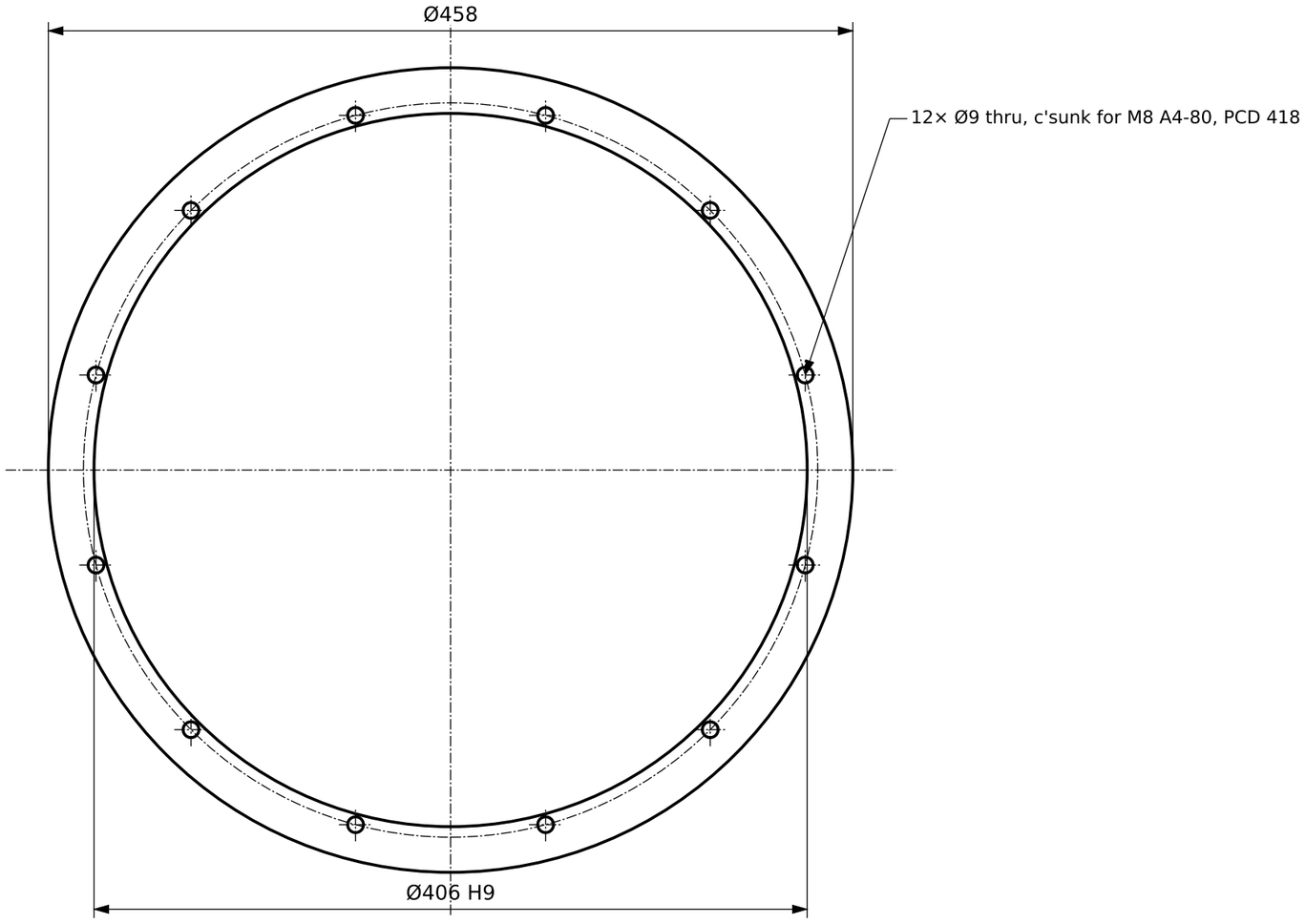
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STATOR TOOTH SEGMENT LAMINATION (45 + 3 PER STATOR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-201C	C · 2026-11-05	2:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
M270-35A, 0.35 mm, self-bonding varnish	1.07 kg per stack	Claude / not checked

VIEW (thickness 8.0, faces parallel 0.05)



NOTES

1. Laser cut from 8 mm plate, faces machined flat 0.05.
2. Must not touch the coils: inner end-winding radius ≥ 232; ring OD 229.

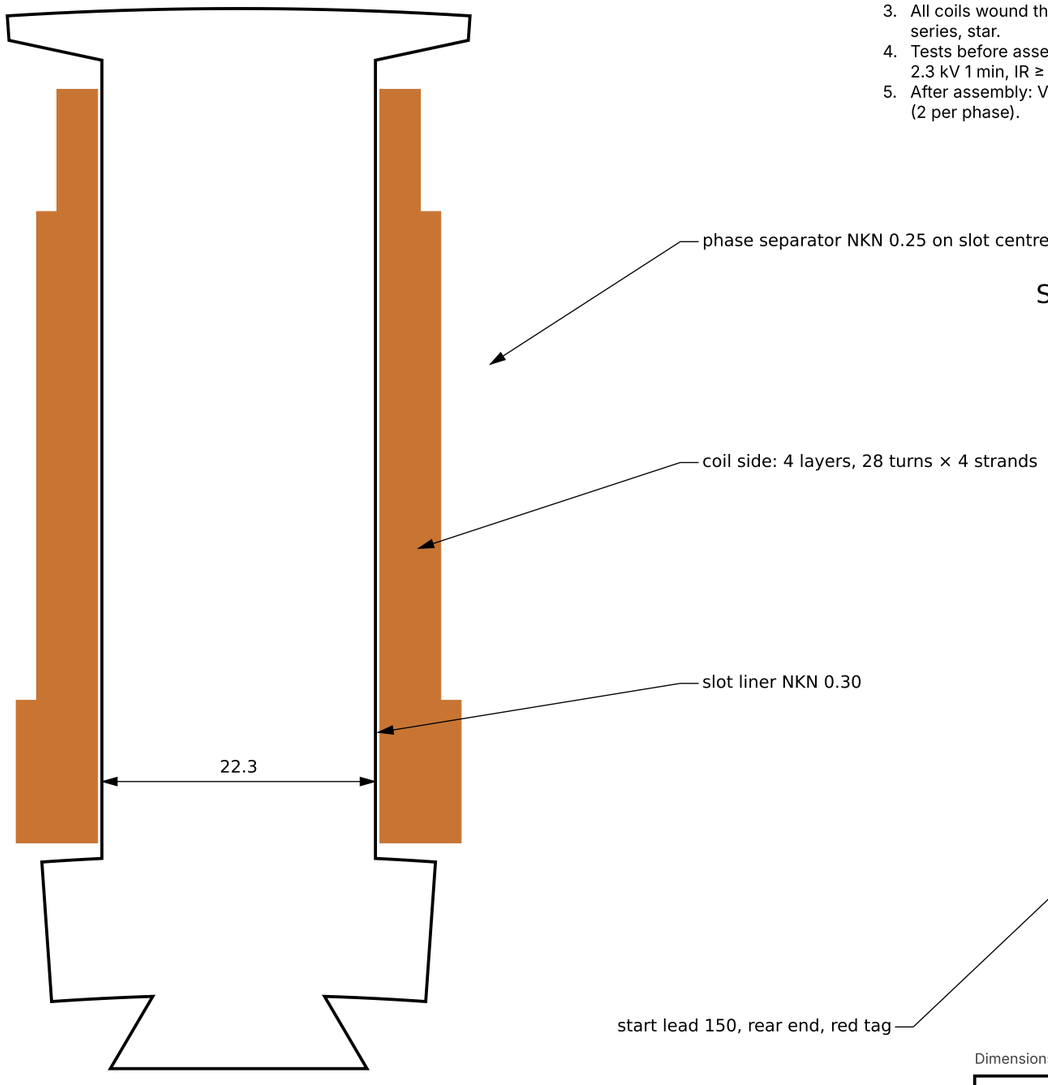
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STATOR FRONT CLAMP RING

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-202	B · 2026-11-05	1:3.3
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
1.4404 (non-magnetic)	2.2 kg	Claude / not checked

COIL SIDES IN THE SLOT (2:1)

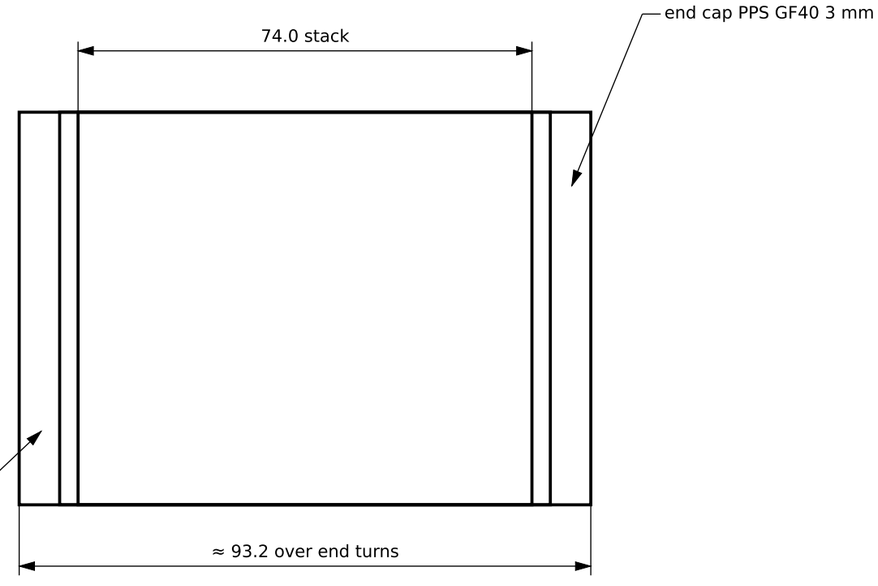


Layer (from tooth flank)	Positions	Used
1	37	37
2	37	37
3	31	31
4	17	7
5	3	0 (reserve)
Total (28 turns × 4 strands)	125	112

NOTES

1. Wind on chuck NP1-T04 by the dovetail, 4 parallel strands Ø1.55 (≤ 1.66 over enamel), 25 to 30 N per strand, layer by layer from the tooth flank, ≤ 60 rpm (doc 18 §5.6).
2. 28 turns per coil (Rev C keeps doc 18): 373 V line at 178 rpm, magnets 100 °C; 389 V at 60 °C. Wire per coil ≈ 6.6 m per strand; R20 per phase 0.217 Ω.
3. All coils wound the same direction; reversal at connection (doc 18 Appendix A): 15 coils per phase in series, star.
4. Tests before assembly (doc 18 §5.7): resistance ±2 % of the lot mean, surge 3.5 kV, hipot to the segment 2.3 kV 1 min, IR ≥ 1 GΩ.
5. After assembly: VPI class H epoxy, end windings potted (doc 18 §5.11, §5.12). PT100 in 6 separator gaps (2 per phase).

SIDE VIEW OF COIL ON TOOTH (1:1, simplified)



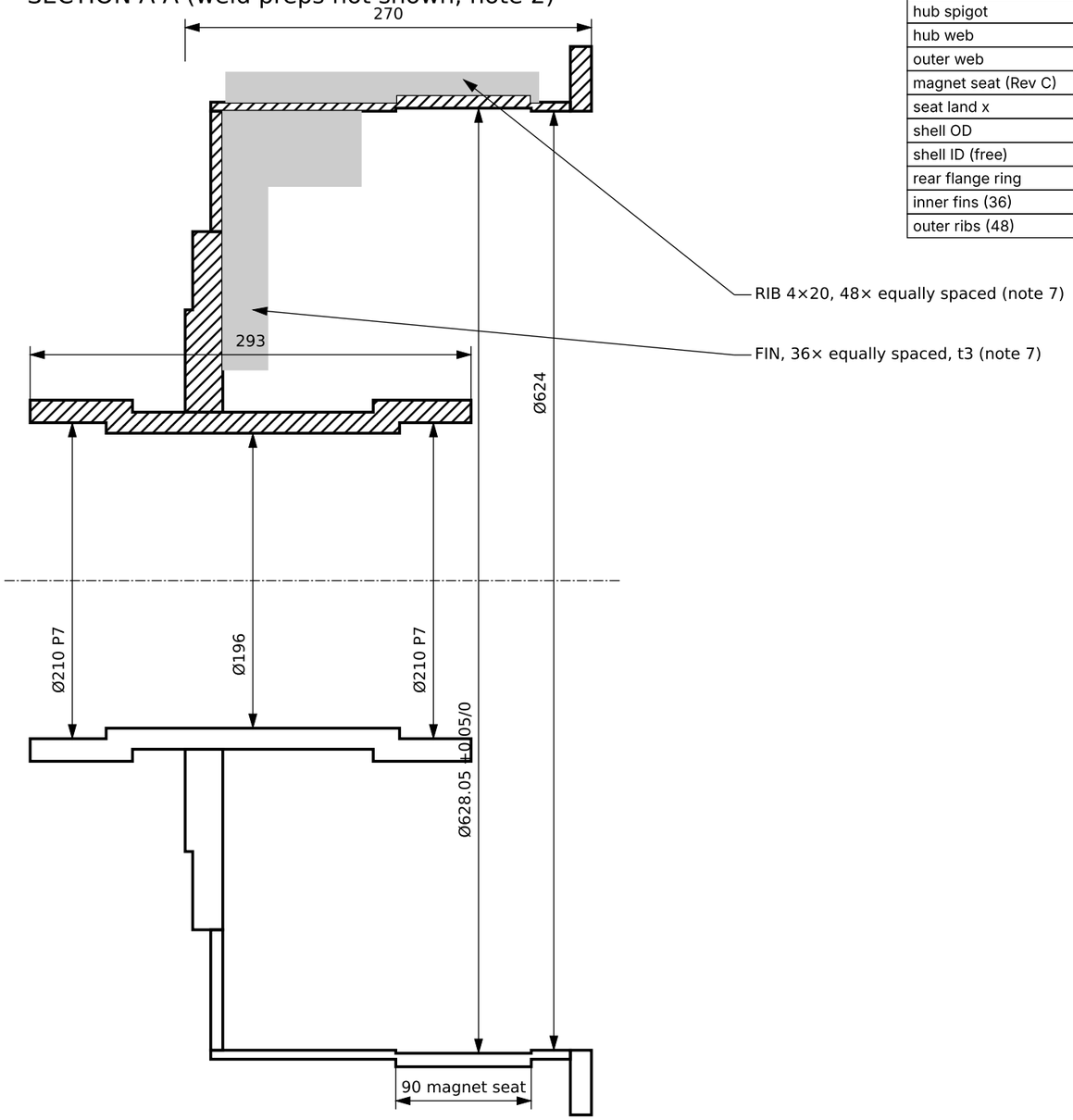
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

COIL ON SEGMENT (WOUND SEGMENT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-203C	C · 2026-11-05	2:1 / 1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
Cu enamelled Ø1.55 grade 2, class 200; NKN 0.30; PPS 4540 caps	≈ 0.45 kg Cu per coil	Claude / not checked

SECTION A-A (weld preps not shown, note 2)



Feature	Size	Tolerance / note
cup seats (2)	Ø210 × 45	P7, Ra 0.8, coax. 0.02
cup shoulders	Ø196	square 0.01 to axis
hub spigot	Ø360 × 5	f7; face flat 0.03
hub web	20 thick, Ø464	12× M16 thru PCD 400
outer web	8 thick	full pen. weld to shell
magnet seat (Rev C)	Ø628.05 × 90	+0.05/0, coax. 0.02, Ra 3.2
seat land x	335 to 425	as Rev B
shell OD	Ø646	steel under seat 8.98
shell ID (free)	Ø624	clearance to magnets
rear flange ring	Ø624/Ø710 × 14	M10 PCD 644 + M10 PCD 678
inner fins (36)	t3, x 220-312, r 140-312	L-plate, web + bore, every 10°
outer ribs (48)	4 × 20, x 222-430	tips Ø676, every 7.5°

NOTES

1. Rev C: magnet seat Ø628.05 +0.05/0 (was Ø618.40) for the 12 mm N45SH blocks; steel under the seat 8.98 (FEA: 1.45 T median, 1.7 T at the pole edges, included in the EMF).
2. Welds full penetration at web/housing and web/shell, ISO 5817 B, 100 % MT, UT of the shell seam (ring-rolled shell preferred, doc 18 §4.2). Stress relieve 580 to 620 °C before machining.
3. Machine in one set-up: cup seats Ø210 P7, magnet seat, hub spigot Ø360 f7, rear flange face; coaxiality ≤ 0.02, shoulders square ≤ 0.01.
4. Hub web 12× M16 thru PCD 400; rear flange ring 24× M10 PCD 644 (rear cover) and 24× M10 PCD 678 (disc).
5. Magnet seat: degrease, grit blast Sa 2½ just before bonding; no paint inside. Outside zinc spray + epoxy (C5).
6. Balance finished rotor (drum, magnets, sleeve, disc, cover) to ISO 21940-11 G2.5 at 250 rpm, correction on the rear flange ring.
7. Passive cooling (doc 23 §4a): 36 inner fins t3 S355, L-shaped, fillet-welded both sides to hub web and shell bore, every 10°, 9 mm clear of the end windings; 48 outer ribs 4 × 20 S355 on the shell outside, every 7.5°, intermittent fillet welds 30/60 both sides, ends rounded. Weld fins and ribs before stress relief and machining.
8. Rear flange ring welded to the shell full round, continuous: with the hub web it keeps the 9 mm shell round against the magnet pull (doc 23 §5). Seat roundness 0.05 after stress relief.

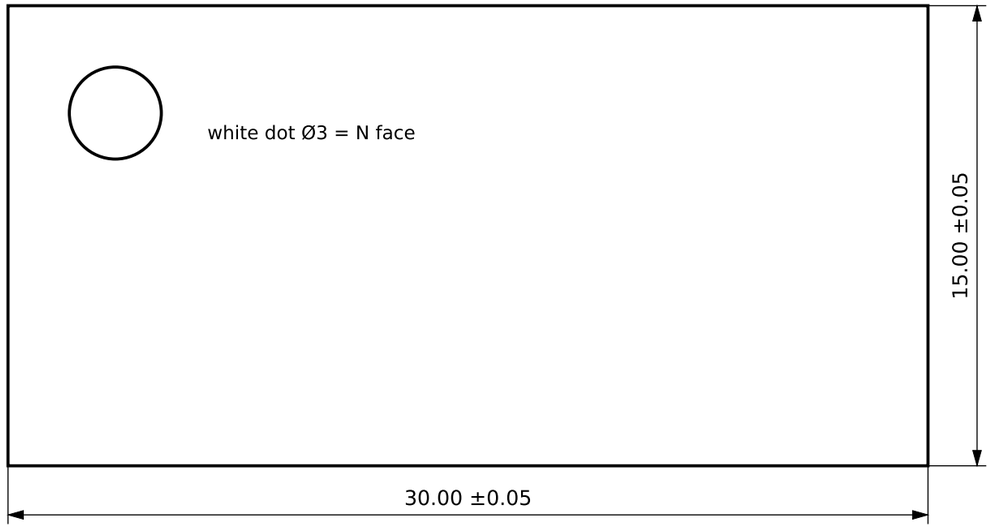
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

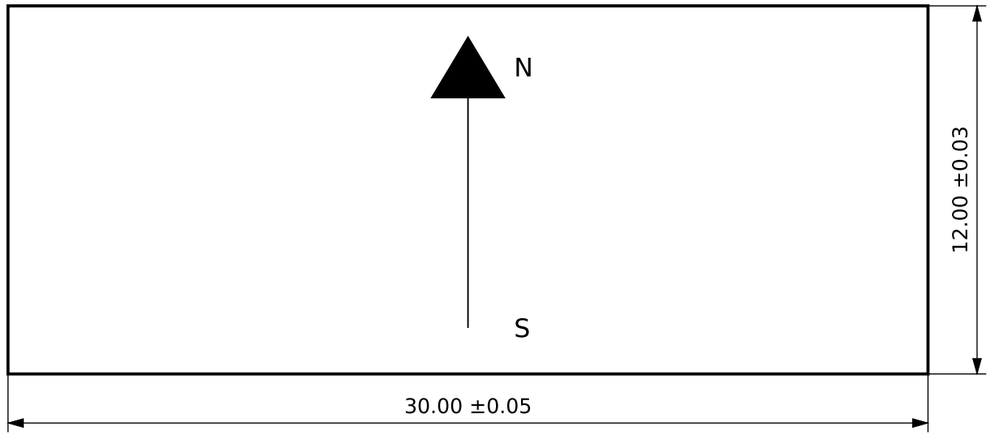
ROTOR DRUM WELDMENT (GENERATOR YOKE + BEARING HOUSING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-210C	C · 2026-11-05	1:3.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, welded, stress relieved	91.8 kg + 12.5 kg fins/ribs	Claude / not checked

VIEW A: N FACE (5:1)



VIEW B: END, magnetised through 12 (5:1)



Item	Requirement
Grade	N45SH: Br ≥ 1.32 T, HcJ ≥ 1,592 kA/m (20 kOe), (BH)max ≥ 342 kJ/m³ at 20 °C
Curves	J-H and B-H per batch at 20, 60, 100, 120 °C; αBr, βHcJ stated
Magnetisation	Through the 12 mm thickness, saturated; axis within 2° of the face normal
Tolerances	30 ±0.05, 15 ±0.05, 12 ±0.03; faces flat 0.02, parallel 0.02
Edges	Chamfer 0.2 to 0.3 × 45° all 12 edges; no chips > 0.3
Coating	NiCuNi 10 to 20 µm; 72 h neutral salt spray clean; adhesion OK with the epoxy
Uniformity	Moment of each piece within ±1.5 % of the lot mean; value per serial
Marking	White dot Ø3 on the N face; serial on the packaging position
Quantity	200 + 10 spare per generator; 8 extra for Q1 tests
Packing	Repelling layout in non-magnetic spacers; air freight per IATA PI 953

NOTES

1. One part for N and S poles: N poles face N towards the stator, S poles are the same block flipped (dot hidden against the seat).
2. Solid block (not two 6 mm pieces bonded). If a maker offers only 30 × 15 × 6, the doc 20 bonded pair is the fall-back with identical magnetics.
3. Five blocks per pole along the stack (15 mm pieces = axial segmentation for eddy loss, 4.7 W at rated).
4. Release of N45SH for production is conditional on the 100 °C design basis (doc 19 §16.3, doc 23 §4): bench magnet temperature ≤ 75 °C, temperature watch, no deliberate phase shorting.

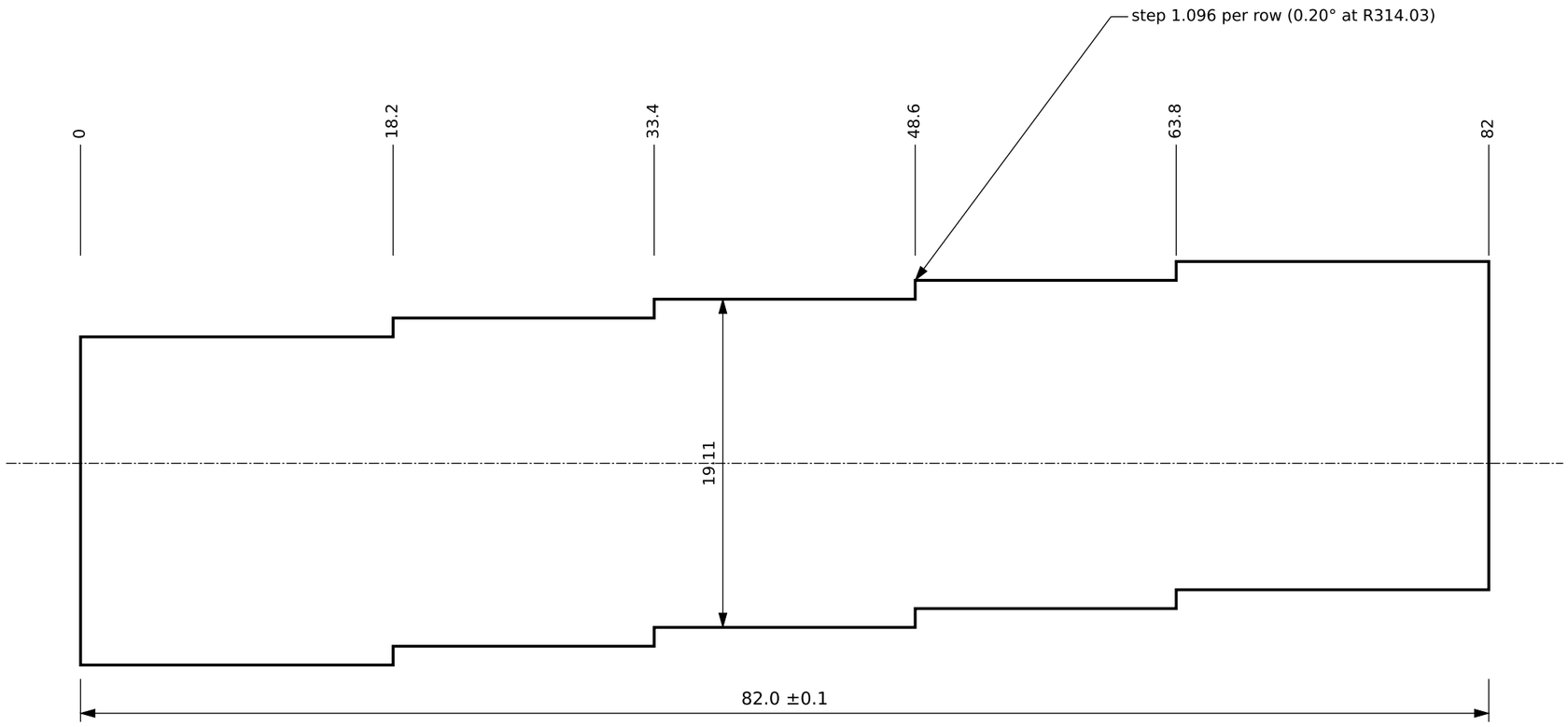
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

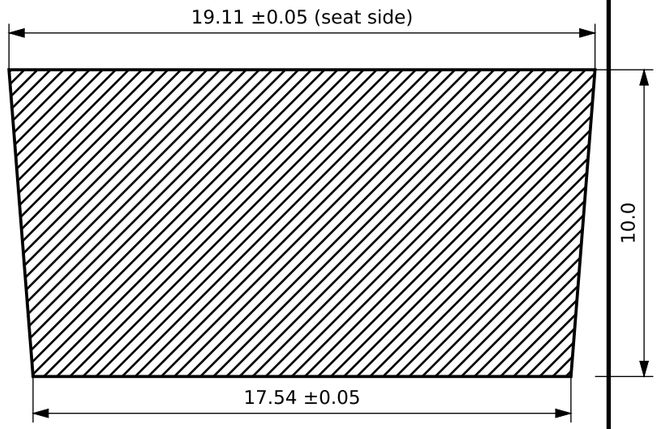
MAGNET BLOCK 30 × 15 × 12 (PURCHASE DRAWING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-220E	E · 2026-11-05	5:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
Sintered NdFeB N45SH, NiCuNi	41.0 g each, 8.21 kg per generator	Claude / not checked

PLAN, seen from the stator (3:1). Steps set the row skew.



SECTION (5:1), flanks follow the block sides



NOTES

1. One bar per pole gap (40 per rotor + 4 spare). Bar sits on the seat (wide face), bonded with the magnet epoxy; ends engage 4 mm pockets in the end rings NP1-222C.
2. Row boundaries (ordinates from the front end): steps of 0.20° each; the bar is identical for every gap. Mark the front end.
3. Side faces are the datum for the blocks: 0.10 glue clearance each side (comb width = gap between neighbouring blocks − 0.20).
4. G11 (class H), ±0.05 on widths; edges deburred, no delamination. Prove on the Q1 panel (doc 18 §10).

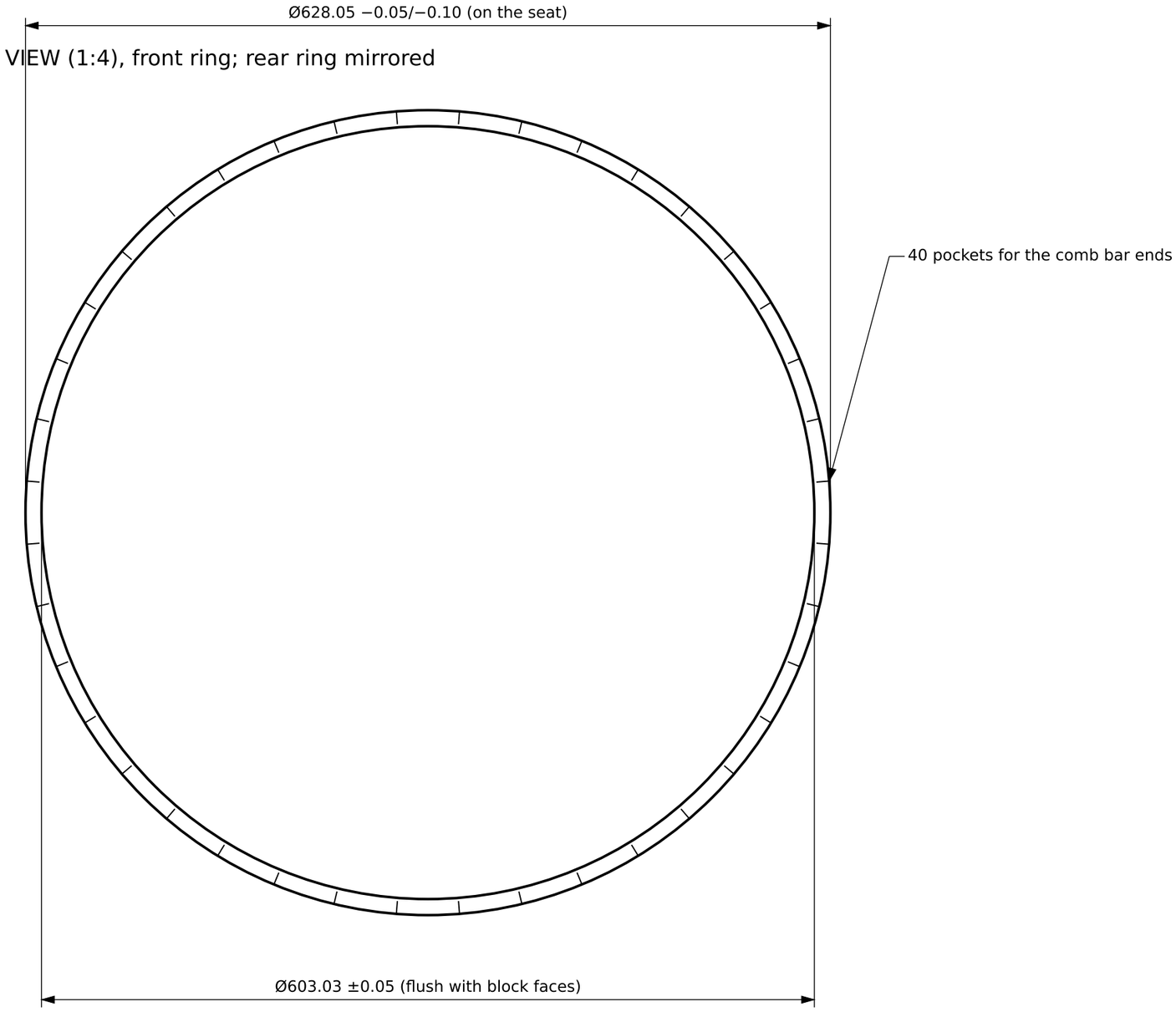
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

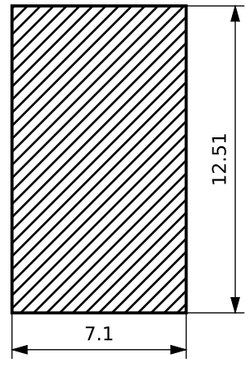
STEPPED POLE COMB BAR (40 + 4 PER ROTOR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-221C	C · 2026-11-05	3:1 / 5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
G11 glass-epoxy sheet, CNC machined	≈ 25 g	Claude / not checked

VIEW (1:4), front ring; rear ring mirrored



SECTION (4:1)



NOTES

1. Front ring at x 335 to 342.1, rear ring at x 417.9 to 425 (both 7.1 wide). They close the magnet pack and hold the comb bar ends.
2. Pockets: 19.11 wide (seat side), 10 deep radially, 4 deep axially, on the gap centre lines (4.5° + n × 9°), front ring rotated −0.40°, rear ring +0.40° (row skew).
3. May be made in 4 bonded arcs if no G11 tube of this size is available; joints on pocket centre lines.
4. Bonded to the seat with the magnet epoxy, then covered by the moulded glass sleeve.

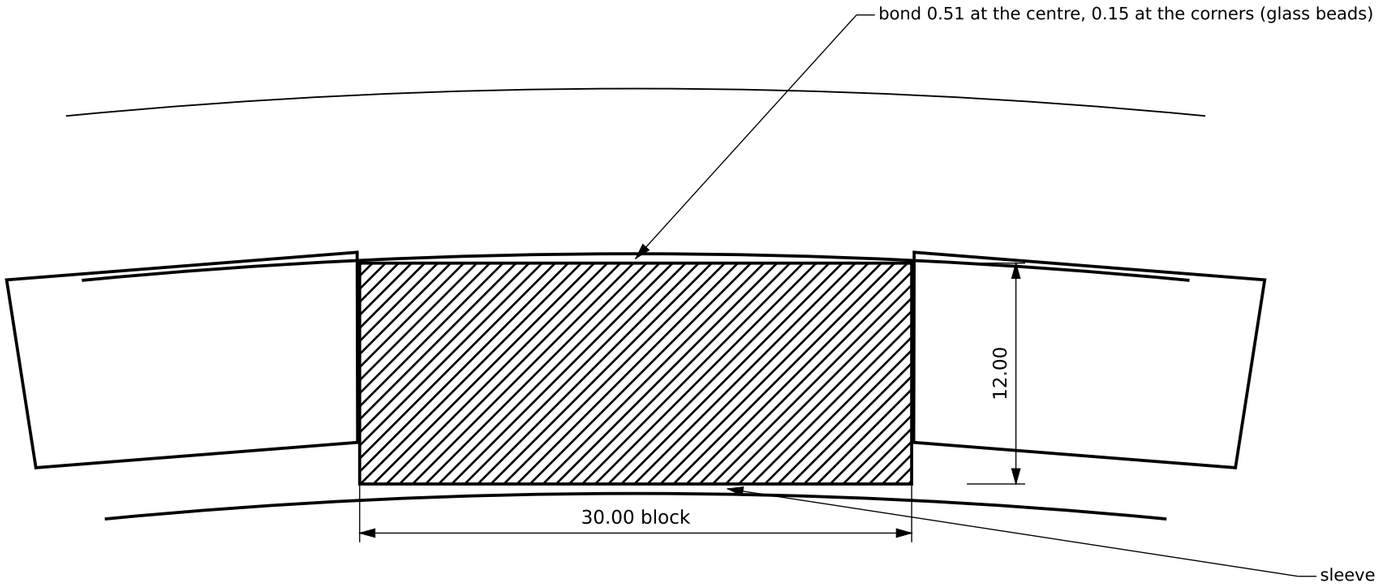
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

MAGNET END RINGS, FRONT AND REAR

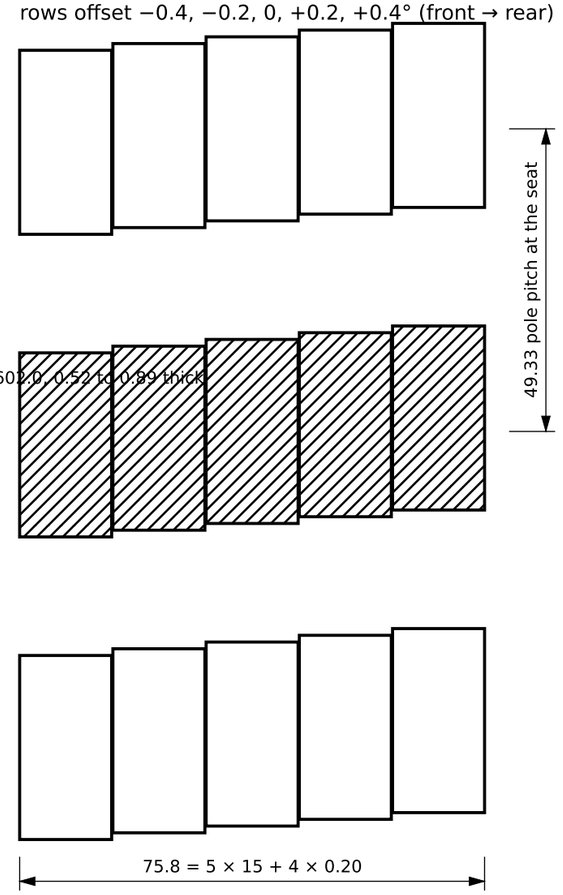
DRAWING NO.	REV / DATE	SCALE (A3)
NP1-222C	C · 2026-11-05	1:4 / 4:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
G11 glass-epoxy, machined from tube or segments	≈ 0.12 kg each	Claude / not checked

POLE SECTION (3:1), centre line = pole N1



LAYOUT, 3 POLES UNROLLED (1:1), front row left

rows offset −0.4, −0.2, 0, +0.2, +0.4° (front → rear)



NOTES

1. Bond: two-component toughened epoxy, RT cure, 80 °C post-cure, with 0.15 glass beads (doc 18 §4.2). Blocks placed with jig NP1-T02 against the comb bar faces; all N poles first, then S.
2. Sequence per pole: comb bars and end rings bonded and cured first; blocks row by row from the front; check each row against the template (skew 0.20 ±0.02° per row).
3. Pole map NP1-T10 after cure: every pole centre within ±0.05° of nominal, flux per pole within ±1.5 %.
4. Sleeve: E-glass/epoxy moulded on the Al mandrel NP1-T06 located on both cup seats; bore Ø602.0 ±0.03, thickness 0.52 at the block centres, 0.89 at the edges; covers the end rings.
5. Q1 check (doc 18 §10): sleeve thickness by section 0.45 to 0.95; air gap feeler check 2.0 ±0.1 all round after joining.

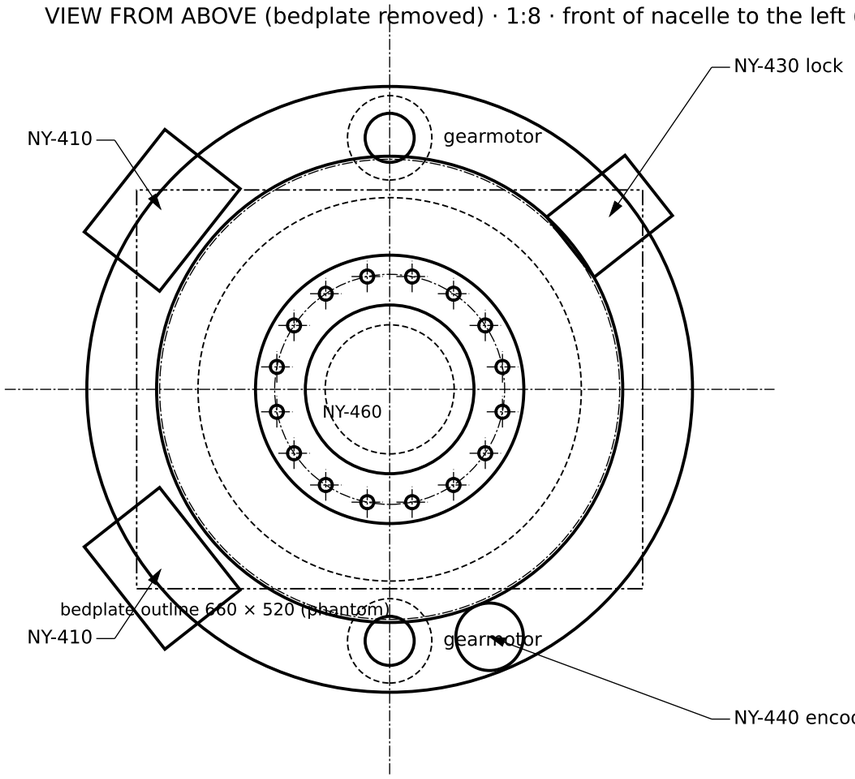
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

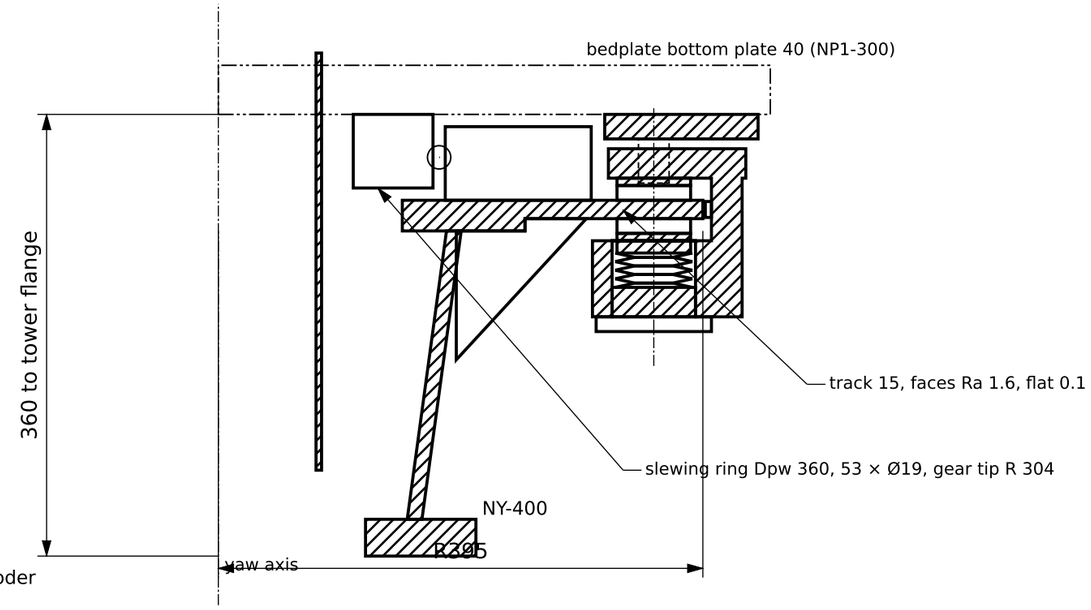
ROTOR MAGNET PACK: POLE SECTION, LAYOUT AND SLEEVE

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-223C	C · 2026-11-05	3:1 / 1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
assembly	8.2 kg magnets	Claude / not checked

VIEW FROM ABOVE (bedplate removed) · 1:8 · front of nacelle to the left (−X)



SECTION A-A through the pad unit at 141.8° · 1:5



Item	Part	Qty	Make/buy	Mass
1	NY-400 tower-top adapter (S355J2+N weldment)	1	make	107 kg
2	Slewing ring Dpw 360, ext. gear z150 m4, ind. hardened teeth	1	buy	≈ 55 kg
3	Gearmotor 0.18 kW 4-pole, 1:1,680 planetary, 2 Nm motor brake	2	buy	≈ 22 kg
4	NP1-420 drive bracket (doc 16)	2	make	7.7 kg
5	NY-410 friction pad unit	2	make	13 kg
6	NY-430 yaw lock	1	make	6.5 kg
7	NY-440 encoder + cam limit switch unit	1	make/buy	6.9 kg
8	NY-460 cable hanger	1	make	10 kg
9	Bolts: 16 M16 inner, 16 M16 outer (from below), 16 M20 tower	48	buy	
10	Heated 2D ultrasonic + heated cup/vane on the nacelle mast	1+1	buy	

NOTES

1. No hydraulics on the yaw system. Holding: 2 spring-applied motor brakes (fail-safe) through the gear train; damping: 2 passive pad units, 2.4 to 3.6 kNm.
2. Design yaw moment held 8 kNm by ONE pinion (backlash) without friction credit; gearbox static rating ≥ 1,500 Nm output.
3. All bolts 10.9, preload with calibrated torque, M16 90 kN, M20 110 kN. Grease the ring per supplier (−40 °C synthetic).
4. Yaw rate 0.5 °/s at 1,500 rpm. One VFD drives both motors; the safety chain opens the brake supply directly.

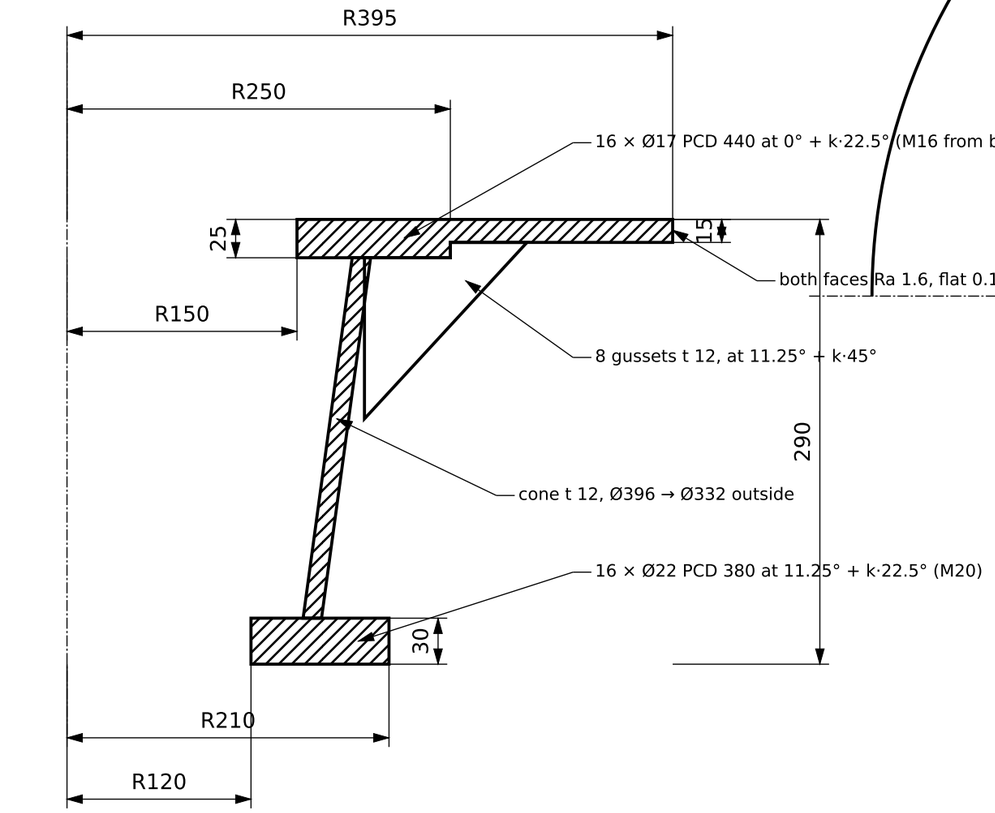
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

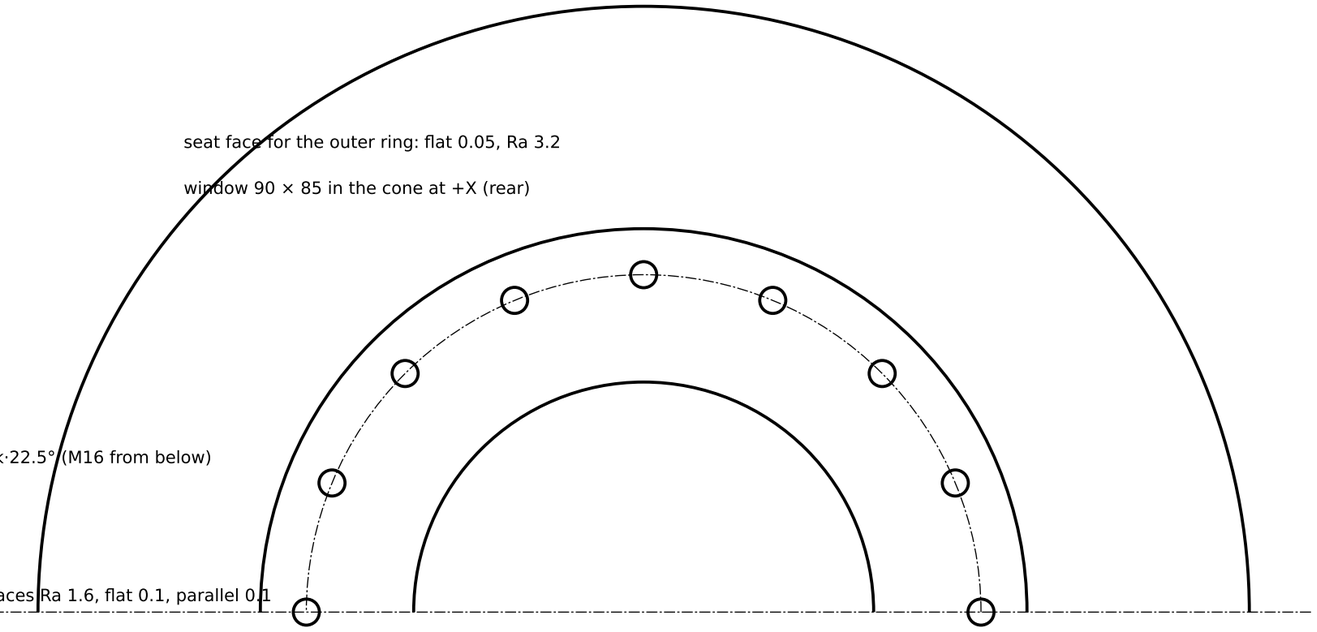
YAW SYSTEM · GENERAL ARRANGEMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NY-000	A · 2026-11-05	1:5 / 1:8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see parts list	≈ 275 kg (yaw incl. adapter)	Claude / not checked

HALF SECTION (between gussets) · 1:4



PLAN FROM ABOVE · 1:4



NOTES

1. Weld cone to seat plate and to flange with full-penetration welds (ISO 5817 B), gussets with 5 mm double fillets; stress relieve, then machine seat, track faces and flange in one set-up.
2. Seat and track are one machined plate: 25 mm blank, track turned to 15 mm (r 250 to 395). Track faces left bare under the pads; all other surfaces C5-VH coating (ISO 12944, zinc-rich primer).
3. Bolt loads at the SLM tilt moment 34.6 kNm: outer ring 19.1 kN per M16 (preload 90 kN), tower flange 32.0 kN per M20 (preload 110 kN).
4. Replaces the NP1-400 proposal of doc 16 (25 mm plate to Ø840, 143 kg): track moved in to r 395 and thinned to 15 mm because the pads only clamp 5.3 kN.
5. Gussets stay inside r 300 and below the track so the lower pad jaws (r ≥ 318) pass all the way round.

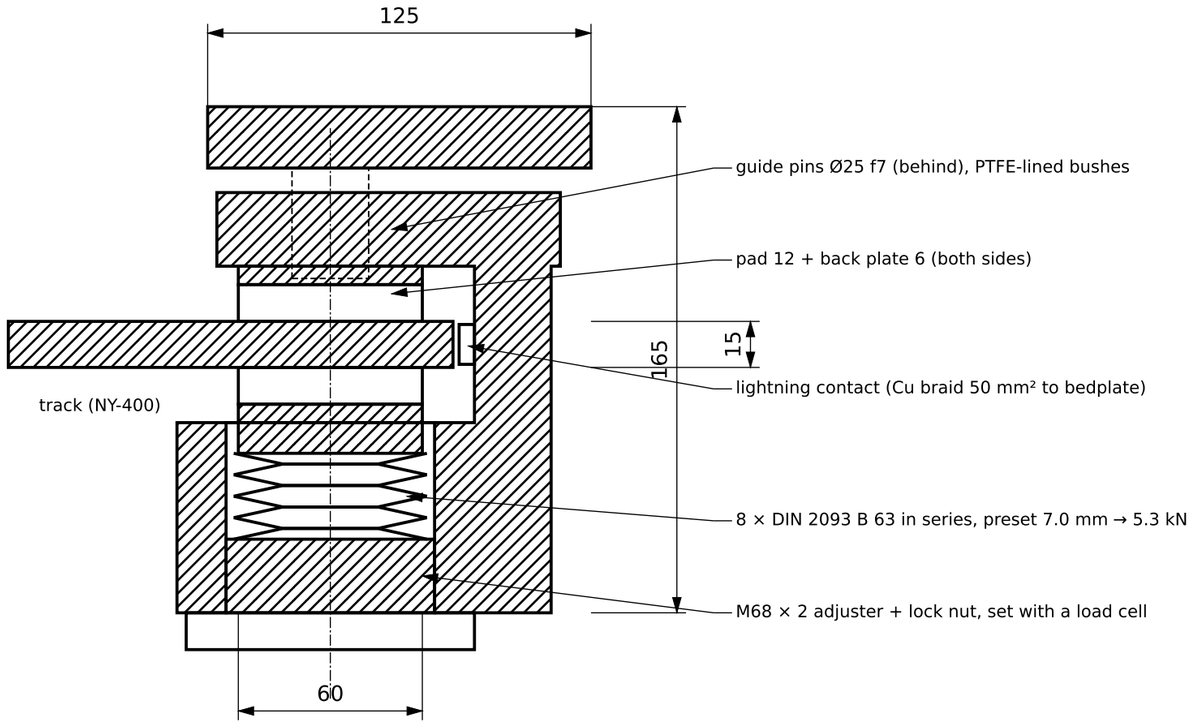
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

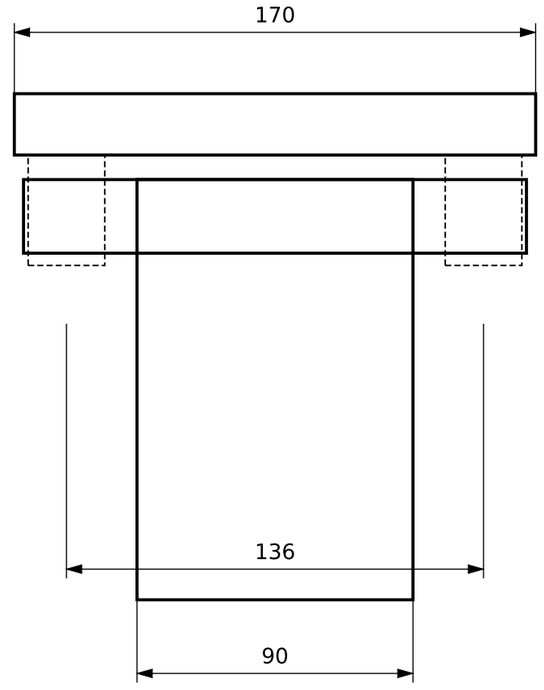
TOWER-TOP ADAPTER WITH FRICTION TRACK

DRAWING NO.	REV / DATE	SCALE (A3)
NY-400	A · 2026-11-05	1:4
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plate and tube, weldment	107 kg (model)	Claude / not checked

SECTION IN THE RADIAL PLANE · 1:2



VIEW FROM OUTSIDE (radial) · 1:2



Item	Part	Qty	Material / spec
1	Bracket 170 × 125 × 20, 4 × Ø17 to the bedplate	1	S355J2+N
2	C-frame (upper jaw, web, lower jaw), machined	1	S355J2+N
3	Guide pin Ø25 f7 × 55, pressed into item 1	2	1.4462 duplex
4	Pad 90 × 60 × 12 on 6 mm back plate	2	organic industrial pad, μ 0.40 ± 20 %
5	Piston Ø60 × 10	1	1.4462
6	Disc spring DIN 2093 B 63 (63 × 31 × 2.5)	8	51CrV4, zinc-flake
7	Adjuster M68 × 2 + lock nut	1+1	1.4462
8	Lightning contact block + Cu braid 50 mm²	1	CuSn / Cu

NOTES

1. Floating C-frame: the springs push the lower pad up; the reaction pulls the frame down onto the upper pad, so both faces clamp 5.3 kN at any pad wear.
2. Two units give 2.4 to 3.6 kNm (μ 0.32 to 0.48), 3.0 kNm nominal, 1.51 × the 95th percentile operating yaw moment.
3. Pad pressure 0.98 MPa, sliding speed 3.1 mm/s, heat 26 W. Estimated wear 6.6 mm in 20 years of 9 mm usable (to confirm by test).
4. Spring stack at 0.50 h0 per disc: one year of wear lowers the clamp by 8 %. Re-set at the yearly service (load cell between the pads).
5. Pad change without lifting the nacelle: back off the adjuster, swing the frame down on the pins, slide the pads out sideways.

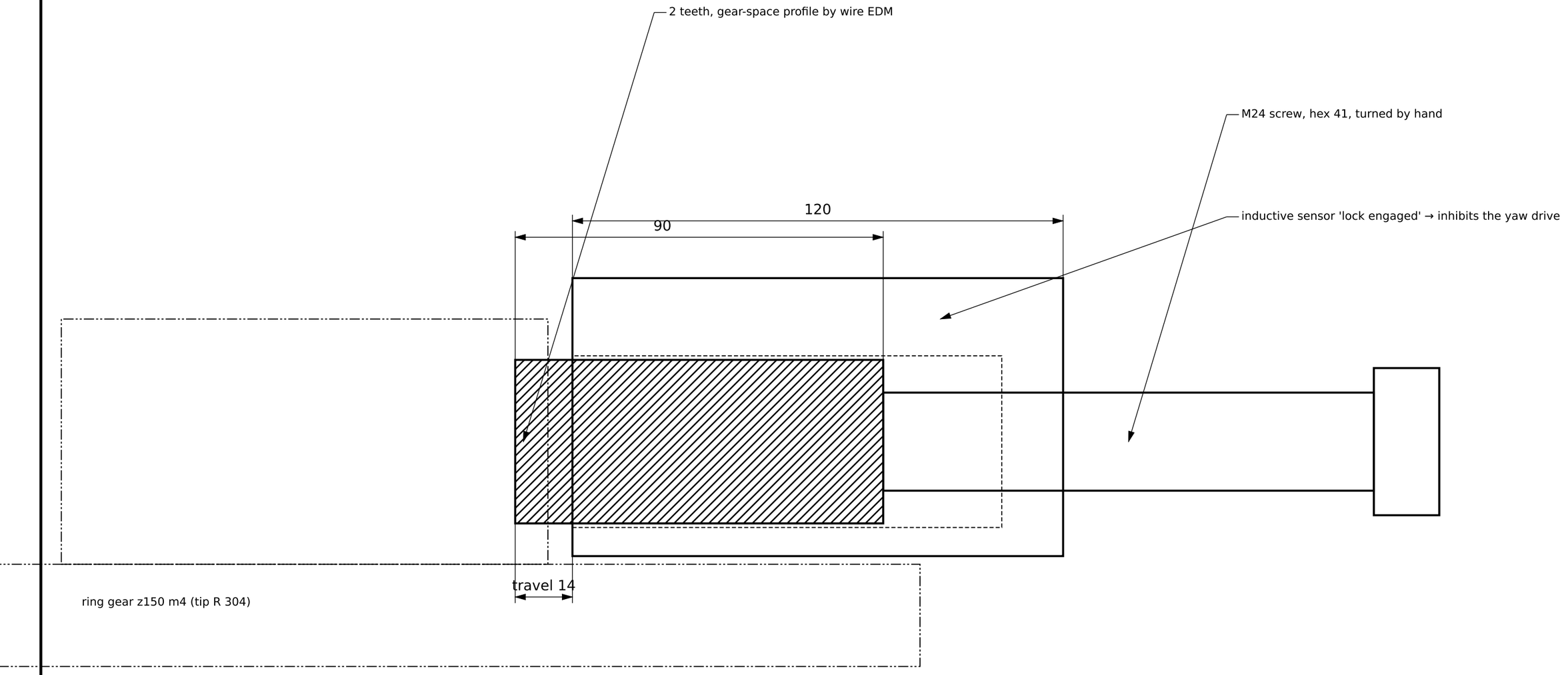
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

FRICTION PAD UNIT (PASSIVE, SPRING-LOADED)

DRAWING NO.	REV / DATE	SCALE (A3)
NY-410	A · 2026-11-05	1:2
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
frame S355J2+N, pins 1.4462	13 kg (model)	Claude / not checked

SECTION IN THE RADIAL PLANE (engaged) · 1:1



NOTES

1. Use: service only (yaw drives locked out). Turn the nacelle until a tooth space is opposite the block (encoder target), then screw the block in.
2. Strength: design yaw moment 8 kNm → 26.7 kN at the gear pitch radius on two teeth; tooth root shear and bending of the block S ≥ 3 (42CrMo4+QT, 900 MPa).
3. The housing bolts to the bedplate underside with 4 × M16 at the rear left corner (38.2°). Grease the slide; boot over the screw.

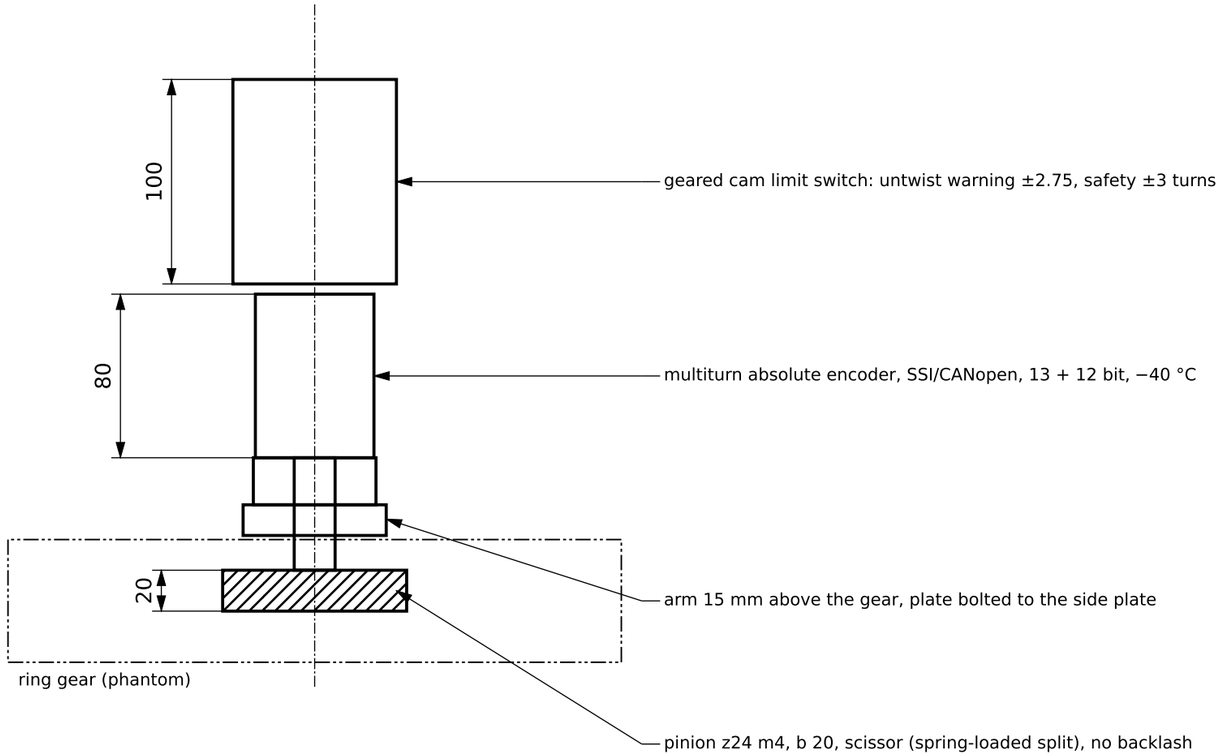
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

YAW LOCK (SERVICE)

DRAWING NO.	REV / DATE	SCALE (A3)
NY-430	A · 2026-11-05	1:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
housing S355J2+N, block 42CrMo4+QT	6.5 kg (model)	Claude / not checked

ELEVATION, VIEW TOWARDS THE SIDE PLATE · 1:3



NOTES

1. Ratio ring/pinion 150/24 = 6.25: ±3 nacelle turns = ±18.75 turns at the switch.
2. Centre distance 348 mm at -68° (right side, behind the drive), outside the side plate so the instruments stand beside it.
3. Cam contacts are positively opened, wired into the hardwired safety chain (PL d). The encoder is only for control (yaw angle, twist count, untwist).
4. Plausibility: proximity switch at 0° on a ring tooth marker checks the encoder every revolution.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

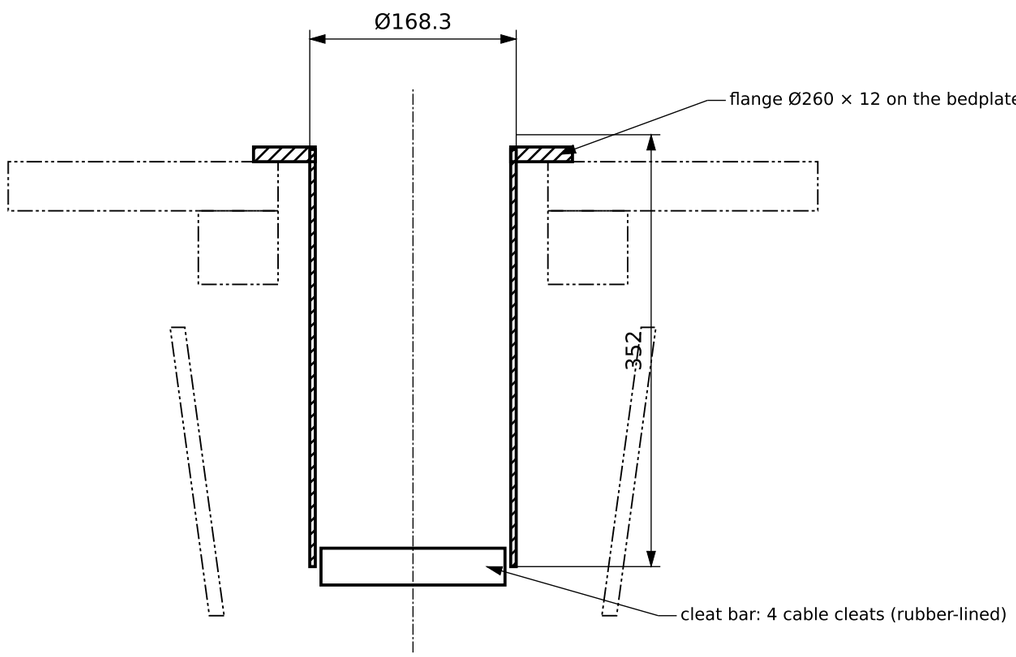
12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

YAW ENCODER AND CAM LIMIT SWITCH UNIT

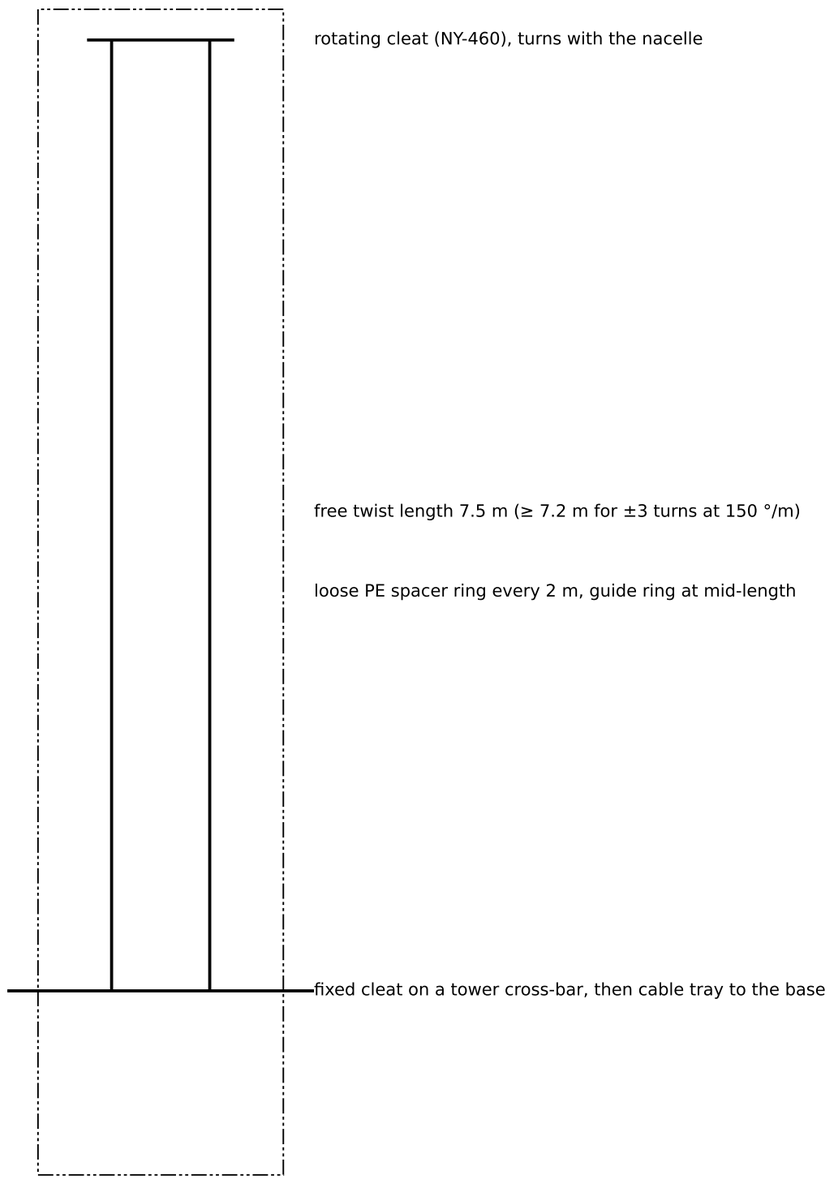
DRAWING NO.	REV / DATE	SCALE (A3)
NY-440	A · 2026-11-05	1:3
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
bracket S355J2+N, pinion 42CrMo4+QT	6.9 kg incl. instruments	Claude / not checked

Cable (all torsion-rated for wind turbine loops, −40 °C)	Ø approx.
Generator 4G6 + screen, 0.6/1 kV (doc 12)	20 mm
48 V auxiliary / backup supply 2 × 10 mm²	14 mm
Safety chain + control signals 12 × 1.0 mm², screened	12 mm
Industrial Ethernet (Cat 6A, torsion) + fibre option	12 mm
Earthing: separate Cu braid via the yaw lightning contact	–

HANGER IN THE YAW BORE · 1:5



TWIST LOOP IN THE TOWER (not to scale)



NOTES

1. Untwist by the controller at ±2.5 turns when the 10 min mean wind is below 5 m/s; safety chain opens at ±3 turns (NY-440 cam switch).
2. No slip ring for power (doc 07). Cables leave the bedplate inside the hanger tube, so they never touch the rotating ring or the adapter.
3. Inspection through the adapter window (NY-400) at every service: sheath wear, cleats, twist count against the encoder.

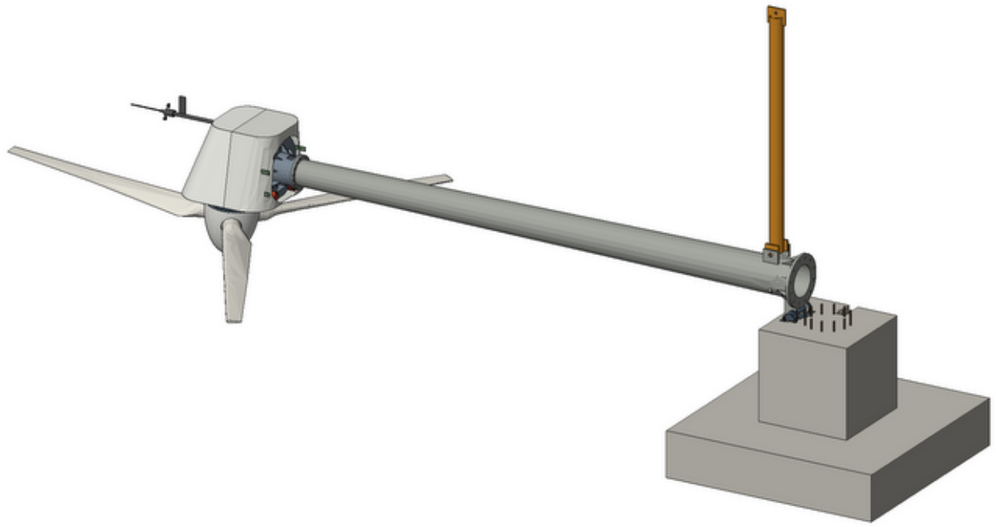
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

CABLE HANGER AND TWIST LOOP

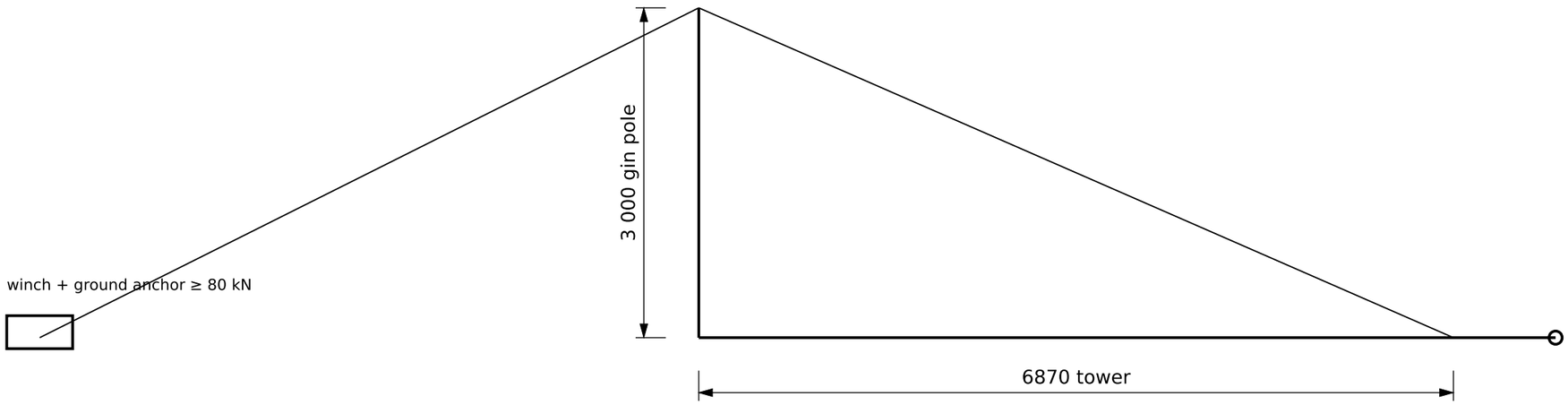
DRAWING NO.	REV / DATE	SCALE (A3)
NY-460	A · 2026-11-05	1:5 / NTS
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
tube S355 Ø168.3 × 4.5, flange S355J2+N	10 kg (model)	Claude / not checked

AXONOMETRIC VIEW (not to scale)



Item	Value
Gin pole	3.0 m, fixed square to the tower on its lugs
Peak line pull	≈ 38 kN (wind ≤ 10 m/s)
Bridle	2 × wire rope Ø16 6×36 IWRC to the top flange lugs
Pull line	Ø16 wire rope, winch ≥ 50 kN with 1:2 block
Ground anchor	screw anchors or concrete block, ≥ 80 kN at 6 m
Alternative	truck crane ≥ 3 t at 9 m reach (lift by the top flange)

TILT-UP (tower horizontal on the hinge, head fitted, rotor locked)



NOTES

1. Two people + winch operator; wind below 10 m/s; nobody inside the fall zone (1.5 × tower + rotor). Rotor locked with pin NP1-320, blade B1 pointing up away from the ground.
2. Lower the tower the same way for service; support the head on a trestle 1.0 m high so the lower blade clears the ground.
3. After erection: fit base nuts and torque (NS6-900), remove the gin pole, keep it with the site kit.

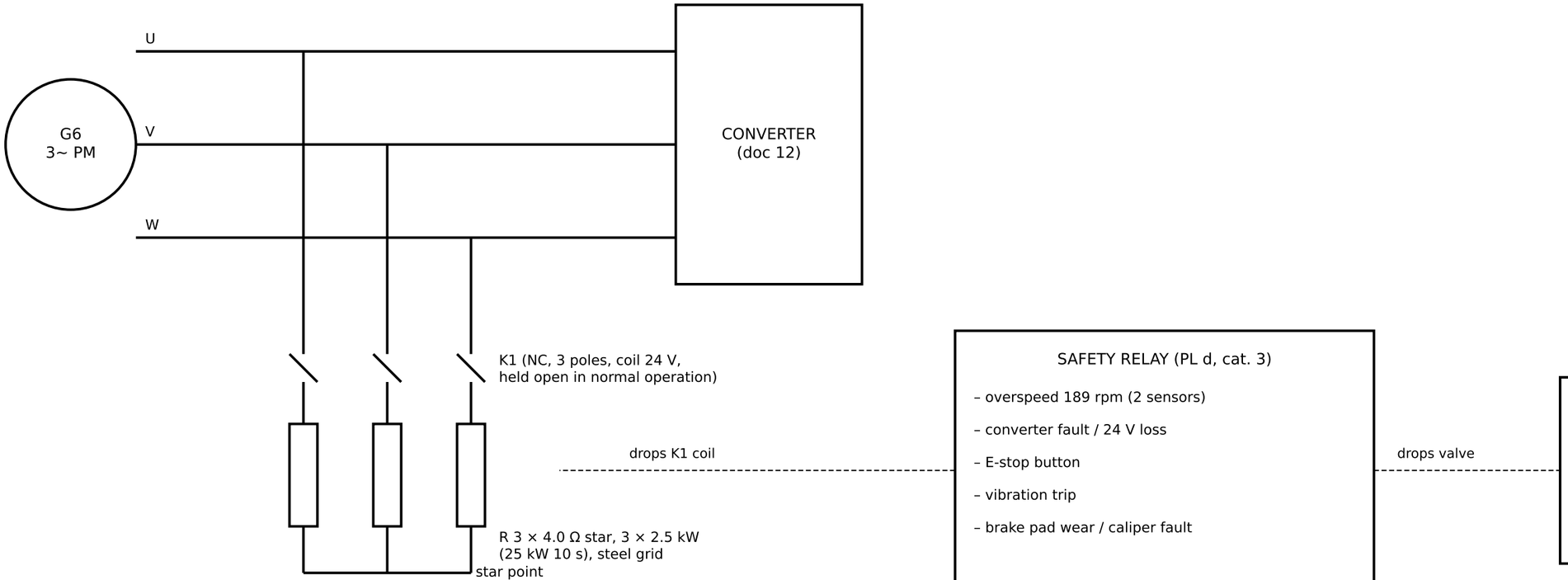
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

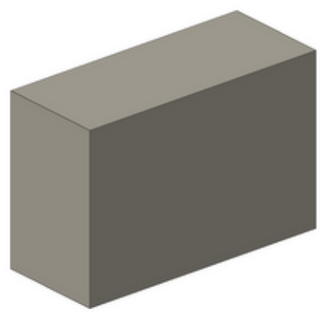
GIN POLE AND TILT-UP RIGGING (ERECTION TOOL)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-420	B · 2026-11-06	1:50
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355 tube 168.3 × 6.3, plates 25	94 kg	Claude / not checked

SINGLE-LINE: BOTH STOPPING CHANNELS DROP TOGETHER ON A TRIP



AXONOMETRIC VIEW (not to scale)



NOTES

1. Doc 29 / 31: K1 is normally closed. While the safety relay is healthy it holds K1 open; any trip or loss of 24 V closes K1 and drops the brake at the same time. Either channel alone keeps the rotor below the trip speed (doc 31: brake alone stops in ≤ 2.4 s, peak 173 rpm; resistor alone slows to a 3–15 rpm crawl).
2. Resistor current 43 A peak at trip speed with hot magnets; energy of one stop ≈ 120 kJ. Grid resistors in a ventilated 1.4301 box (600 × 400 × 250) on the tower base cabinet wall, not in the nacelle.
3. K1: 3-pole contactor with mechanically linked mirror contact (EN 60947-4-1 Annex F) for monitoring; AC-3 rating ≥ 63 A at 690 V.
4. Brake fault alarm: with the brake failed the rotor only crawls, so a failed or worn brake keeps the turbine parked and yawed out until repaired (doc 31 §6).
5. Spring pack stays 1.5 kNm (Martin, 2026-11-06). Parked yaw-lost 50-year peak aero torque 1.45 kNm; if exceeded the rotor creeps slowly against the brake and the closed K1 (doc 32 §1). Use the lock pin NP1-320 for long unattended outages.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

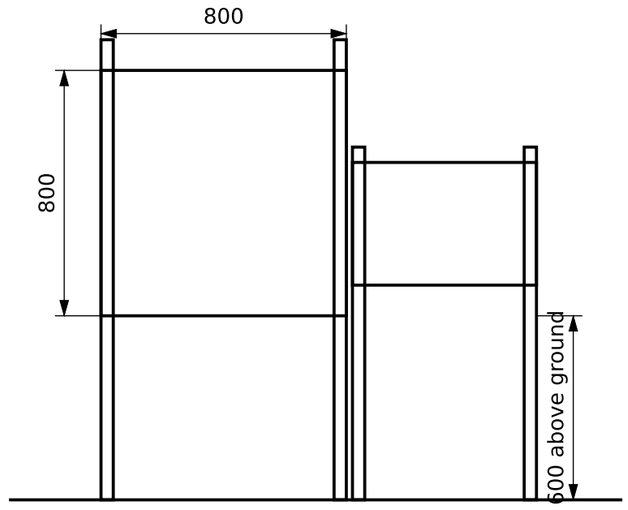
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BRAKING RESISTOR BOX AND SAFETY CHAIN

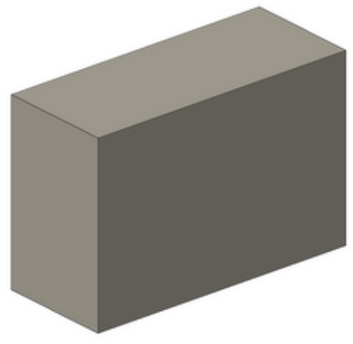
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-600	B · 2026-11-06	schematic / 1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
Box 1.4301, IP54; resistors steel-grid	≈ 18 kg	Claude / not checked

Cabinet content (doc 12 modules)	Note
SiC active rectifier 1 200 V, 750 V DC link	6 kW firmware limits
Dump chopper 12 kW + heater feed	hardware trigger 820 V
AC module 6.5 kW (grid / heater)	single- or three-phase
48 V battery module + 200 Wh aux battery	yaw-out without grid
Safety relay PL d, K1 contactor	NS6-600 chain
Controller, modem, surge arresters T1+T2	earth bar to the tower lug

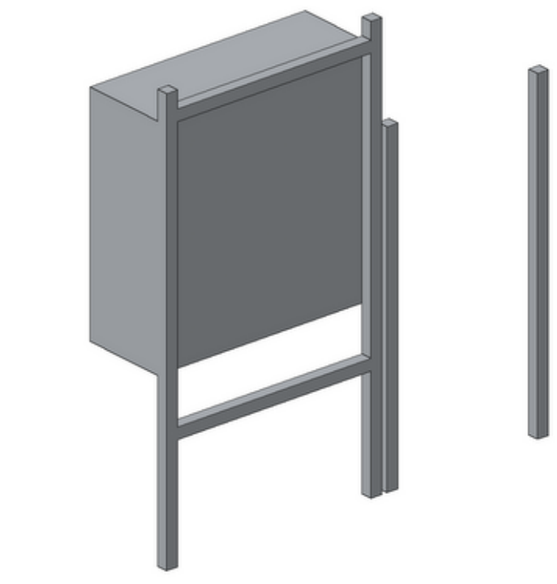
FRONT (from +Y): cabinet left, braking resistor box right (vented)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Stand on two concrete pads 400 × 400 × 600 (or bolted to the pedestal side); cabinet door faces away from the tower; minimum 1.0 m from the tower axis, outside the rotor disc projection is not required (tip at 5 m).
2. Resistor box NS6-600 mounted 20 mm proud of the stand, louvres down, ≥ 300 mm from the cabinet (heat).
3. Cables enter from below through the two PE ducts in the footing (NS6-800).

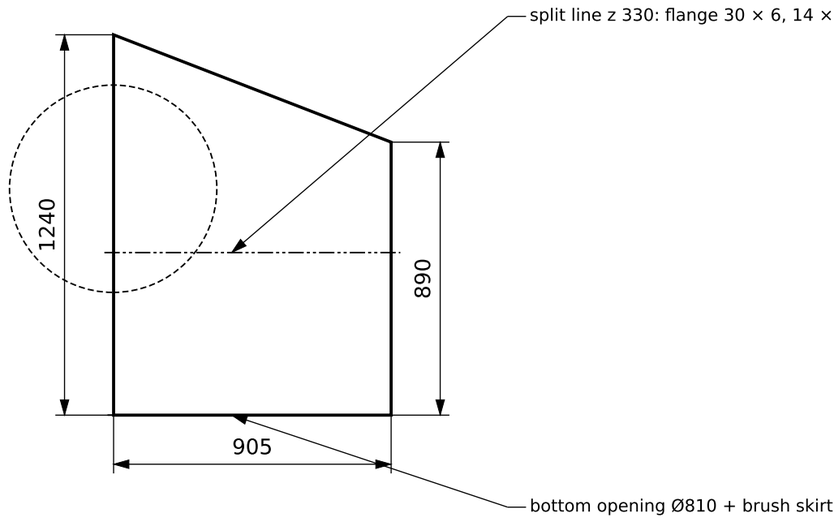
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

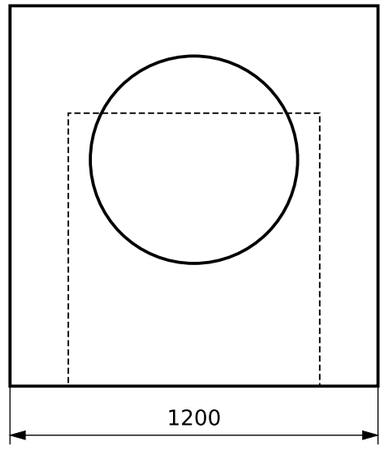
TOWER BASE CABINET, STAND AND RESISTOR BOX POSITION

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-620	B · 2026-11-06	1:20
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
enclosure 1.4301 IP55; stand S235 galv.	95 kg + 18 kg	Claude / not checked

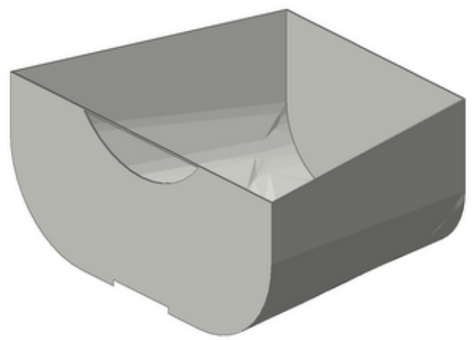
SIDE VIEW (yaw axis at x 0)



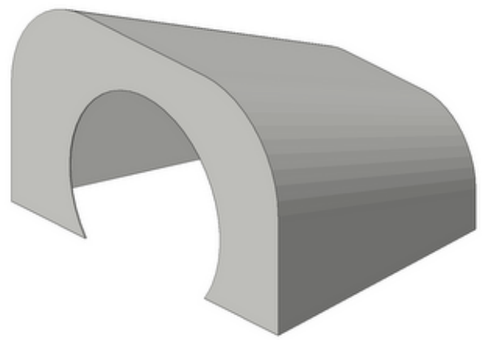
FRONT (opening Ø676 round the drum, 15 gap)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Lofted rounded-rectangle shell, front 1 200 × 1 240 (R330) to rear 820 × 890 (R230); encloses the disc, caliper, bedplate, yaw drives and pad units; drum and spinner stay outside (drum cools in the free air).
2. Lay-up 4 mm: gelcoat 0.5 + 3 × 450 g CSM + 1 × 300 g woven; PU foam hat-section ribs every 300 mm; lid has 2 lifting points and a gas-strut-held service hatch 500 × 400 over the junction box.
3. Lower tray hangs on the NS6-710 arms (4 × M10 rubber mounts); the lid bolts to the tray flange. Air in at the bottom skirt, out through a labyrinth vent at the rear top (no forced cooling).

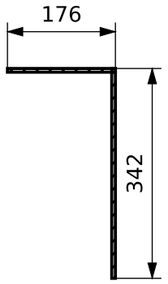
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

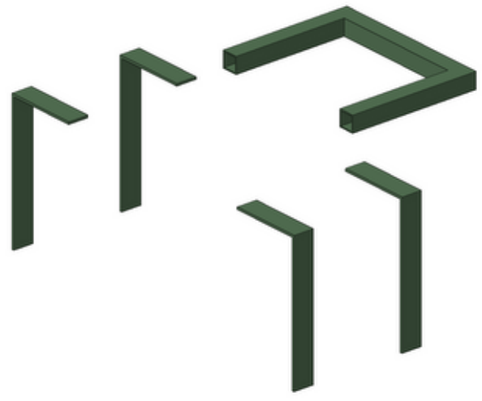
NACELLE COVER · LOWER TRAY AND UPPER LID (GRP)

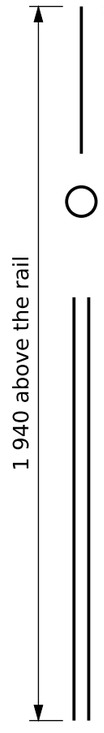
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-700	B · 2026-11-06	1:20
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
glass/polyester 4 mm, gelcoat RAL 9010, PU foam ribs	≈ 22 kg	Claude / not checked

ARM (4 off): flat 50 × 6, bolted 2 × M10 to the side plate



AXONOMETRIC VIEW (not to scale)

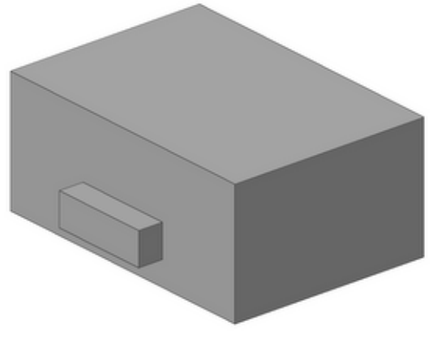




AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Rear rail RHS 40 × 40 × 3 on the bedplate rear tie carries the junction box NS6-730 (IP66, 400 × 340 × 150) and the met mast clamp.
2. Met mast: wind sensors 0.5 m above the cover top (doc 25), air terminal bonded with 50 mm² Cu to the bedplate; the bedplate–tower path goes through the yaw pad brush (NY-410).
3. Junction box: phase terminals 3 × 16 mm², brake/sensor terminals, surge arresters for the sensor lines, 24 V supply; cable glands at the bottom only.

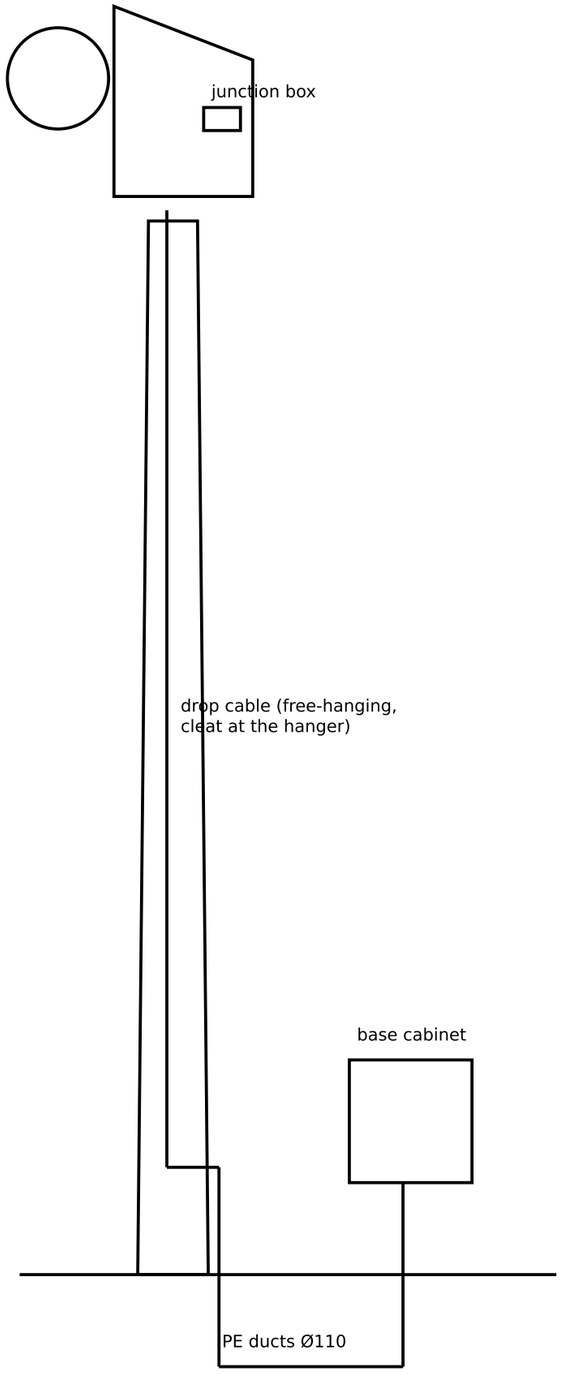
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

COVER ARMS, REAR RAIL, MET MAST NS6-720, JUNCTION BOX NS6-730

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-710	B · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S235 galv.; mast 1.4404 Ø42.4 × 2	22 kg; 4.5 kg; 9 kg	Claude / not checked

ROUTE (nacelle → loop → tower → cabinet)

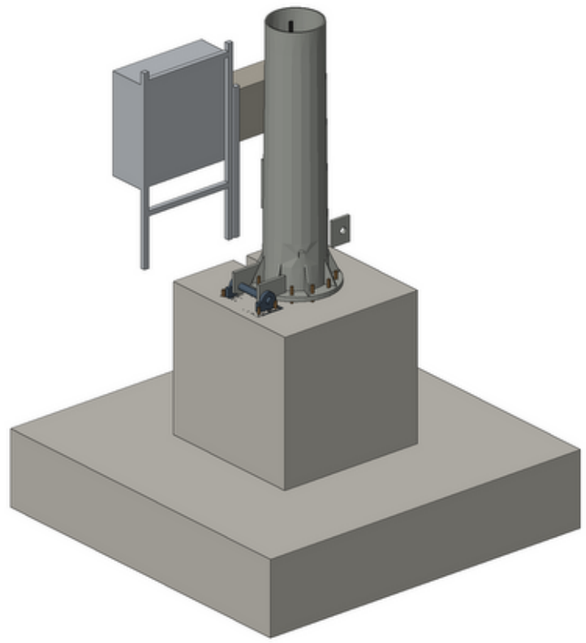


Cable	Type	From → to	Length m
W1 power	H07RN-F 4G10 (3 ph + PE)	generator terminals → junction box	3
W2 power	H07RN-F 4G10, 2.5 turns loop	junction box → hanger loop → hand hole	12
W3 power	NYY-J 4 × 10 in duct	hand hole → cabinet	4
W4 brake	ÖLFLEX 4G1.5 shielded	caliper → junction box → cabinet	16
W5 sensors	LiYCY 12 × 0.5	overspeed, encoder, cam, met, PT100 → cabinet	16
W6 yaw motors	ÖLFLEX 7G1.5	2 gearmotors + brakes → cabinet	16
W7 earth	Cu 50 mm² green/yellow	air terminal → bedplate; tower lug → earth ring	6
W8 data	CAT6 outdoor or fibre	cabinet → house / grid meter	site

NOTES

1. Cable loop NY-460: power and control cables hang free in the tower below the cleat with 2.5 turns of slack; cable-twist limit ±2.5 turns from the encoder/cam switch (doc 25), auto-untwist when idle.
2. Lightning (LPL I, doc 02): air terminal → bedplate → yaw pad brush → adapter → tower → base flange → 2 earthing lugs → ring earth (≤ 10 Ω). Generator winding star point not earthed (IT, insulation monitor in the cabinet).
3. Surge protection: T1+T2 arresters at the cabinet entry for W2/W3, data-line arresters for W5/W8; shields bonded at both ends via the junction box gland plate.
4. All cables UV and cold resistant to −40 °C; drip loops at the hand hole; glands IP68; label every core at both ends.

AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

CABLING ROUTES, EARTHING AND LIGHTNING

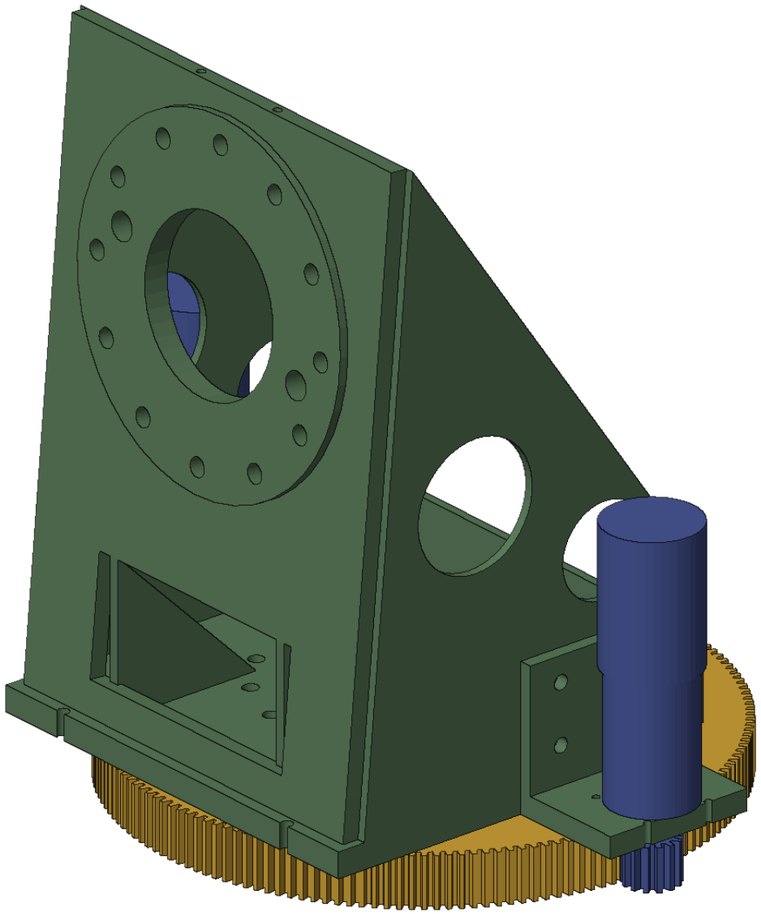
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-800	B · 2026-11-06	schematic
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see cable schedule	–	Claude / not checked

6 Assembly sequence

Workshop steps 1–11, site steps 12–13

6 Assembly sequence

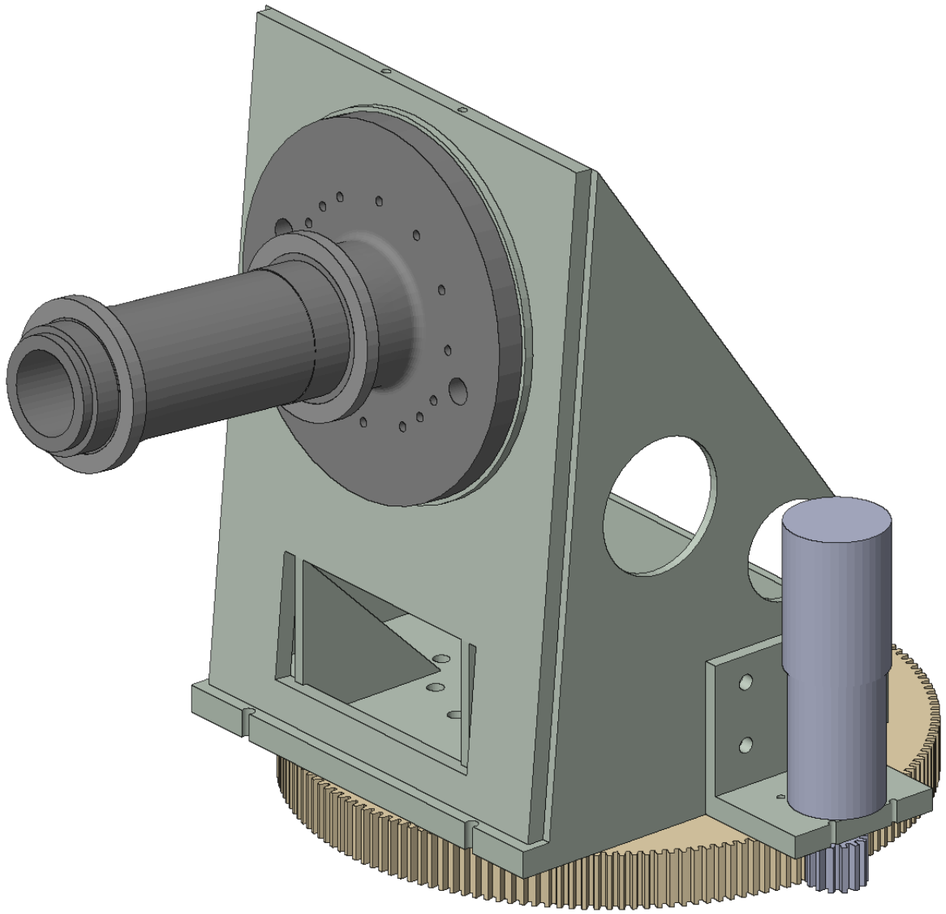
Step 1 Bedplate NP8-300 on the stand; yaw slewing ring Dpw 410, gearmotors and their brackets



- Set the NP8-300 bedplate upside-down on the assembly stand; check the yaw flange flatness (0.05) and the front-face angle (5.00° ± 0.05°) and record them.
- Fit the Dpw 410 slewing ring inner ring: 16 × M16 HV on PCD 350, cross pattern, 280 Nm (soft gearing mark towards the lock side).
- Fit the gearmotor brackets NP1-420 and the two gearmotors; set the backlash 0.15–0.30 mm at the soft point with the bracket shims.
- Check: ring turns freely by hand with both motor brakes released; record the turning torque.

Parts (new in colour, earlier steps light): NP8-300, NM8-911-I, NM8-911-O, NM8-912-L, NM8-912-R, NP1-420-L, NP1-420-R. Torques: NM8-900.

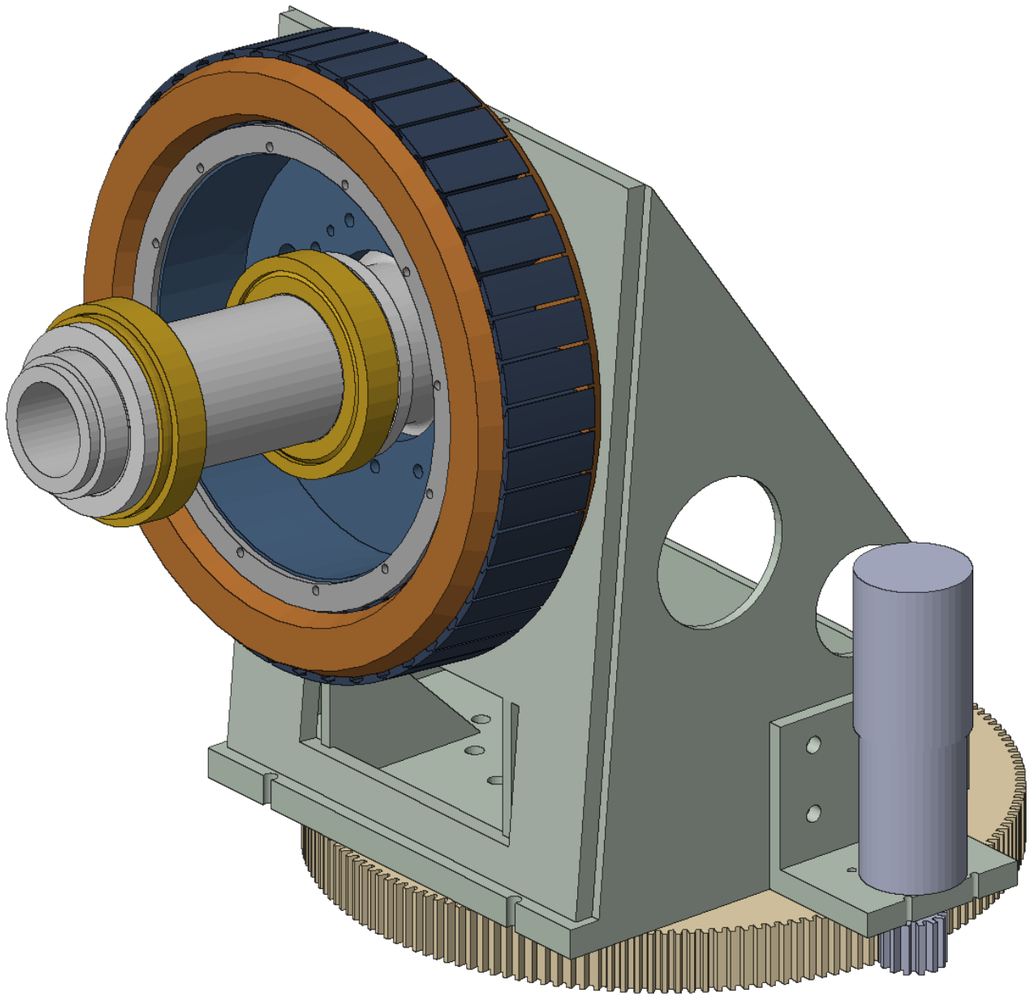
Step 2 Spindle Rev D with front spacer and rear seal sleeve bolted to the bedplate front plate



- Clean the spindle flange and pad; fit the Ø200 spigot; 12 × M20 10.9 from inside, 540 Nm in three passes.
- Fit the front spacer NP1-101 and rear seal sleeve NP1-102 (heat to 120 °C); cables through the two Ø28 holes before the stator goes on.
- Check: flange face square to the yaw axis (5° tilt) within 0.05/400. Spindle Rev D is 30 mm longer: check the length 466 ± 0.1 before fitting.

Parts (new in colour, earlier steps light): NP1-100D, NP1-101, NP1-102. Torques: NM8-900.

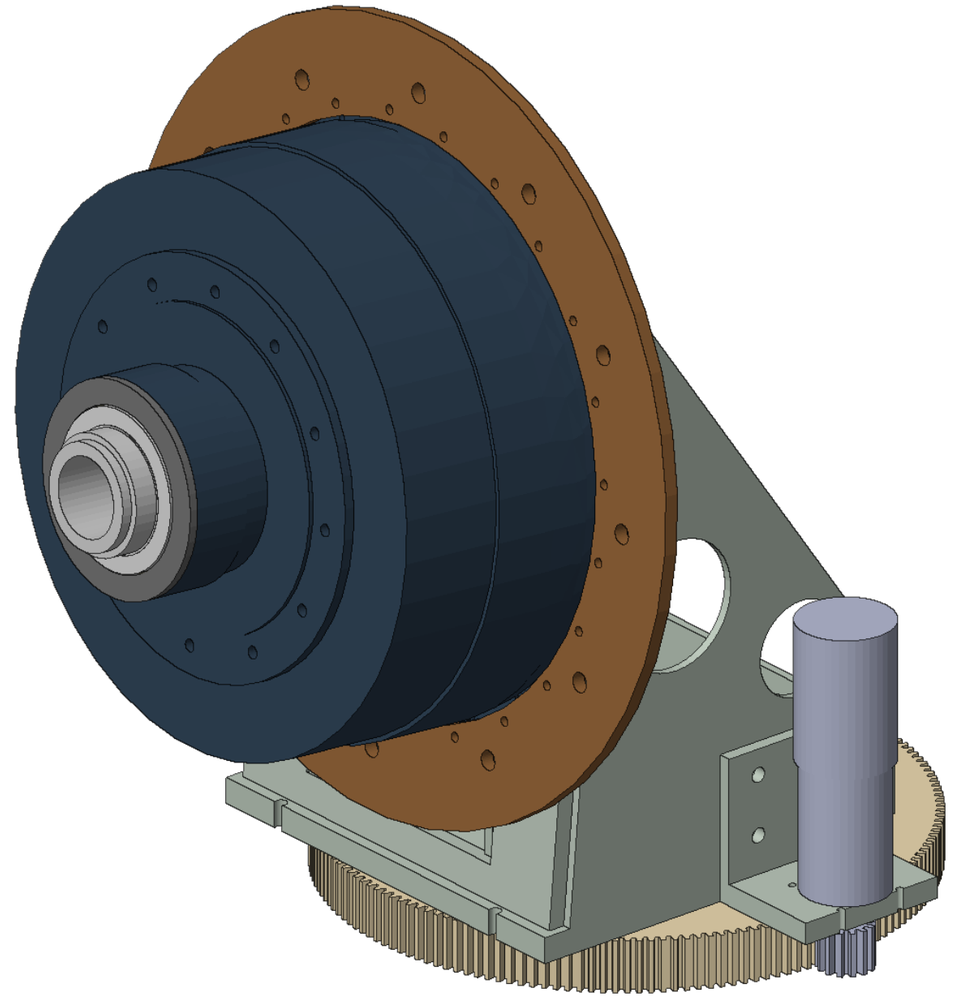
Step 3 Stator Rev D: carrier, wound tooth segments (104 mm), clamp ring; bearing cones on the spindle



- Assemble the stator Rev D: 45 wound tooth segments (104 mm stacks) slid into the carrier dovetails, clamp ring NP1-202 (doc 18 instructions, Rev C with the Rev D lengths).
- Insulation 1 kV, winding resistance balance ≤ 2 %, surge test. Carrier to spindle: 12 × M12, 110 Nm (thermal paste).
- Heat the bearing cones (110 °C) and fit them; grease 60 % fill.

Parts (new in colour, earlier steps light): NP1-200D, NP1-201D, NP1-203D, NP1-202, NM8-913-F, NM8-913-R. Torques: NM8-900.

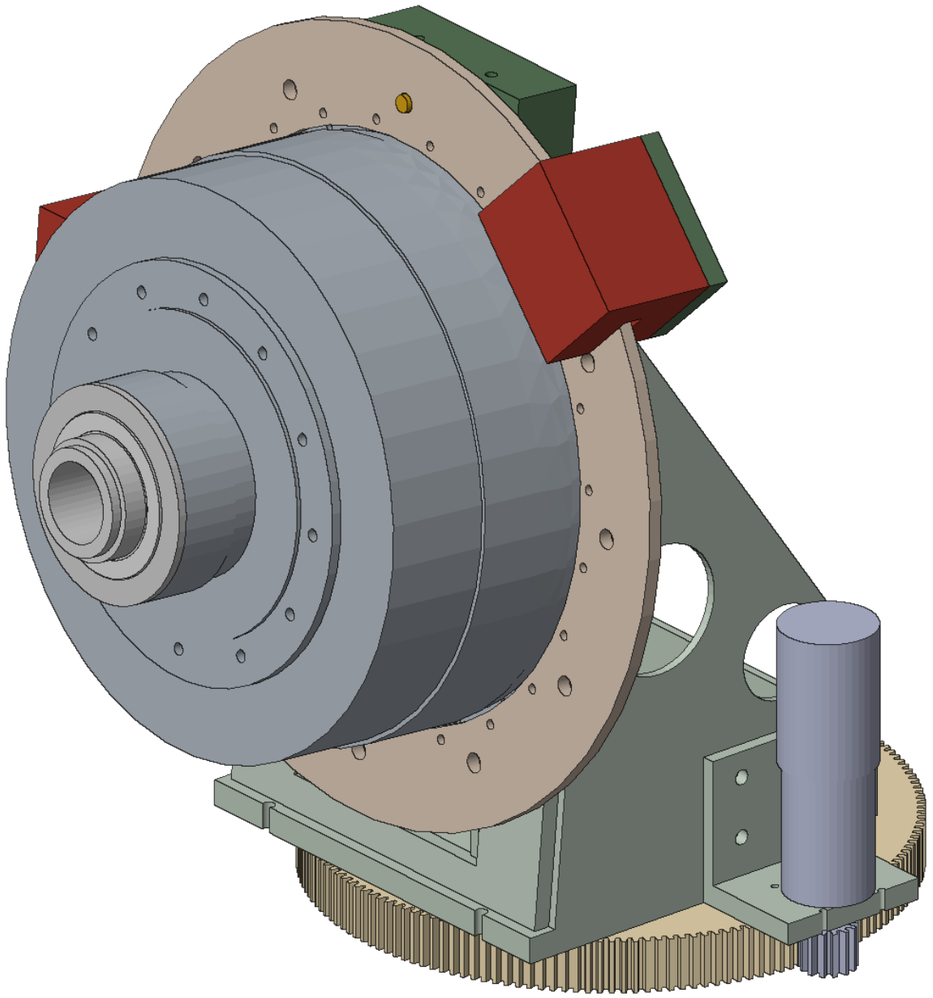
Step 4 Rotor drum Rev D with 7 magnet rows lowered over the stator on the guide tool; disc, seal covers, rear cover



- Magnet drum: use the drum guide NP1-T01 (lengthened 30 mm) and the magnet safety rules of doc 18 (no steel tools within 1 m, two people). 280 blocks in 7 rows, skew steps ±0.4°.
- Lower the drum on the guide screw; set the bearing preload with the KMT 27 locknut (rolling torque 6–8 Nm).
- Fit the disc (24 bolts, NM8-900), seal covers and the Rev C rear cover with fan blades. Remove the guide.
- Check: air gap ≥ 1.4 mm at 8 positions, drum run-out ≤ 0.2 mm, back-EMF at 60 rpm vs generator_m8 (+3.4 % on Rev C).

Parts (new in colour, earlier steps light): NP1-210D, NP1-220F, NP1-230, NP1-211-F, NP1-211-R, NP1-212C. Torques: NM8-900.

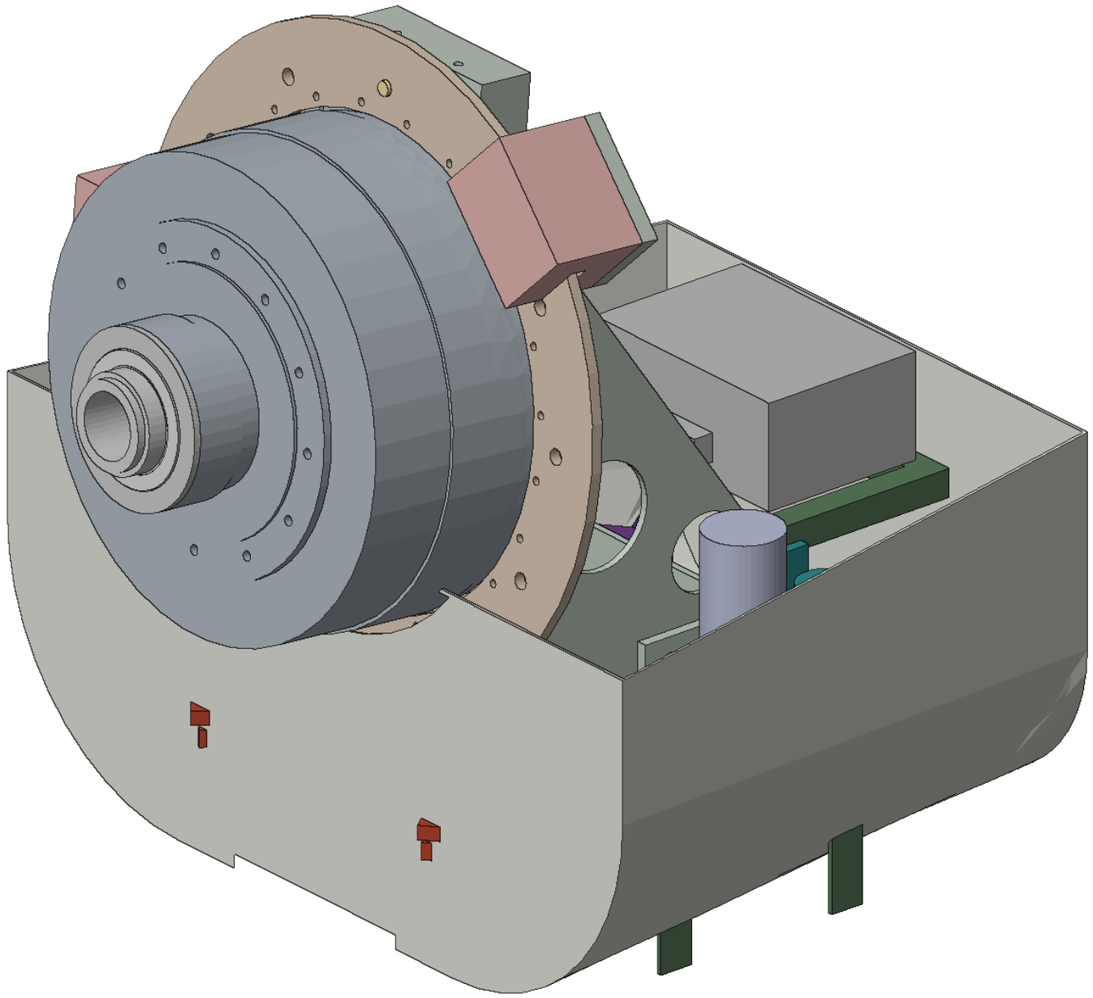
Step 5 Three brake calipers on their brackets, lock pin holder with overspeed sensors



- Fit the two side caliper brackets NP1-310 and the third bracket NM8-311 (4 × M16 through the front plate). Fit the three calipers; set pad gaps; check release on 24 V.
- Fit the lock pin holder with the two overspeed sensors; test the lock pin in all 12 disc holes.
- Check: holding torque test 14.4 kNm (torque arm on the drum), each caliper alone ≥ 4.8 kNm; sensor pulses.

Parts (new in colour, earlier steps light): NP1-310, NM8-915, NM8-311, NM8-915-T, NP1-320, NP1-321, NM8-914. Torques: NM8-900.

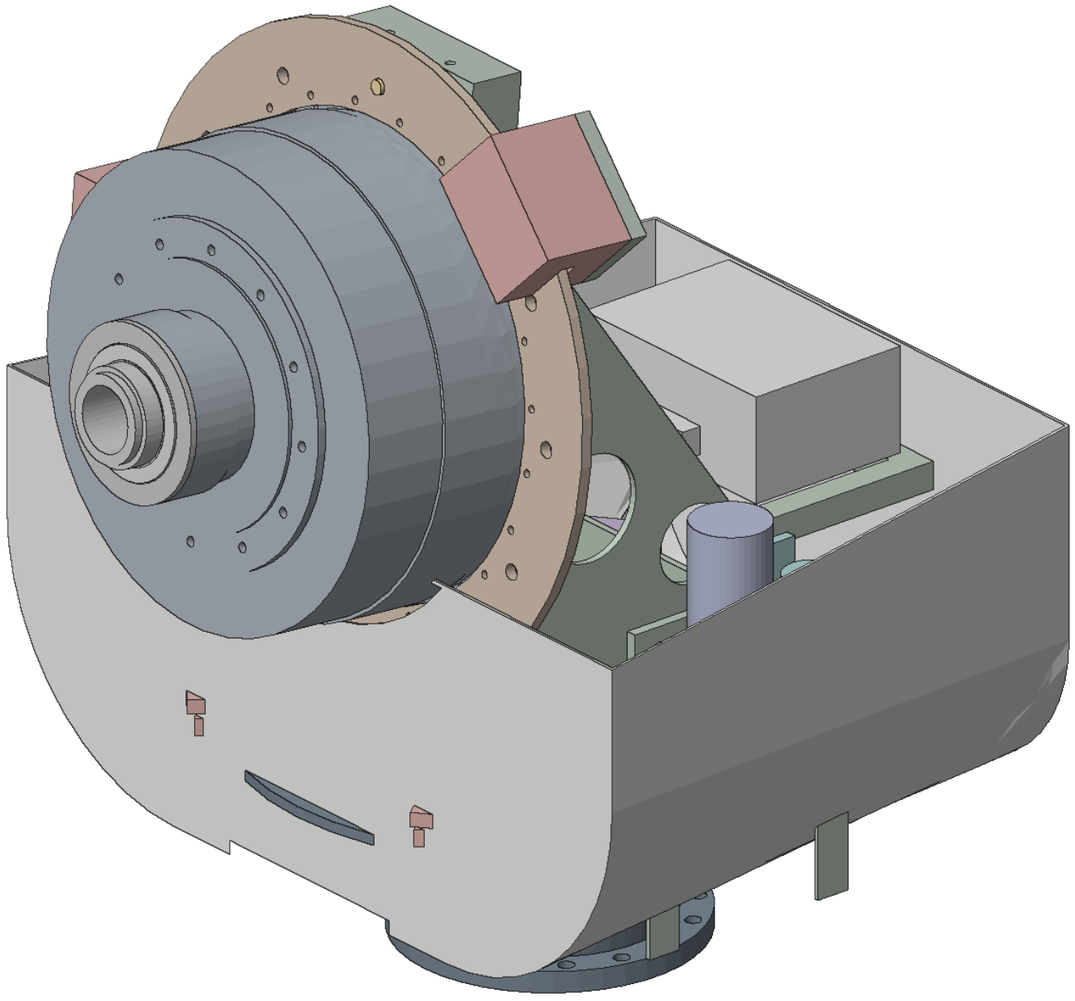
Step 6 Yaw pad units, lock, encoder and cable hanger; cover frame, lower cover tray and junction box



- Fit the pad units NY-410 (radius +25), yaw lock NY-430, encoder unit NY-440 and the cable hanger NY-460.
- Fit the cover arms, the lower GRP tray, the rear rail and the junction box; wire W1, W4, W5, W6.
- Check: pad clamping force, encoder zero at yaw 0°, cam switch at ±2.5 turns.

Parts (new in colour, earlier steps light): NY-410-L, NY-410-R, NY-430, NY-440, NY-460, NM8-710, NM8-700-1, NM8-730. Torques: NM8-900.

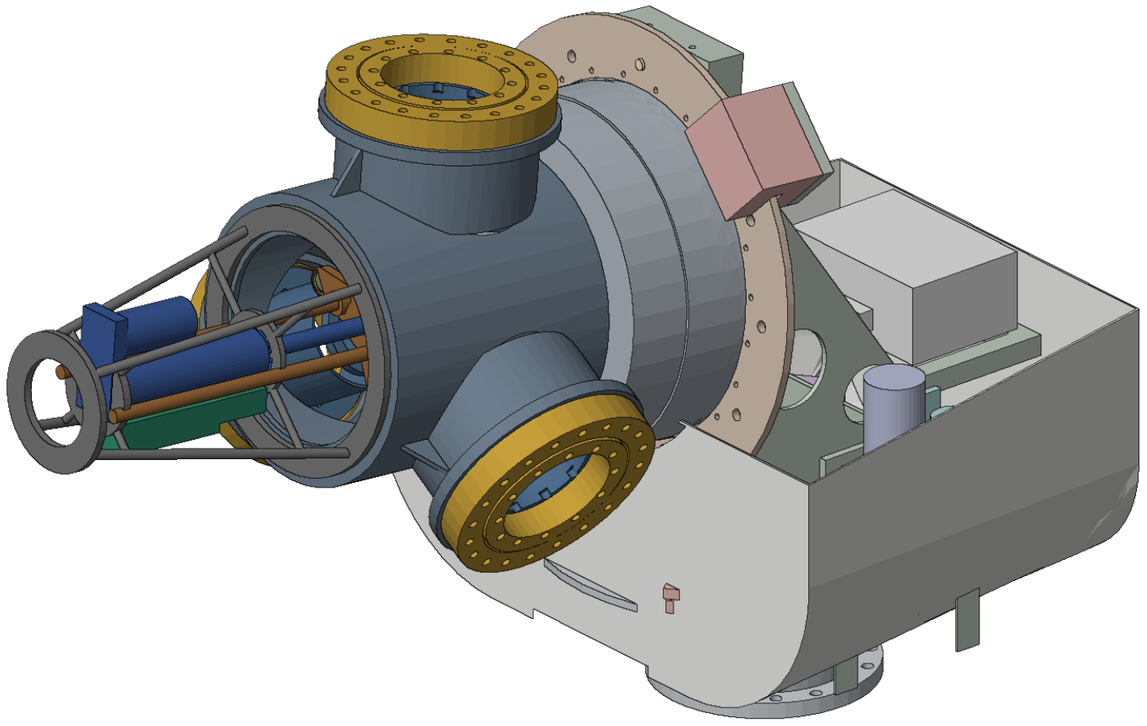
Step 7 Tower-top adapter on the yaw outer ring (nacelle now complete for transport)



- Lower the nacelle onto the tower-top adapter NY-400 (outer ring 16 × M16 PCD 490, 280 Nm). The nacelle is a transport unit with lifting points on the bedplate.
- Check: yaw turns 360° with the drives; pads slip at the doc 25 torque (scaled to the Dpw 410 ring).

Parts (new in colour, earlier steps light): NY-400. Torques: NM8-900.

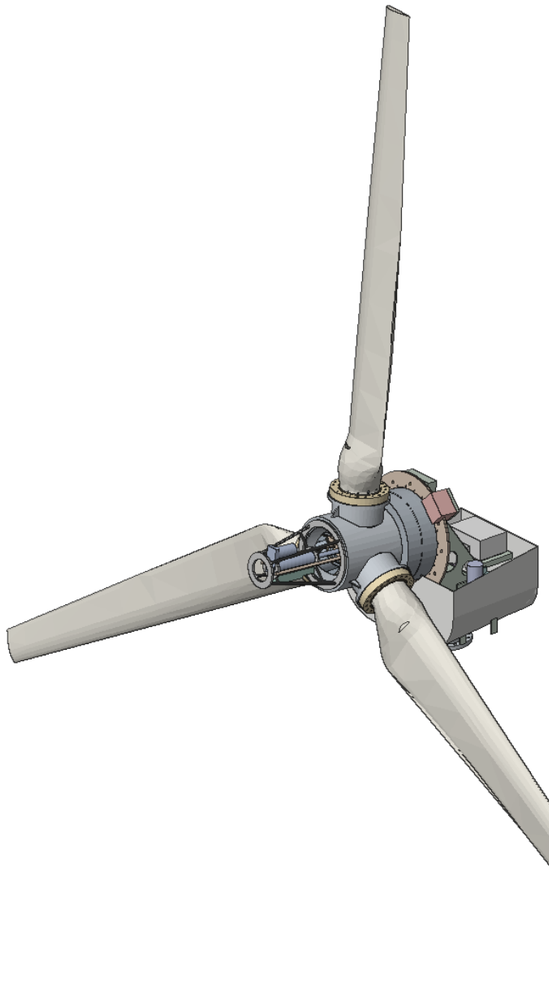
Step 8 Pitch hub pre-assembled on the bench (weldment, pitch bearings, cranks, links, yoke, cylinder, supercaps, nose frame), then on the drum flange (12 × M16 10.9)



- Bench: weld-checked hub NM8-200; fit the three pitch bearings (20 × M16 each, 280 Nm), cranks r140 with pins, links, yoke and guide rods, the cylinder and motor, supercap module and drive, nose frame.
- Run the pitch system on the bench from −2° to 92° and back (limit switches, end stops, supercap feather test, time ≤ 4 s).
- Fit the hub on the drum spigot (Ø360), 12 × M16 10.9, 280 Nm in 3 passes; mark blade 1 at the lock-pin azimuth; connect the slip ring.

Parts (new in colour, earlier steps light): NM8-200, NM8-250, NM8-260, NM8-261, NM8-262, NM8-270, NM8-280, NM8-290. Torques: NM8-900.

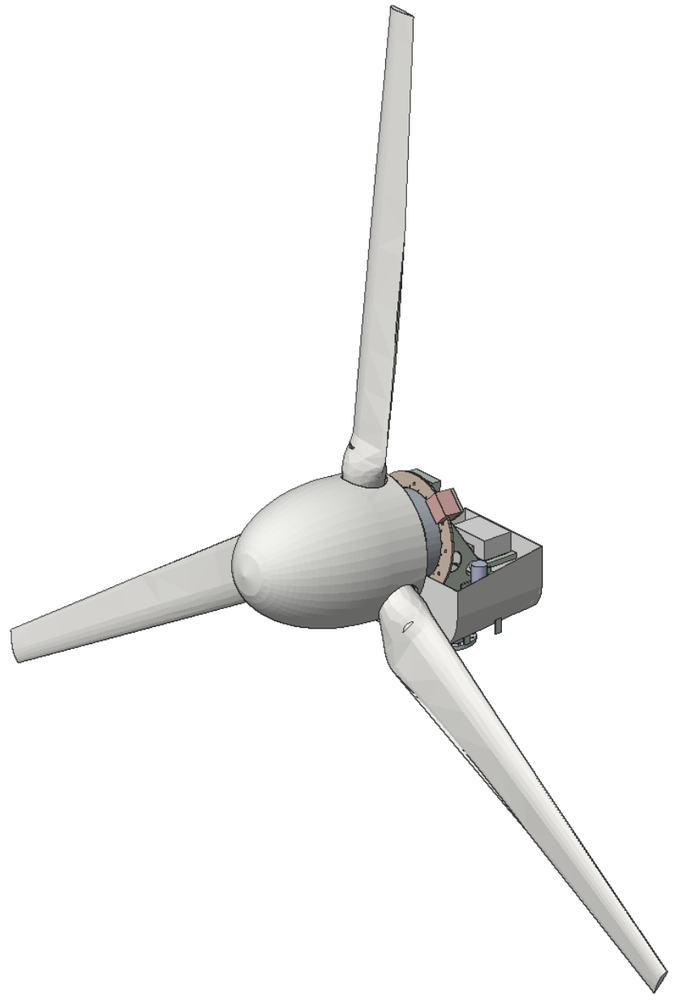
Step 9 Blades on the pitch bearings (12 × M16 each), pitch zero set with the cylinder at its −2.0° end stop



- Cylinder at the −2° end stop. Mount B1 on bearing 1, B2, B3 by their static-moment record: 12 × M16 10.9, 280 Nm star pattern.
- Check with the inclinometer at r 2 000 that all three blades read −2.0° ± 0.1° (linkage adjusted at the rod ends); tip track ±15 mm.

Parts (new in colour, earlier steps light): NM8-100-1, NM8-100-2, NM8-100-3, NM8-105. Torques: NM8-900.

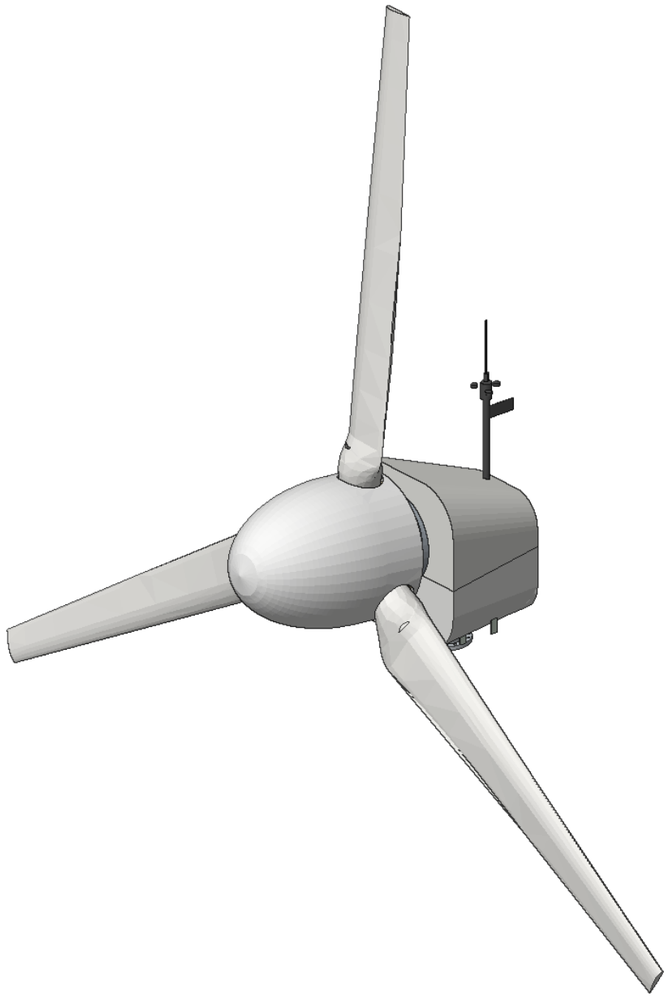
Step 10 Spinner with blade boots



- Fit the three spinner segments with the blade boots (9 brackets, A4 M8, 12 Nm). Check 10 mm clearance to all pitch parts over the full stroke.

Parts (new in colour, earlier steps light): NM8-210. Torques: NM8-900.

Step 11 Upper cover lid and met mast



- Fit the upper cover lid and the met mast; bond the air terminal to the bedplate (50 mm² Cu).
- Factory test: pitch, brake, yaw, sensors; record insulation and EMF.

Parts (new in colour, earlier steps light): NM8-700-2, NM8-720. Torques: NM8-900.

Step 12 On site: footing with anchor cage, hinge brackets, tower lying on the hinge; head bolted to the tower top



- Site: excavate to 1.6 m, blinding, bottom mesh, anchor cage NM8-510 with the template ring (± 3 mm, level ± 1 mm), hinge anchors, ducts, earthing ring.
- Pour C30/37 XF4, cure ≥ 7 days. Fit the hinge brackets NM8-410 and levelling nuts; lay the tower on trestles on the hinge pin.
- Bolt the head to the tower top (16 × M20 HV, 540 Nm) on the ground; pull W2/W5; rotor locked with B1 up, blades feathered.

Parts (new in colour, earlier steps light): NM8-500, NM8-510, NM8-410, NM8-400. Torques: NM8-900.

Step 13 Tilt-up with the gin pole and winch, base nuts torqued; cabinet, resistor box, cabling, hand-hole cover



- Gin pole and rigging; winch line pull ≈ 66 kN (use a 2-part or 4-part line); wind below 8 m/s; nobody in the fall zone.
- Land on the levelling nuts, fit top nuts, grout 30 mm; after 7 days tension the anchors (NM8-900). Remove the gin pole.
- Cabinet, resistor box, cabling W3, earthing ≤ 10 Ω, hand-hole cover. Commissioning checklist §7.

Parts (new in colour, earlier steps light): NM8-420, NM8-620, NM8-600, NM8-800, NM8-401. Torques: NM8-900.

7 Erection, commissioning and inspection

Before power-up

- All torque stripes present; anchor, adapter, hub, pitch bearing and blade bolt records signed (NM8-900).
- Tower plumb within 0.3 % in two directions; grout cured; hand-hole cover fitted.
- Earthing ≤ 10 Ω; continuity air terminal → bedplate → pad brush → tower → earth ring ≤ 0.2 Ω; hub bonded across the slip ring.
- Insulation: generator ≥ 100 MΩ at 1 kV; cables ≥ 100 MΩ. Pitch supercap charged, heater on below 0 °C.
- Rotor lock pin out; yaw lock retracted; cover lid closed; nobody inside 24 m.

Functional tests (wind below 8 m/s)

- Safety chain: each trip input (overspeed × 2, E-stop, 24 V loss, vibration, converter fault, pitch fault) feathers the blades on supercap power and drops the brake after the delay; record the stop time.
- Feather on the supercap alone (mains off) from rated speed; brake alone from 100 rpm with the blades at −2°.
- 8 m controller settings loaded: speed loop 6 rad/s, normal pitch rate 20°/s, overspeed trip 165 rpm (18 m: 4 rad/s, 15°/s, 155 rpm). Trip checked by lowering the set point (test mode).
- Yaw ±360° with the cam stop at ±2.5 turns; exclusion zone 48–61 rpm set in the controller (passed under torque).
- Rotor balance at 120 rpm: nacelle 1P vibration, trial mass to ≤ 0.2 kg·m; pitch angle check (all three within 0.3°) with a camera at the tips.
- Tower frequency (tap test): f1 must be ≥ 2.65 Hz; if lower, widen the exclusion zone.
- Power curve: 48 h supervised operation; pitch regulation above 10 m/s; storm feather test on the next windy day.

Inspection plan

Interval	Inspection
50 h	Re-torque blade root, pitch bearing, hub, adapter and anchor bolts; pitch zero check
3 months	Blades (LE erosion, root cracks), brake pads, pitch linkage play, cover fixings
Yearly	Bolts 10 % sample; pitch bearing grease; cylinder and rod ends; supercap capacity test; slewing ring and yaw gear grease; main bearing grease; earthing; safety chain; brake torque
5 years	Lower the tower: full blade inspection, main bearing play, magnet sleeve, cable loop, pitch bearing play
After a storm > 50 m/s or an ice event	Visual check, pitch and yaw logs, brake slip log

Lowering the tower for service

- Stop, feather, yaw to the hinge direction, lock the rotor with B1 up, lock the yaw; rig the gin pole and winch; slacken the base nuts on the hinge side last.
- Lower to the trestle: nacelle, hub and pitch system are at working height.

Records to keep with the turbine

- Blade set static moments; magnet certificates; weld and UT records (hub, tower, bedplate, caliper bracket); torque records; commissioning results; f1 measured.

8 Open items before the prototype release

- Martin's decisions D1–D5 (doc 35): this book uses the recommended defaults.
- FE checks: hub neck-to-shell welds, NP8-300 ring seat and front plate with the third caliper bracket, spindle Rev D flange fillet.
- Supplier drawings replace the envelopes (pitch bearings, cylinder, slewing ring Dpw 410, gearmotors, calipers, main bearings); then rerun clash8.py.
- Blade first-article static test to SLM H × 1.35, insert pull-out, mould acceptance.
- Pitch system bench test (feather on supercap at −30 °C).
- Site investigation: soil class, frost depth, earthing, obstacle survey at 8 m; snow-drift height under the rotor.
- Tower frequency on site (≥ 2.65 Hz) and the fatigue at the base: measured strain on the prototype for one winter.
- Energy at 8 m: the 8 m tower is for open coast, highland and test sites; sheltered sites need the 18 m tower.
- All drawings NOT CHECKED: independent check of every sheet before parts are ordered.